ISTANBUL TECHNICAL UNIVERSITY ★ GRADUATE SCHOOL

Topological Data Analysis of Wearable Sensor Signals: Development and Evaluation of Recall-Oriented Subject-General and Subject-Personalized Models

M.Sc. THESIS

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Department of Mathematical Engineering

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Thesis Advisor: Prof. Dr. Atabey KAYGUN

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İSTANBUL TEKNİK ÜNİVERSİTESİ ★ LİSANSÜSTÜ EĞİTİM ENSTİTÜSÜ

Giyilebilir Sensör Sinyalleriyle Topolojik Veri Analizi: Düşme Vakalarını Kaçırmamayı Önceleyen Genel ve Kişisel Modellerin Geliştirilmesi ve Değerlendirilmesi

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FOREWORD

Bu tez boyunca gösterdiği rehberlik ve destek için danışmanım Prof. Dr. Atabey KAYGUN'a içtenlikle teşekkür ederim. Çalışmada kullanılan hesaplama kaynakları, Türkiye Ulusal Yüksek Başarım Hesaplama Merkezi'nin (UHeM) 4025462026 numaralı tahsisi kapsamında sağlanmıştır.

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ABBREVIATIONS

ADL : Activities of Daily Living
CI : Confidence Interval
CNN : Convolutional Neural Network
CPU : Central Processing Unit
CV : Cross-Validation
EI : Expected Improvement
FAD : FallAllD 40 Hz Derived Variant
FN : False Negative
FP : False Positive
GP : Gaussian Process
GPU : Graphics Processing Unit
GUDHI : Geometry Understanding in Higher Dimensions
HAR : Human Activity Recognition
HPC : High Performance Computing
KKT : Karush–Kuhn–Tucker
LOSO : Leave-One-Subject-Out
LR : Logistic Regression
LSTM : Long Short-Term Memory
PD : Persistence Diagram
PH : Persistent Homology
RBF : Radial Basis Function
SMBO : Sequential Model-Based Optimisation
SMV : Signal Magnitude Vector
SVM : Support Vector Machine
TDA : Topological Data Analysis
TN : True Negative
TP : True Positive
TPE : Tree-structured Parzen Estimator

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Topological Data Analysis of Wearable Sensor Signals: Development and Evaluation of Recall-Oriented Subject-General and Subject-Personalized Models

SUMMARY

Among older adults, falls are a principal driver of traumatic injury, progressive loss of independence, and excess mortality; wearable inertial sensors offer a low-infrastructure path to continuous automatic detection. The asymmetry between error types is stark: a missed fall may leave a person unattended on the ground for hours, whereas a false alarm costs at most a brief unnecessary check. This asymmetry makes fall detection inherently recall-oriented. The question driving this thesis is whether persistent homology, applied to short wearable accelerometer windows through a delay-embedding pipeline, can support such recall-oriented detection while remaining rigorous in evaluation, principled in feature construction, and light enough to run on edge hardware.

Each three-axis recording is first reduced to an orientation-independent signal magnitude vector, then lifted into a multidimensional point cloud by delay embedding in the sense of Takens, and finally summarised by persistent homology in degrees zero and one. Ten statistics of each persistence diagram, combined with four signal-level descriptors, yield a compact twenty-four-dimensional feature vector passed to two deliberately simple classifiers—logistic regression and a support vector machine. The hyperparameters of the topological representation are chosen by Bayesian optimisation with a tree-structured Parzen estimator, running a separate search per dataset with a classifier-agnostic objective. Three public benchmark datasets are used for evaluation—MobiFall, SisFall, and FAD_40Hz— under a subject-aware framework that reports naive, leave-one-subject-out, and subject-personalised protocols alongside a decision-threshold sweep.

Under leave-one-subject-out evaluation the mean F_2 score reaches 0.97 on SisFall, 0.96 on FAD_40Hz, and 0.90 on MobiFall, with recall above 0.95 across all dataset-classifier combinations. The gap between the two classifiers is consistently below 0.02, while tuning the topological hyperparameters moves F_2 by an order of magnitude more—evidence that performance is governed by the representation map rather than the classifier family. All three datasets converge to structurally distinct optimal configurations, reflecting genuine differences in fall dynamics and point-cloud geometry. Post-hoc threshold personalisation, which requires no model retraining and runs in under one second, yields gains whose distribution is strongly right-skewed: most subjects benefit only marginally while a small minority gain more than 0.10 units of F_2. Throughout, the Cohen–Steiner stability theorem provides a formal noise-robustness certificate for the topological features, distinguishing them from competing feature families that offer no analogous theoretical guarantee.

Keywords: fall detection, wearable sensors, topological data analysis, persistent homology, delay embedding, recall-oriented classification.

Giyilebilir Sensör Sinyalleriyle Topolojik Veri Analizi: Düşme Vakalarını Kaçırmamayı Önceleyen Genel ve Kişisel Modellerin Geliştirilmesi ve Değerlendirilmesi

ÖZET

Düşmeler, yaşlı bireylerde yaralanmanın, bağımsızlık kaybının ve ölümün önde gelen nedenlerinden biridir. Dünya genelinde yaşlı nüfusun hızla artmasıyla birlikte düşme olaylarının zamanında ve güvenilir biçimde tespit edilmesi bir kamu sağlığı önceliğine dönüşmüştür. Giyilebilir ataletsel sensörler bu soruna pratik bir yanıt sunar: bireyin günlük yaşamını kısıtlamadan sürekli izleme imkânı tanıyan ivmeölçer ve jiroskop, otomatik tespit için gereken donanım altyapısını neredeyse sıfır ek maliyetle sağlar. Ancak bu problemde iki hata türü eşit ağırlıkta değildir. Tespit edilemeyen bir düşme, kişinin yerde uzun süre yardımsız kalmasına ve buna bağlı ciddi komplikasyonlara yol açabilirken yanlış bir alarm en fazla gereksiz bir kontrol maliyeti doğurur. Düşme tespiti bu asimetri nedeniyle özü itibarıyla geri-çağırma odaklı bir sınıflandırma problemidir.

Araştırma Sorusu ve Genel Yaklaşım. Bu tez, gecikme gömme (delay embedding) tabanlı bir işlem hattı aracılığıyla kısa giyilebilir ivmeölçer pencerelerine uygulanan kalıcı homolojinin (persistent homology); değerlendirmesi titiz, öznitelik inşası ilkeli ve uç cihazlarda çalışabilecek kadar hafif bir geri-çağırma odaklı düşme tespitini destekleyip destekleyemeyeceğini sorgular. Temel iddia şudur: başarım büyük ölçüde sınıflandırıcı seçiminden değil, sinyali topolojik bir öznitelik uzayına taşıyan temsil haritasından kaynaklanır.

Yöntem. İşlem hattı, üç eksenli her kaydı önce yönelimden bağımsız bir sinyal büyüklük vektörüne (SMV) dönüştürür; ardından tüm kayıtlar ortak bir örnekleme frekansı olan 50 Hz'e yeniden örneklenir ve sabit uzunlukta pencerelere bölünür. Her pencerenin faz uzayı, Takens'in gömme teoremi anlamında gecikme gömmeyle yeniden kurulur; tek boyutlu skaler sinyalden çok boyutlu bir nokta bulutu elde edilir. Bu nokta bulutu üzerine bir filtrasyon kurularak sıfırıncı ve birinci derecede kalıcı homoloji hesaplanır. Her kalıcılık diyagramı on istatistikle özetlenir; bu istatistikler dört sinyal düzeyi tanımlayıcısıyla birleşerek yirmi dört boyutlu kompakt bir öznitelik vektörü (ϕ_λ) oluşturur. Vektör, kasıtlı olarak basit tutulan iki sınıflandırıcıya verilir: lojistik regresyon ve destek vektör makinesi. Her ikisi de sınıf dengesizliğini gidermek üzere dengeli sınıf ağırlıklarıyla, destek vektör makinesi ayrıca Platt kalibrasyonuyla eğitilir.

Topolojik temsilin davranışı; pencere uzunluğu, gömme boyutu, gecikme, filtrasyon türü (Vietoris–Rips, Alpha, SparseRips), uzaklık metriği ve ölçek gibi parametrelerden oluşan bir demete (λ) bağlıdır. Bu yüksek boyutlu koşullu arama uzayını ızgara aramasıyla taramak hesaplama açısından olanaksız olduğundan, hiperparametreler ağaç yapılı Parzen kestiricisi (TPE) ile yürütülen Bayesçi

optimizasyonla seçilir. Her veri kümesi için ayrı bir arama yürütülmüş ve sınıflandırıcıdan bağımsız bir amaç fonksiyonu en iyi yapılandırmayı belirlemiştir.

Veri Kümeleri ve Değerlendirme. Yöntem üç açık karşılaştırma veri kümesi üzerinde sınanmıştır: MobiFall, SisFall ve FallAllD veri kümesinin 40 Hz türetilmiş çeşidi (FAD_40Hz). Bu veri kümeleri sensör konumu, örnekleme frekansı, denek profili ve düşme türleri bakımından birbirinden ayrılarak yöntemin farklı edinim koşullarındaki genellenebilirliğini sınar. Değerlendirme denek-duyarlı bir çerçevede üç protokolle yürütülür: rastgele bölünmeye dayanan ve üst sınır referansı sağlayan naif protokol; her deneğin bütünüyle dışarıda bırakıldığı birincil protokol olan LOSO; ve her deneğin yalnızca kendi verisiyle eğitilip test edildiği kişisel protokol. Bu üçünü, karar eşiğini geniş bir aralıkta tarayan bir eşik analizi tamamlar. Birincil başarım ölçütü, geri çağırmayı kesinliğe göre dört kat ağırlıklandıran F_2 skorudur; deneklerarası dağılım yüzde doksan beş güven aralıklarıyla raporlanır. Tüm ölçekleme ve eşik parametreleri yalnızca eğitim bölümünden kestirilerek bilgi sızıntısı önlenir.

Bulgular. LOSO protokolü altında ortalama F_2 skoru SisFall'da 0.97'ye, FAD_40Hz'de 0.96'ya, MobiFall'da ise 0.90'a ulaşmış; geri çağırma değeri tüm veri kümesi–sınıflandırıcı bileşimlerinde 0.95'in üzerinde kalmıştır. Temsilin bu etkinliğinin kuramsal dayanağı Cohen–Steiner kararlılık teoremidir: iki kalıcılık diyagramı arasındaki darboğaz uzaklığının, girdi sinyallerindeki bozulmanın supremum normuyla sınırlı olduğu güvence altına alınmakta ve böylece elle tasarlanmış istatistiksel, spektral ya da derin öğrenme tabanlı öznitelik ailelerinde benzeri bulunmayan biçimsel bir gürültü-dayanıklılığı sertifikası sağlanmaktadır.

Üç veri kümesi yapısal olarak birbirinden farklı en iyi yapılandırmalara yakınsamıştır. MobiFall geniş bir pencerede SparseRips kompleksini, SisFall kısa bir pencerede Alpha kompleksini, FAD_40Hz ise kısa bir pencerede standart Rips kompleksini seçmiştir. Evrensel bir yapılandırmanın bulunmadığı bu bulgu, en iyi topolojik geometrinin her veri kümesindeki düşme dinamiklerini ve nokta-bulutu yapısını gerçek anlamda yansıttığına işaret eder. Öte yandan iki sınıflandırıcı arasındaki F_2 farkı tüm veri kümelerinde 0.02'nin altında kalırken, hiperparametrelerin ayarlanması F_2'yi bunun çok daha geniş bir aralığında değiştirmektedir. Bu gözlem, başarımın sınıflandırıcı sınıfından çok temsil haritası tarafından belirlendiği tezin temel iddiasını güçlü biçimde destekler.

Model yeniden eğitimi gerektirmeyen eşik kişiselleştirmesinin etkisi de incelenmiştir. Denek düzeyindeki kazanım dağılımı tüm bileşimlerde sağa çarpıktır: deneklerin büyük çoğunluğu yalnızca marjinal fayda sağlarken küçük bir azınlık F_2'de 0.10'ı aşan kazanımlar elde etmektedir. FAD_40Hz'de bir denek, eşik ayarlamasıyla F_2 skorunu 0.80'den 0.92'ye taşımıştır; üstelik bu iyileşme model güncellenmeksizin gerçekleşmiştir. Bu heterojenlik, tek bir sabit eşik kaydırmasının tüm denekleri eşit biçimde iyileştireceği varsayımını çürütür ve en uygun eşiğin gerçekten denek-özgü olduğunu ortaya koyar.

Sonuç. Bu tezin temel katkıları birbirine bağlı dört bileşenden oluşmaktadır. İlki, giyilebilir sinyallerden geri-çağırma odaklı düşme tespiti için teorik güvence

taşıyan, hesaplama açısından hafif bir topolojik temsil haritasının geliştirilmesidir. İkincisi, bu temsili üç farklı veri kümesinde tutarlı biçimde iyileştiren Bayesçi arama çerçevesidir. Üçüncüsü, naif, LOSO ve kişisel protokolleri bir arada raporlayan ve böylece değerlendirmenin adilliğini güvence altına alan denek-duyarlı değerlendirme çerçevesidir. Dördüncüsü ise model yeniden eğitimi gerektirmeyen, çıkarım zamanında tek bir parametre olarak uygulanabilen ve denek bazındaki heterojenliği sayısal olarak ortaya koyan eşik kişiselleştirme analizidir. Bulgular bir bütün olarak, topolojik temsilin basit sınıflandırıcılarla dahi güçlü ve gürültüye dayanıklı düşme tespiti sağlayabileceğini; ancak kişiselleştirmenin her denek için aynı ölçüde yarar sağlamadığını ve bu nedenle dağılımsal bir perspektiften ele alınması gerektiğini göstermektedir.

Anahtar Kelimeler: düşme tespiti, giyilebilir sensörler, topolojik veri analizi, kalıcı homoloji, gecikme gömme, geri-çağırma odaklı sınıflandırma.

1. Introduction

1.1 Clinical and Societal Motivation

Among all accidental injury deaths worldwide, falls rank second only to road traffic injuries (World Health Organization, 2021). Each year roughly 684,000 people die from falls, and older adults—those aged 60 and above—account for the overwhelming majority of that toll (World Health Organization, 2021). Beyond fatalities, roughly 37 million fall-related injuries per year require clinical care, imposing a sustained load on health systems that demographic trends will only amplify: WHO projections place the over-60 population at 2.1 billion by 2050 (World Health Organization, 2022), making timely detection an increasingly urgent public-health priority.

What makes an undetected fall particularly dangerous is the cascade it sets off. Once a person is on the ground and unable to summon help, they enter what clinicians call a "long lie"—exposure to dehydration, hypothermia, pressure sores, and aspiration pneumonia that accumulates with every passing hour. A long lie of more than one hour, even following an otherwise minor injury, doubles the probability of subsequent institutionalisation and is independently associated with increased one-year mortality (Wild, Nayak, & Isaacs, 1981). There is also a psychological toll: the *fear of falling* (Tinetti, Speechley, & Ginter, 1988), a well-documented post-fall syndrome, causes affected individuals to restrict their own activity, which accelerates physical deconditioning and—paradoxically—elevates future fall risk still further. Timely detection, then, is not a comfort measure. It is a clinical intervention.

Wearable inertial sensors are well suited to closing this detection gap. A single accelerometer worn at the waist or carried in a trouser pocket records three-axis body acceleration continuously throughout the day, imposing no fixed infrastructure and preserving the user's privacy and freedom of movement. More practically, the necessary hardware is already present in the smartphones and smartwatches carried by a large share of the target population, so passive fall detection can be deployed at scale with minimal additional hardware cost.

These two error types—missed fall and false alarm—are far from symmetric in consequence. A missed fall risks the long-lie complications described above; a false alarm costs at most a brief, unnecessary check. This asymmetry is the premise that shapes the rest of the thesis. We pursue a *recall-oriented* objective and score every model with the F_2 measure, which weights recall four times as heavily as precision. The detector is deliberately tuned to give precedence to the fall event—tolerating the occasional false alarm rather than risk overlooking a genuine one—and this design choice is reflected in the title of the thesis. The formal definition of F_2 and the reasoning behind the four-to-one weighting are developed in the Methodology.

1.2 Limitations of Existing Approaches

Despite the practical appeal of wearable fall detection, a set of persistent technical difficulties stands between sensor signal and reliable deployment. The existing literature falls into three broad families, and each carries structural limitations that are worth spelling out before motivating the approach taken here.

Threshold-based detectors. The earliest and most widely deployed family compares the signal magnitude vector $\|\mathbf{a}(t)\|_2$ against a fixed threshold T, declaring a fall whenever $\|\mathbf{a}(t)\|_2 > T$. Computationally trivial and requiring no training data, these detectors nonetheless suffer from the absence of any principled calibration mechanism: T is set empirically, the output is binary with no associated uncertainty, and there is no statistical framework for adjusting the operating point to the false-alarm tolerance of a given deployment. In practice they can achieve high recall on controlled datasets, but the false-alarm rates under continuous wear are clinically unacceptable—vigorous daily activities such as running, stair descent, or jumping routinely produce magnitude excursions indistinguishable from a fall impact.

Statistical and spectral feature methods. From the mid-2000s onward, a substantial literature moved to constructing handcrafted feature vectors from short accelerometer windows—means, variances, spectral energies, signal-magnitude area, zero-crossing rates—and training standard classifiers on the resulting representations (de la Hoz Franco, Ariza-Colpas, Quero, & Espinilla, 2014; Igual, Medrano, & Plaza, 2013). Calibrated probability outputs and exploitation of multi-dimensional signal structure are genuine improvements over hard thresholds. The underlying problem, however, is that these features are *ad hoc*: assembled by domain intuition rather than derived from any geometric theory of the signal. No formal stability guarantee exists—a small perturbation of the input that happens to cross a feature threshold can cause an arbitrarily large change in the feature vector—and the empirical consequence is that trained models transfer poorly across sensors, sampling rates, and subject populations.

Deep learning approaches. Convolutional and recurrent networks have achieved competitive benchmark performance by learning end-to-end representations directly from raw inertial data (Batchuluun, Lee, Nguyen, Pham, Arsalan, & Park, 2022; Wang, Ellul, & Azzopardi, 2020), and the gains over handcrafted features are real. The trade-offs, however, are severe. Large labelled datasets, GPU compute, and expensive hyperparameter searches are non-trivial prerequisites, and the resulting models are opaque: there is no geometric or mathematical account of why a given window is classified as a fall. Deployment on edge hardware— smartwatches, embedded sensors—is complicated by memory footprint and energy budget. Most critically, deep models are typically trained once on a fixed corpus and deployed without adaptation, a "trains once, deploys forever" paradigm that systematically ignores the substantial variation in motion dynamics across individuals.

The personalization gap. All three families share a structural weakness: the decision rule is calibrated at the population level and then applied uniformly to every individual. Statistically, this is equivalent to assuming the optimal decision boundary is the same for all subjects. In reality, individuals differ in gait dynamics, body geometry, sensor placement, and fall style, so the population-level optimum will in general be suboptimal for any specific subject s. Some recent work reports personalized evaluation results, but typically through isolated case studies rather than a systematic distributional characterisation—leaving open the question of whether subject-level adaptation is uniformly beneficial or only selectively so.

1.3 Topological Data Analysis for Wearable Signals

Persistent homology, the central tool of topological data analysis (TDA) (Carlsson, 2009; Edelsbrunner & Harer, 2010), offers a different geometric framework for characterising signal windows. Rather than fixing a set of scalar statistics in advance, it builds a sequence of simplicial complexes—a *filtration*—that tracks how the topology of the signal's point-cloud representation evolves across spatial scales. The output is a *persistence diagram*: a multiset of (b, d) pairs recording the scales at which topological features (connected components, loops, voids) are born and die. The diagram encodes the intrinsic geometry of the window without committing to any particular feature basis.

Two properties make this framework particularly attractive for fall detection.

Provable stability. The Cohen–Steiner stability theorem (Cohen-Steiner, Edelsbrunner, & Harer, 2007) bounds the bottleneck distance between two persistence diagrams by the supremum norm of the perturbation between the corresponding input functions:

$$d_B(D(f),\, D(g)) \;\leq\; \|f - g\|_\infty.$$

For inertial windows this means small sensor-noise perturbations produce at most equally small perturbations of the topological descriptor—a guarantee that is a *mathematical theorem*, not an empirical observation, and that holds regardless of the training dataset or subject population. No existing handcrafted feature family possesses a comparable certificate.

Fixed-dimensional vectorization. The persistence diagram is summarised as a fixed 24-dimensional feature vector $\phi_\lambda(w) \in \mathbb{R}^{24}$ by computing statistics of the birth-death pairs in homological degrees $k = 0$ and $k = 1$. Crucially, the dimension 24 does not depend on the window length, sampling rate, or filtration size—unlike spectral or autoregressive features whose dimension grows with signal length, or deep-network embeddings whose dimension is a design parameter coupled to model size.

The geometric link between the scalar sensor signal and the point cloud is provided by the *delay embedding* (Takens, 1981): the orientation-invariant signal magnitude vector $\mathrm{SMV}(t) = \|\mathbf{a}(t)\|_2$ is lifted to an m-dimensional point cloud by taking overlapping m-tuples of lagged values. Fall windows and ADL windows produce qualitatively different topologies under this construction. A fall impact—a sharp impulsive excursion— generates a spread-out cluster with pronounced H_0 structure, whereas periodic ADL motion (walking, stair climbing) yields richer H_1 structure in the form of loops. Persistent homology captures this distinction in a principled, geometry-grounded way.

TDA has been applied to time-series classification across multiple domains (Gidea & Katz, 2018; Pereira & de Mello, 2015; Umeda, 2017), including activity recognition from inertial sensors (Ott, Tomforde, & Willis, 2015; Xu, Yao, Li, & Zhang, 2019). What distinguishes the present work is the combination of a recall-oriented objective, systematic dataset-specific hyperparameter optimisation, and a statistical treatment of subject-level personalisation as a distributional phenomenon rather than a collection of case studies.

1.4 The Threshold Decision as a First-Class Component

Fall detection is an instance of binary classification under class imbalance: falls are rare relative to activities of daily living, so the positive class is significantly underrepresented in any realistic deployment. Under this imbalance the decision threshold τ in the classification rule $\widehat{y}(w) = \mathbf{1}\{f_\theta(\phi_\lambda(w)) \geq \tau\}$ is not a cosmetic implementation detail. It is a critical design parameter that directly controls the precision-recall trade-off, and choosing it thoughtfully is inseparable from choosing to optimise F_2. Lowering τ raises recall at the cost of precision; the operating point therefore must be set deliberately rather than left at the conventional default of 0.5.

Beyond a single population-level threshold, we investigate *subject-level threshold personalisation.* The posterior probability function $f_\theta \circ \phi_{\lambda^*}$ may be systematically miscalibrated differently for different subjects: a subject whose fall windows tend to yield lower-than-average posterior scores needs a lower threshold to maintain high recall, while a subject whose ADL windows yield elevated scores needs a higher threshold to avoid excessive false alarms. A sweep of τ over the grid $\mathcal{T} = \{0.05, 0.10, \ldots, 0.95\}$ identifies the per-subject optimal value $\tau^*(s)$ retrospectively.

The computational cost of this sweep is $O(|\mathcal{T}| \cdot |W_s|)$ comparisons, where $|\mathcal{T}| = 19$ and $|W_s|$ is the window count for subject s. In practice this runs in under one second for all subjects in any of the three datasets used here, against a Bayesian representation search that takes approximately 75 minutes per dataset—a ratio exceeding 10^3. Threshold personalisation is therefore a computationally negligible add-on to an already-trained model. It requires no retraining, no access to the original training data, and no server-side computation: it is executable on-device at inference time from a small per-subject calibration set. Near-zero cost, privacy preservation, and measurable F_2 gain together make this step a first-class engineering contribution, not merely an evaluation convenience.

1.5 Contributions

This thesis makes five contributions.

Contribution 1: A mathematically grounded TDA pipeline for wearable fall detection. We develop a feature map $\phi_\lambda \colon \mathbb{R}^{n\times 3} \to \mathbb{R}^{24}$ that transforms a raw three-axis accelerometer window into a fixed 24-dimensional topological descriptor via delay embedding, simplicial filtration, persistent homology computation, and vectorization. The map is parameterised by a hyperparameter tuple $\lambda = (w_{\mathrm{sec}}, m, \tau, r, c, d_{\mathrm{met}}, \varepsilon, q)$ governing the geometric construction, and the Cohen–Steiner stability theorem provides a formal noise-robustness certificate for the resulting features.

Contribution 2: A classifier-agnostic Bayesian hyperparameter optimisation framework. A single Optuna/TPE study (Akiba, Sano, Yanase, Ohta, & Koyama, 2019) with a classifier-averaged F_2 objective identifies λ^*_K for each dataset K in 200 trials. Averaging over classifiers ensures the recovered representation supports both logistic regression and support vector machines without being over-fitted to either. We show that this optimisation recovers dataset-specific configurations (H2) and that the variation in F_2 across trials—up to 0.18 units on MobiFall—far exceeds the variation attributable to classifier choice (H3).

Contribution 3: A three-protocol subject-aware evaluation framework. Rather than reporting a single accuracy figure, we evaluate each configuration under three complementary protocols: the Naive (random split, inflated upper bound), the General (leave-one-subject-out, primary result), and the Personal (per-subject split, idealised personalisation reference). The LOSO protocol with $R = 15$ independent repetitions provides repetition-level confidence intervals via a t-statistic, and the joint reporting of all three protocols makes the train–test leakage implicit in naive evaluation visible quantitatively.

Contribution 4: A statistical characterisation of threshold heterogeneity as a distributional phenomenon. The per-subject sweep gain $\Delta F_2(s) = F_2(\tau^*(s), s) - F_2(\hat{\tau}_0(s), s)$ is treated as a distribution rather than a scalar average. The distribution proves to be right-skewed and dataset-dependent: MobiFall shows a mean gain of ≈0.04–0.05 with a coefficient of variation exceeding 50%, while SisFall shows gains near zero for most subjects. This analysis confirms Hypothesis 4 and replaces the anecdotal personalisation accounts common in the prior literature.

Contribution 5: A deployment pathway to consumer devices. The pipeline is CPU-only, requires no GPU, and runs persistent homology at the window sizes considered in tens of milliseconds on a desktop-class CPU—a profile compatible with ARM Cortex-A smartphone processors and modern smartwatch SoCs. Threshold personalisation runs in under one second on device. We describe a prospective on-device calibration procedure that requires only a short wearing period and no server-side computation, and identify one-class anomaly detection from ADL data as the natural extension that would eliminate the cold-start problem entirely.

1.6 Thesis Organisation

Chapter 2 surveys the three families of prior wearable fall detection methods—threshold-based, statistical-feature, and deep-learning—and positions the TDA approach with respect to each. Prior applications of persistent homology to time-series classification outside of fall detection are also reviewed.

Chapter 3 develops the mathematical foundations: the signal-magnitude-vector transformation and delay embedding, the theory of simplicial complexes, persistent homology, and the Cohen–Steiner stability theorem, the feature vectorization and Bayesian optimisation procedure, and the three evaluation protocols and the statistical inference framework used throughout.

Chapter 4 describes the three benchmark datasets, the experimental setup, and the numerical results—dataset-specific optimal configurations, full performance tables for all classifier-protocol combinations, and the threshold sweep analysis.

Chapter 5 interprets the experimental outcomes against the four research hypotheses, presents the per-subject threshold gain distributions, synthesises findings across datasets, discusses the deployment pathway, and identifies limitations and directions for future work.

Chapter 6 summarises the contributions at three levels: mathematical, statistical, and engineering.

Notation

Throughout this thesis $\mathbb{R}$ denotes the real line and $\mathbb{R}^d$ the d-dimensional Euclidean space. Scalars are denoted by lowercase letters, vectors by bold lowercase, and sets by calligraphic uppercase. The indicator function is written $\mathbf{1}\{\cdot\}$, the Euclidean norm $\|\cdot\|_2$, and the supremum norm $\|\cdot\|_\infty$. All probabilistic statements are with respect to the randomness of the class-balancing seed unless stated otherwise.

2. Related Work

Wearable fall detection has been tackled from three broad directions: rule-based threshold detectors, classical machine-learning pipelines built on handcrafted features, and end-to-end deep-learning models. Each has produced systems with strong scores on controlled benchmark datasets, yet each carries structural limitations that surface whenever the goal shifts from benchmark performance to robust, personalised deployment. A fourth strand—separate from fall detection proper—applies persistent homology to time-series classification in other domains, and it is from this strand that the present thesis draws its core methodology. What follows surveys all four lines of research, identifies the limitations that motivated the choices made here, and closes with a positioning statement.

2.1 Threshold-Based Fall Detectors

2.1.1 Method family

The oldest and most widely deployed wearable fall detectors work directly on the signal magnitude vector $\mathrm{SMV}(t) = \|\mathbf{a}(t)\|_2$, raising an alarm whenever $\mathrm{SMV}(t)$ exceeds a fixed scalar threshold T:

$$\widehat{y}(t) = \mathbf{1}\{\mathrm{SMV}(t) > T\}.$$

(Kangas, Konttila, Lindgren, Winblad, & Jämsä, 2008) established the archetypal version of this approach on a waist-mounted accelerometer and reported high sensitivity on laboratory falls. Later work added a second stage: rather than triggering on a high-magnitude spike alone, a fall is declared only if the spike is followed by a prolonged period of low activity, filtering out vigorous daily actions that produce a similar impulse without a subsequent motionless phase (Igual, Medrano, & Plaza, 2013). Multi-axis variants replace the scalar magnitude with component-specific thresholds or combine vertical, horizontal, and resultant readings in a logical rule (de la Hoz Franco, Ariza-Colpas, Quero, & Espinilla, 2014).

2.1.2 Mathematical limitations

Three limitations of the rule $\mathbf{1}\{\mathrm{SMV}(t) > T\}$ are mathematical rather than incidental to any particular implementation.

No probabilistic output. The rule returns a hard binary decision with no posterior probability. There is no continuous operating parameter that smoothly trades false negatives for false positives: adjusting sensitivity means shifting the hard threshold T, and this shift applies identically to all individuals and all activities.

No principled threshold selection. The value of T is determined by empirical cross-validation or clinical convention. No statistical framework allows selecting T that is optimal under a specified loss function—in particular, the F_2-optimal threshold depends on the class-imbalance ratio of the deployment population in a way that cannot be identified without a probabilistic model.

No subject-specific adaptation. A single T is applied to everyone. An active person with a vigorous gait may produce baseline SMV values that approach T during normal walking, resulting in chronic false alarms. A frail elderly person with a low-energy fall may never reach T at all. The detector has no mechanism to adapt to individual motion dynamics because it has no subject-level probabilistic model.

2.1.3 Comparison with the present work

In this thesis the scalar hard decision is replaced by a calibrated posterior probability $f_\theta(\phi_\lambda(w)) \in [0, 1]$, making the decision threshold τ a continuous parameter that can be set to maximise F_2 on a per-subject basis. A sweep over $\mathcal{T} = \{0.05, 0.10, \ldots, 0.95\}$ identifies the subject-optimal $\tau^*(s)$ retrospectively. Gains of up to $+0.12$ units of F_2 on individual subjects demonstrate that this kind of subject-specific calibration is structurally inaccessible to any global threshold detector.

2.2 Statistical Feature Methods

2.2.1 Method family

From the mid-2000s through the mid-2010s the dominant paradigm combined handcrafted feature extraction with a standard supervised classifier. A window w of length L is mapped to a fixed-dimensional vector $\phi_{\text{hand}}(w) \in \mathbb{R}^d$ by computing scalar statistics of the raw signal or its frequency-domain representation. Common time-domain choices include window mean, variance, range, peak value, root-mean-square, and signal-magnitude area; common frequency-domain choices include spectral energies in low-frequency bands, peak frequency, and spectral entropy (de la Hoz Franco, Ariza-Colpas, Quero, & Espinilla, 2014; Igual, Medrano, & Plaza, 2013). These vectors are then fed to a classifier—most often an SVM, k-nearest neighbours, or a decision tree—trained on labeled windows from a fixed dataset.

Surveys by Igual et al. (Igual, Medrano, & Plaza, 2013) and Delahoz and Labrador (de la Hoz Franco, Ariza-Colpas, Quero, & Espinilla, 2014) catalogue dozens of systems built on this template and report accuracy in the range 90%–99% on controlled datasets. Interestingly, these figures are broadly consistent across methods with very different feature sets, which suggests that feature choice matters less than evaluation protocol: the studies reporting the highest scores tend to use random train-test splits that ignore subject membership, allowing models to memorize individual quirks rather than generalize to new persons.

2.2.2 Mathematical limitations

Absence of a stability certificate. For handcrafted statistics the relationship between input perturbation $\|w - w'\|$ and feature perturbation $\|\phi_{\text{hand}}(w) - \phi_{\text{hand}}(w')\|$ is not bounded in any useful way. A small noise perturbation can produce an arbitrarily large change in a zero-crossing count, an extremum detection, or a threshold-based spectral band energy. This instability is inconsequential when training and test come from the same sensor and subject, but becomes significant under sensor substitution, mounting-position variation, or cross-dataset evaluation.

No geometric grounding. The feature set is assembled by domain intuition and validated empirically. There is no underlying theory explaining why the chosen statistics should separate falls from activities of daily living, nor any guarantee that the same statistics remain discriminative when the population, sensor, or activity set changes. In learning-theory terms, the inductive bias of the feature extractor is implicit and uncontrolled.

Fixed dimension with no adaptation. The dimension d is fixed by the feature recipe and cannot be adjusted to the dataset geometry. Different datasets exhibit qualitatively different fall dynamics—controlled laboratory versus naturalistic community falls, elderly versus young-adult populations—but the feature extractor is invariant to these differences. In effect, the only adaptable hyperparameter is the classifier, not the representation itself.

Conflation of variability sources. Most papers in this family report a single aggregate accuracy figure, making it impossible to distinguish within-subject repeatability from across-subject generalisation. Whether a reported score reflects genuine generalisation to new individuals or memorisation of subjects seen during training is typically unclear.

2.2.3 Comparison with the present work

The feature map ϕ_λ introduced here differs from handcrafted features in three substantive ways. First, the Cohen–Steiner stability theorem provides a formal Lipschitz bound: small noise perturbations of the input window produce at most equally small perturbations of the persistence diagram and hence of ϕ_λ. This holds for any signal, any subject, and any sensor. Second, the extractor is parameterised by $\lambda \in \Lambda$ and the optimal λ^*_K is identified by Bayesian search—different datasets converge to different configurations (SparseRips at 5 s for MobiFall; Alpha at 1 s for SisFall; Rips at 1 s for FAD_40Hz), which is evidence that dataset-specific geometry is being captured. No handcrafted feature family has an analogous adaptation mechanism. Third, evaluation strictly separates subject-level generalisation from within-subject repeatability by using a leave-one-subject-out protocol with $R = 15$ repetitions, reporting both the mean and the 95% confidence interval for every combination of dataset, classifier, and protocol.

2.3 Deep Learning Approaches

2.3.1 Method family

Convolutional networks, LSTMs and GRUs, and more recently transformer architectures have been applied to wearable fall detection with the goal of learning discriminative representations directly from raw or minimally processed inertial data. Wang et al. (Wang, Ellul, & Azzopardi, 2020) showed that a multi-scale CNN applied to three-axis accelerometer windows achieves competitive sensitivity and specificity on MobiFall and SisFall without manual feature engineering. The systematic review by Batchuluun et al. (Batchuluun, Lee, Nguyen, Pham, Arsalan, & Park, 2022) covering the years 2018–2022 identifies CNN-LSTM hybrids and attention-based models as the current state of the art, with reported F_1 scores above 0.95 on several controlled datasets.

Part of the apparent performance advantage of deep models over classical methods comes from their ability to capture temporal structure across multiple scales simultaneously. Another part, however, is methodological: deep-learning papers tend to use larger datasets and more favourable evaluation protocols than their classical counterparts, which makes direct comparison difficult.

2.3.2 Engineering and scientific limitations

GPU dependency. Training a competitive deep network on inertial windows requires GPU compute for gradient-based optimisation. Inference may be feasible on a smartphone CPU with quantized models, but retraining for personalisation is not—a practical barrier in battery-constrained clinical or community settings.

Opacity. Features learned by intermediate layers are not interpretable in terms of the physical signal. There is no account of why a particular window is classified as a fall that a clinician or system designer can evaluate independently of the training data. TDA features, by contrast, have a direct geometric meaning: H_0 statistics measure the spread of the delay-embedded cloud, and H_1 statistics capture the presence of loops from periodic motion.

No formal stability guarantee. Adversarial-robustness research has repeatedly shown that imperceptibly small perturbations can reverse a deep classifier's output. For a safety-critical system, the absence of a stability certificate is a principled concern: a sensor hardware revision, a firmware update affecting the analog front-end, or a change in mounting position can degrade performance in unpredictable ways. The Cohen–Steiner bound of TDA provides a certificate that is independent of the training distribution.

The "trains once, deploys forever" paradigm. Deep models are trained on a fixed corpus and deployed without adaptation. Achieving subject-specific personalisation requires either fine-tuning—which demands labeled fall examples from the target individual—or meta-learning frameworks that add substantial complexity while still requiring labeled falls at deployment time. In this thesis subject-level adaptation is achieved by a threshold sweep that requires no labeled falls, no update to model parameters, and no access to the training corpus.

Computational cost. The Bayesian representation search used here consumed roughly 75 minutes of CPU time on a 40-core server—a cost paid once per dataset. Training a competitive deep model requires comparable or greater wall-clock time and additionally needs GPU hardware. The CPU-only profile of the TDA pipeline is a genuine engineering advantage for resource-limited deployments.

2.3.3 Comparison with the present work

This thesis does not claim to match or exceed state-of-the-art deep-learning scores. The goal is different: to determine whether a mathematically well-founded, computationally lightweight, and interpretable pipeline can achieve strong F_2 scores under rigorous subject-aware evaluation. On two of the three datasets that question has an affirmative answer ($F_2 \geq 0.93$ on SisFall and FAD_40Hz), and on the third the result is still satisfactory ($F_2 \geq 0.88$ on MobiFall). These results are achieved on a CPU alone, with formally certified feature stability, and with an interpretable feature space—a combination that deep models do not jointly offer.

2.4 Topological Data Analysis for Time-Series Classification

2.4.1 Persistent homology in applied settings

The use of persistent homology as a feature extractor for time-series classification grew substantially in the 2010s, driven by efficient boundary matrix reduction algorithms (Edelsbrunner, Letscher, & Zomorodian, 2002) and open-source implementations such as GUDHI (GUDHI Editorial Board, 2015) and Ripser.

Pereira and de Mello (Pereira & de Mello, 2015) applied persistence diagrams to time-series classification across several benchmark domains, finding that topological features are competitive with or superior to classical statistics when the discriminative signal is encoded in the *shape* of short segments rather than amplitude or frequency content. Umeda (Umeda, 2017) extended this to multivariate settings, showing that combining H_0 and H_1 features provides complementary information not captured by either degree alone.

Gidea and Katz (Gidea & Katz, 2018) applied TDA to financial time series and found that topological features of sliding-window point clouds detect market instability with a lead time inaccessible to standard spectral methods. Their work established the delay-embedding framework—building a point cloud from lagged copies of a scalar signal and computing its persistent homology—as a broadly applicable TDA pipeline. The present thesis follows this framework but extends it with a systematic search over the embedding and filtration parameters rather than fixing them a priori.

2.4.2 TDA for activity recognition and wearable signals

(Ott, Tomforde, & Willis, 2015) applied TDA features to human activity recognition (HAR) from smartphone accelerometer data and showed that persistence statistics achieve accuracy comparable to handcrafted features on a standard HAR benchmark, using a fixed sliding window and pre-selected embedding parameters. Xu et al. (Xu, Yao, Li, & Zhang, 2019) similarly applied TDA to distinguish activity classes in a multiclass HAR setting, reporting improvements over baseline statistical features on several datasets.

These works establish the viability of persistent homology for wearable inertial classification, but several differences limit their direct relevance to the fall detection problem addressed here.

Objective. Both studies optimise accuracy in a balanced multiclass setting. Fall detection is a binary problem with severe class imbalance, and the appropriate objective is recall-oriented (F_2) rather than accuracy. A classifier that maximises accuracy on imbalanced data can do so by predicting the majority class (non-fall) for almost every window, achieving high accuracy at the cost of near-zero recall—which is precisely the wrong trade-off for a safety-critical detector.

Evaluation protocol. The cited HAR studies use random train-test splits that do not control for subject membership, inflating performance estimates by allowing individual-specific patterns to leak from training to test. Here, subject separation is strictly enforced via leave-one-subject-out cross-validation.

Hyperparameter treatment. Prior TDA-for-wearables works fix the embedding dimension m, delay τ, and filtration type either by convention or by a coarse manual grid. We treat the full hyperparameter tuple $\lambda = (w_{\text{sec}}, m, \tau, r, c, d_{\text{met}}, \varepsilon, q)$ as the primary object of optimisation and identify the optimal tuple per dataset via 200 Bayesian iterations. That all three datasets converge to different complex types (SparseRips, Alpha, and Rips respectively) is a direct consequence of this systematic search and could not have emerged from fixing the complex type by convention.

Personalisation. Neither Ott et al. nor Xu et al. report subject-level personalisation results or analyse the distribution of per-subject gains. We treat the sweep gain $\Delta F_2(s)$ as a scientific object in its own right, showing that it is right-skewed, dataset-dependent, and has a coefficient of variation exceeding 50% in all cases.

2.5 Positioning of the Present Work

Table 2.1 summarises how the four approaches surveyed above compare with the present thesis across seven dimensions.

Table 2.1: Structural comparison of fall detection and TDA-for-wearables approaches. ✓ = property satisfied; × = property absent; ∼ = partially satisfied.

Property	Threshold	Stat. features	Deep learning	TDA-HAR	This work
Probabilistic output	×	✓	✓	✓	✓
Formal stability certificate	×	×	×	✓	✓
Dataset-specific tuning	×	×	∼	×	✓
Subject-aware evaluation	×	∼	∼	×	✓
Recall-oriented objective	×	∼	∼	×	✓
CPU-only deployment	✓	✓	×	✓	✓
Subject-level personalisation	×	×	×	×	✓

The table makes clear that no prior approach combines a formal stability certificate, dataset-specific parameterisation, subject-aware evaluation, a recall-oriented objective, and subject-level personalisation within a single unified pipeline. The TDA-for-HAR works come closest in terms of formal guarantees and computational profile, but satisfy none of the last four evaluation properties.

The gap that this thesis directly targets is precisely that intersection: a principled, recall-oriented, subject-aware, personalised fall detector that carries a mathematical stability guarantee and can run on consumer hardware without a GPU.

3. Methodology

At its core, the pipeline of this thesis chains three mathematical maps: a raw labelled inertial recording is first reduced to an orientation-invariant scalar window, then lifted to a finite point cloud in $\mathbb{R}^m$ via delay embedding, then summarised as a fixed-length vector $\phi_\lambda(w) \in \mathbb{R}^{24}$ by persistent homology and a compact set of signal-level statistics. That vector is passed to a probabilistic classifier $f_\theta \colon \mathbb{R}^{24} \to [0, 1]$; a threshold on the output completes the binary decision. The treatment here is deliberately mathematical. The central object of study is the representation map ϕ_λ itself—not the engineering of any particular sensor deployment—and making that map and its inferential context precise is the purpose of this chapter.

Sections 3.1–3.2 establish notation and signal preprocessing. Section 3.3 constructs the delay embedding and states the Takens and Sauer–Yorke–Casdagli embedding theorems that motivate the geometric interpretation. Sections 3.4–3.6 develop the topological background: point-set topology, simplicial homology (including $\partial^2 = 0$, the Euler–Poincaré formula, and functoriality of $H_k(\,\cdot\,; \mathbb{F})$), and persistent homology as a functor on filtrations together with the structure theorem and the algebraic stability theorem. Section 3.7 specifies the vectorisation, Section 3.8 the two classifier families and Platt calibration, Section 3.9 the Bayesian hyperparameter search and the tree-structured Parzen estimator, and Sections 3.10–3.12 the evaluation protocols, statistical-inference machinery, and computational environment.

3.1 Formal Problem Setting

Let $K \in \{\text{MobiFall, SisFall, FAD\_40Hz}\}$ index one of the three benchmark datasets. Each dataset consists of labelled inertial recordings together with subject identifiers, drawn from an unknown joint distribution

$$(\mathcal{X}_K,\, \mathcal{Y}_K,\, \mathcal{S}_K) \sim \mathbb{P}_K,$$

where $\mathcal{X}_K$ is the recording-valued random element, $\mathcal{Y}_K \in \{0,1\}$ the binary fall label, and $\mathcal{S}_K$ the subject index, taking values in a finite set of cardinality $|\mathcal{S}_K|$. The marginal class prior $\pi_K := \mathbb{P}_K(\mathcal{Y}_K = 1)$ is strictly between 0 and 1/2 for every K, reflecting the empirical fact that falls are minority events.

3.1.1 Resampled discrete signal

After resampling to the target frequency $f_s = 50$ Hz (justified in Section 3.2.2), a single recording is the discrete signal

$$\mathbf{x}\colon \{0,1,\ldots,T\} \longrightarrow \mathbb{R}^p, \qquad p = 2,$$

where the two coordinates are the accelerometer and gyroscope magnitudes, defined in Section 3.2.1. A binary label $y \in \{0,1\}$ indicates whether the recording contains a fall. A subject index $s \in \mathcal{S}_K$ records the individual to whom the signal belongs. The subject-aware nature of the data set is essential: independence of training and test samples is asserted at the level of subjects, not at the level of individual windows.

3.1.2 Window operator

For a window length $L \in \mathbb{N}$ and a starting index $t_0 \in \{0,\ldots,T-L\}$, the *window operator* extracts a contiguous segment:

$$W_{t_0,L}\colon \mathbb{R}^{(T+1)\times p} \longrightarrow \mathbb{R}^{L\times p}, \qquad W_{t_0,L}(\mathbf{x}) = (\mathbf{x}_{t_0},\, \mathbf{x}_{t_0+1},\, \ldots,\, \mathbf{x}_{t_0+L-1}). \tag{3.1}$$

The window length is determined by a hyperparameter $w_{\text{sec}} > 0$ measuring window duration in seconds:

$$L := \lfloor w_{\text{sec}} \cdot f_s \rfloor.$$

3.1.3 The hyperparameter tuple

The topological feature extractor depends on a hyperparameter tuple

$$\lambda = \big(w_{\mathrm{sec}},\ m,\ \tau,\ r,\ c,\ d_{\mathrm{met}},\ \varepsilon,\ q\big) \in \Lambda, \tag{3.2}$$

where w_{sec} is the window length in seconds, $m \geq 2$ the embedding dimension, $\tau \geq 1$ the embedding lag, $r \geq 1$ the stride factor, $c \in \{\text{Alpha, Rips, SparseRips}\}$ the filtered complex family, d_{met} a metric on $\mathbb{R}^m$, $\varepsilon \in (0, 1)$ a filtration-scale percentile, and q a subsampling rule used by the SparseRips construction. The search space Λ is a finite Cartesian product, specified in Section 3.9.5.

3.1.4 Representation map and classifier

Given a window w, the pipeline constructs a finite point cloud

$$P_\lambda(w) \subset \mathbb{R}^m$$

by the delay embedding of Definition 3.2 below, builds the filtered simplicial complex $\mathcal{K}_\lambda(P_\lambda(w))$ specified by c, computes the persistent homology in degrees $k \in \{0, 1\}$, and vectorises the resulting diagrams together with a small set of signal-level features into a fixed feature vector

$$\phi_\lambda(w) \in \mathbb{R}^{24}. \tag{3.3}$$

The dimension is fixed at 24 regardless of λ so that distinct configurations are compared under a uniform classifier interface.

A probabilistic classifier is a map

$$f_\theta \colon \mathbb{R}^{24} \longrightarrow [0, 1],$$

where $f_\theta(\phi_\lambda(w))$ is interpreted as the posterior probability $\widehat{\mathbb{P}}(\mathcal{Y} = 1 \mid w; \theta, \lambda)$. The model class is restricted to the two families

$$\mathcal{F}_{\mathrm{lr}} = \{\text{balanced logistic regression models with } \ell_1 \text{ or } \ell_2 \text{ penalty}\},$$

$$\mathcal{F}_{\mathrm{svm}} = \{\text{balanced } C\text{-SVM models with Platt-calibrated probabilities}\}.$$

This restriction is methodological. The aim is to test the predictive value of the representation ϕ_λ in the linear regime, not to maximise score by increasing classifier complexity.

For a threshold $\tau_0 \in (0,1)$ the *subject-general decision rule* is

$$\widehat{y}(w) = \mathbf{1}\{f_\theta(\phi_\lambda(w)) \geq \tau_0\}. \tag{3.4}$$

The default value of τ_0 in primary evaluations is not the nominal 0.5 but the PR-curve optimal threshold $\hat{\tau}_0$ identified on the training fold; its precise construction is described in §3.10.2.

For subject-personalised evaluation a subject-specific threshold τ_s is used:

$$\widehat{y}_s(w) = \mathbf{1}\{f_\theta(\phi_\lambda(w)) \geq \tau_s\}. \tag{3.5}$$

The personalised threshold τ_s must be estimated from data disjoint from the segment of subject s on which the metric is finally reported; otherwise the protocol is contaminated by leakage.

3.1.5 Performance functional

The primary metric is the F_β-score evaluated at $\beta = 2$,

$$F_2(\widehat{y}, y) \;=\; 5\,\frac{\mathrm{Prec}\cdot\mathrm{Rec}}{4\,\mathrm{Prec}+\mathrm{Rec}}, \tag{3.6}$$

with $\mathrm{Prec}, \mathrm{Rec}$ computed on the test sample. The training and selection protocol prioritises recall over precision, which is the appropriate objective for fall detection when false-negative cost exceeds false-positive cost; the asymmetry is quantified in Section 3.10.1.

3.2 Signal Representation

3.2.1 Orientation invariance: the signal magnitude vector

A three-axis inertial sensor outputs a vector-valued signal $\mathbf{x}(t) = (x_1(t), x_2(t), x_3(t))^\top \in \mathbb{R}^3$. The numerical values of the axis components depend on the orientation of the device frame relative to the body frame, an $\mathrm{SO}(3)$-rotation that varies across subjects and even across sessions for a fixed subject. An axis-wise representation therefore conflates motion content with nuisance orientation.

Definition 3.1 (Signal magnitude vector)**.** *The* signal magnitude vector *of a three-axis signal* $\mathbf{x}\colon \mathbb{Z}_{\geq 0} \to \mathbb{R}^3$ *is the scalar process*

$$\mathrm{SMV}(t) = \|\mathbf{x}(t)\|_2 = \sqrt{x_1(t)^2 + x_2(t)^2 + x_3(t)^2}. \tag{3.7}$$

Proposition 3.1 (SO(3)-invariance)**.** *For every rotation* $\mathbf{R} \in \mathrm{SO}(3)$ *and every* $t \geq 0$, $\|\mathbf{R}\mathbf{x}(t)\|_2 = \|\mathbf{x}(t)\|_2$. *Hence the map* $\mathbf{x} \mapsto \mathrm{SMV}$ *is invariant under reorientation of the sensor frame.*

Proof. $\|\mathbf{R}\mathbf{x}\|_2^2 = \mathbf{x}^\top \mathbf{R}^\top \mathbf{R}\mathbf{x} = \mathbf{x}^\top \mathbf{x} = \|\mathbf{x}\|_2^2$, using $\mathbf{R}^\top \mathbf{R} = I$. □

The accelerometer and the gyroscope are processed independently, producing two scalar time series $\mathrm{SMV}_{\mathrm{acc}}$ and $\mathrm{SMV}_{\mathrm{gyr}}$ per recording. All subsequent analysis is carried out on these scalar processes; the loss of directional information is the cost of orientation invariance.

3.2.2 Common sampling rate

The three benchmark datasets were originally recorded at different rates: 87.5 Hz (MobiFall), 200 Hz (SisFall), and 40 Hz (FAD_40Hz). All signals are resampled to the common rate $f_s = 50$ Hz before any further processing. The choice of 50 Hz rests on three independent arguments.

Spectral sufficiency. The fall-discriminative content of inertial wearable signals is concentrated below 20 Hz. The pre-impact acceleration peak and the impact transient are dominated by frequencies in the 0–15 Hz band; energy above 20 Hz consists predominantly of high-frequency sensor noise and mechanical vibration (Kangas, Konttila, Lindgren, Winblad, & Jämsä, 2008; Saleh, Abbas, & Le Bouquin Jeannès, 2021; Sucerquia, López, & Vargas-Bonilla, 2017).

Theorem 3.1 (Nyquist–Shannon (Shannon, 1948))**.** *A real-valued continuous-time signal whose Fourier transform is supported in* $[-B, B]$ *is exactly recovered from its samples taken at any rate* $f_s \geq 2B$.

With $B = 20$ Hz, the rate $f_s = 50$ Hz exceeds $2B = 40$ Hz by a margin of 25%. The corresponding Nyquist frequency $f_s/2 = 25$ Hz dominates the relevant signal bandwidth, so resampling does not erase any fall-relevant spectral information.

Integer divisibility. For SisFall the conversion ratio 200:50 = 4:1 is an integer, so resampling reduces to decimation by a factor of four. For FAD_40Hz the ratio is 40:50 = 4:5 and for MobiFall the ratio is 87.5:50 = 7:4; in both cases the rational ratio admits a finite-impulse-response polyphase resampler at moderate cost.

Computational tractability. The persistent homology computation dominates the cost of the pipeline; on the Vietoris–Rips complex the worst-case complexity is super-polynomial in the size of the point cloud (Section 3.6.5). A five-second window contains $L = 250$ samples at 50 Hz versus $L = 1000$ samples at 200 Hz; the resulting reduction in computational load is what makes a 200-iteration Bayesian search across three datasets feasible.

3.2.3 Window extraction

The classification unit is a fixed-length window of $L = \lfloor w_{\text{sec}} \cdot f_s \rfloor$ samples. Fall and non-fall trials are windowed by different rules to reflect the structural difference between their respective signal regimes.

For a fall trial $\mathbf{x}$, a single window is extracted around the index of the global SMV maximum:

$$t^* = \arg\max_t \mathrm{SMV}_{\mathrm{acc}}(t), \qquad w^* = W_{t^* - \lfloor L/2 \rfloor,\, L}(\mathbf{x}). \tag{3.8}$$

The centring strategy guarantees that the impact transient is contained in the window regardless of the recording length.

For an ADL trial, consecutive non-overlapping windows of length L are extracted from the start of the recording with stride L:

$$w_j = W_{jL,\, L}(\mathbf{x}), \qquad j = 0, 1, \ldots, \lfloor T/L \rfloor - 1. \tag{3.9}$$

This produces multiple windows per ADL trial, which is appropriate because daily activities are quasi-stationary and substantially longer than fall events.

3.2.4 Standard scaling

After the feature vector $\phi_\lambda(w) \in \mathbb{R}^{24}$ has been computed for every window, each of its coordinates is centred and rescaled. Let $\mathcal{Z} = \{\phi_\lambda(w_i)\}_{i\in\mathcal{I}_{\mathrm{tr}}} \subset \mathbb{R}^{24}$ be the collection of feature vectors over the training partition. The empirical mean and variance per coordinate are

$$\mu_j = \frac{1}{|\mathcal{I}_{\mathrm{tr}}|}\sum_{i\in\mathcal{I}_{\mathrm{tr}}} \phi_\lambda(w_i)_j, \qquad \sigma_j^2 = \frac{1}{|\mathcal{I}_{\mathrm{tr}}|}\sum_{i\in\mathcal{I}_{\mathrm{tr}}} \big(\phi_\lambda(w_i)_j - \mu_j\big)^2,$$

and the standardised feature vector is $z_j = (\phi_\lambda(w)_j - \mu_j)/\sigma_j$. The parameters $\{\mu_j, \sigma_j\}$ are estimated from the training partition only and applied to the test partition; this protocol avoids information leakage from held-out subjects or windows into the normalisation procedure.

3.3 Phase-Space Reconstruction via Delay Embedding

3.3.1 Definition

Definition 3.2 (Delay embedding). *Let $s = (s_0, s_1, \dots, s_{L-1}) \in \mathbb{R}^L$ be a finite scalar sequence. For integers $m \geq 2$ (embedding dimension), $\tau \geq 1$ (lag), and $r \geq 1$ (stride), the (m,τ,r)-*delay embedding *of s is the finite point cloud*

$$P_{m,\tau,r}(s) = \big\{\mathbf{p}_i \in \mathbb{R}^m : i = 0, 1, \dots, N-1\big\}, \tag{3.10}$$

where

$$\mathbf{p}_i = \big(s_{ir},\ s_{ir+\tau},\ s_{ir+2\tau},\ \dots,\ s_{ir+(m-1)\tau}\big),$$

and $N = \lfloor (L-(m-1)\tau - 1)/r \rfloor + 1$ is the number of embedded points.

Each point $\mathbf{p}_i$ collects m samples of s separated by lag τ, and consecutive points are separated by stride r. As i increases the sliding window traces a discrete trajectory in $\mathbb{R}^m$. The cardinality N scales linearly in L and inversely in r, which is decisive for the computational cost of the subsequent homology computation.

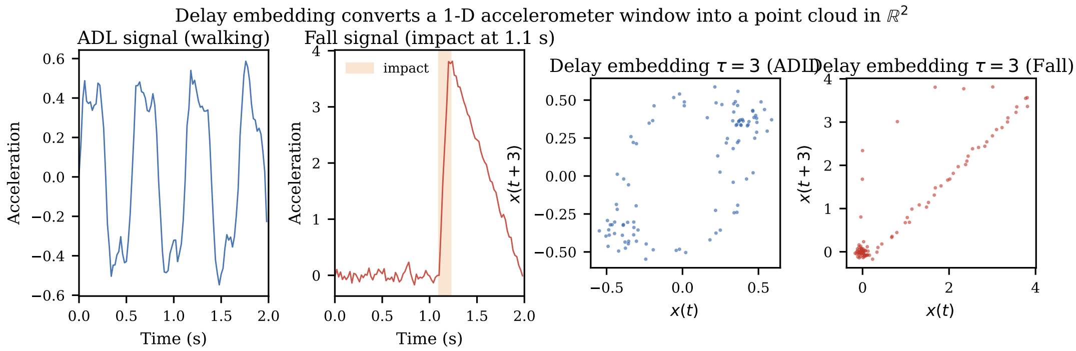

Figure 3.1: Left: synthetic ADL (walking) and fall accelerometer windows. Right: the corresponding two-dimensional delay embeddings ($m = 2$, $\tau = 3$). The ADL trajectory forms a compact, roughly periodic arc; the fall trajectory produces a sparser, elongated point cloud due to the large-amplitude impact spike. This geometric difference motivates the use of persistent homology to distinguish the two signal classes.

3.3.2 The Takens embedding theorem

The use of delay coordinates is motivated by a foundational theorem of dynamical systems theory. Let M be a smooth compact d-dimensional manifold and $\Phi\colon M \to M$ a smooth diffeomorphism (the time-Δt flow map of an autonomous dynamical system). Let $h\colon M \to \mathbb{R}$ be a smooth scalar observable. The associated *delay map* of order m is

$$\Psi_m^{h,\Phi}\colon M \longrightarrow \mathbb{R}^m, \quad x \longmapsto \big(h(x), h(\Phi(x)), h(\Phi^2(x)), \ldots, h(\Phi^{m-1}(x))\big).$$

Theorem 3.2 (Takens (Takens, 1981))**.** *Let M be a smooth compact d-dimensional manifold. For $m \geq 2d+1$, it is a generic property of the pair (Φ, h) that $\Psi_m^{h,\Phi}\colon M \to \mathbb{R}^m$ is a smooth embedding, i.e., a smooth injective immersion onto its image. In particular, $\Psi_m^{h,\Phi}(M)$ is a smooth d-dimensional submanifold of $\mathbb{R}^m$ diffeomorphic to M.*

Generic means that the set of pairs (Φ, h) for which the conclusion fails is contained in a meagre subset of $C^k(M, M) \times C^k(M, \mathbb{R})$ in the appropriate C^k-topology, i.e., a countable union of nowhere-dense sets. Theorem 3.2 states that, for m sufficiently large relative to the intrinsic dimension of the attractor, the reconstructed trajectory is topologically and differentially faithful to the underlying state space; topological invariants computed on $P_{m,\tau,r}(s)$ therefore reflect intrinsic properties of the dynamical system that produced the signal.

Theorem 3.3 (Fractal extension, Sauer–Yorke–Casdagli (Sauer, Yorke, & Casdagli, 1991))**.** *Let $A \subseteq \mathbb{R}^N$ be a compact invariant set of a dynamical system with box-counting (Minkowski) dimension $\dim_B(A) = d_A$. For $m > 2d_A$, almost every pair (Φ, h) (with respect to prevalence in $C^k(\mathbb{R}^N, \mathbb{R}^N) \times C^k(\mathbb{R}^N, \mathbb{R})$) gives an injective delay map $\Psi_m^{h,\Phi}$ on A.*

Theorem 3.3 relaxes the smooth-manifold assumption of Theorem 3.2: fractal invariant sets, which are typical for real chaotic systems, can be embedded provided m exceeds twice their box-counting dimension. This is the relevant statement for inertial sensor signals, whose attractors need not be smooth manifolds.

Remark 3.1 (Heuristic status)**.** *Theorems 3.2 and 3.3 require (i) a deterministic underlying dynamical system, (ii) infinite observation, and (iii) generic observable and dynamics. Wearable fall signals satisfy none of these hypotheses strictly: they are finite, noisy, and not the deterministic output of an autonomous flow on a fixed manifold. The theorems are therefore applied in a* heuristic *rather than a rigorous sense: they motivate the use of delay coordinates as a geometrically richer representation than the raw scalar sequence, without guaranteeing that the resulting point cloud faithfully reconstructs any specific attractor. The parameters (m, τ, r) are treated as hyperparameters and selected empirically by the search of Section 3.9.*

3.3.3 Geometric interpretation for fall detection

What makes the delay embedding useful for fall detection is a structural difference in how the two signal classes look in reconstruction space.

A *fall* produces a brief high-amplitude transient in the SMV: a sharp pre-impact rise, a peak at ground contact, and a partial decay. Embedded in $\mathbb{R}^m$, this trajectory shoots rapidly into an extreme region and does not return within the window—the resulting point cloud is *elongated and non-recurrent*, a quasi-one-dimensional sweep with few or no closed loops.

A *daily activity* such as walking is approximately periodic, with a gait period of roughly $T_{\text{walk}} \approx 1$ s. In reconstruction space the trajectory traces a structured orbit that revisits nearby regions; the point cloud is *recurrent and loop-forming.*

Persistent homology in degrees $k = 0$ and $k = 1$ detects exactly this dichotomy. H_0 records connected components (spread of the cluster); H_1 records independent one-dimensional cycles (loops from periodic structure). Fall windows are expected to show pronounced H_0 and weak H_1; ADL windows, richer H_1. The scalar statistics of Section 3.7 are designed to quantify these expectations.

3.3.4 Filtration scale control

For the Rips and SparseRips families, the filtration is truncated at a data-adaptive scale to bound the size of the simplicial complex. Define the multiset of inter-point distances

$$D(P) = \left\{ d_{\text{met}}(\mathbf{p}, \mathbf{p}') : \mathbf{p}, \mathbf{p}' \in P,\ \mathbf{p} \neq \mathbf{p}' \right\}$$

and set

$$\varepsilon_{\max} = \text{Quantile}_q(D(P)), \qquad q \in \{0.20, 0.40, 0.60, 0.80\}. \tag{3.11}$$

The percentile q controls the scale at which the filtration is truncated: a low percentile concentrates the analysis on fine-scale local geometry; a high percentile admits coarser global structure at the cost of additional simplices. The Alpha complex requires no such cutoff because its size is intrinsically polynomial in $|P|$ via the Delaunay triangulation.

3.4 Topological Preliminaries

Persistent homology rests on objects from point-set and algebraic topology that are not standard in signal processing. This section collects the necessary background.

3.4.1 Topological spaces and homotopy

Definition 3.3 (Topological space). *A* topological space *is a pair* $(X, \mathcal{T})$ *where* X *is a set and* $\mathcal{T} \subseteq 2^X$ *is a family of subsets, the* open sets*, satisfying*

(i) $\emptyset \in \mathcal{T}$ *and* $X \in \mathcal{T}$*;*

(ii) $\mathcal{T}$ *is closed under arbitrary unions;*

(iii) $\mathcal{T}$ *is closed under finite intersections.*

A map $f\colon (X, \mathcal{T}_X) \to (Y, \mathcal{T}_Y)$ *is* continuous *if* $f^{-1}(U) \in \mathcal{T}_X$ *for every* $U \in \mathcal{T}_Y$*. A continuous bijection with continuous inverse is a* homeomorphism*, written* $X \cong Y$*.*

Every metric space (X, d) carries the *metric topology*, generated by the open balls $B(x, r) = \{y \in X : d(x, y) < r\}$.

Definition 3.4 (Homotopy and homotopy equivalence). *Continuous maps* $f, g\colon X \to Y$ *are* homotopic *(*$f \simeq g$*) if there exists a continuous map* $H\colon X \times [0,1] \to Y$ *with* $H(\cdot, 0) = f$ *and* $H(\cdot, 1) = g$*. A continuous map* $f\colon X \to Y$ *is a* homotopy equivalence *if there exists* $g\colon Y \to X$ *with* $g \circ f \simeq \mathrm{id}_X$ *and* $f \circ g \simeq \mathrm{id}_Y$*. A space is* contractible *if it is homotopy equivalent to a point.*

Homotopy equivalence is strictly coarser than homeomorphism. Every convex subset of $\mathbb{R}^m$ is contractible (via the straight-line homotopy to any interior point). Homology is a homotopy invariant: $X \simeq Y$ implies $H_k(X; \mathbb{F}) \cong H_k(Y; \mathbb{F})$ for all $k \geq 0$.

3.4.2 Abstract and geometric simplicial complexes

Definition 3.5 (Abstract simplicial complex). *Let* V *be a finite set. An* abstract simplicial complex *on* V *is a collection* $K \subseteq 2^V \setminus \{\emptyset\}$ *closed under non-empty subset:* $\sigma \in K$ *and* $\emptyset \neq \rho \subseteq \sigma$ *imply* $\rho \in K$*. An element* $\sigma \in K$ *with* $|\sigma| = k+1$ *is a* k-simplex *of dimension* k*. The* k-skeleton *of* K *is* $K^{(k)} = \{\sigma \in K : \dim \sigma \leq k\}$*.*

Definition 3.6 (Standard k-simplex). *The* standard k-simplex *is*

$$\Delta^k = \mathrm{conv}\{e_0, e_1, \ldots, e_k\} = \Big\{(t_0, \ldots, t_k) \in \mathbb{R}^{k+1} : t_i \geq 0,\ \sum_{i=0}^{k} t_i = 1\Big\},$$

where $\{e_i\}$ is the standard basis of $\mathbb{R}^{k+1}$. Its open interior is $\operatorname{int}(\Delta^k) = \{t \in \Delta^k : t_i > 0 \;\forall i\}$.

Definition 3.7 (Geometric realisation)**.** *Let K be an abstract simplicial complex with vertex set V. An injection $\iota\colon V \hookrightarrow \mathbb{R}^N$ is in* general position *for K if no $d+2$ vertices lie in a common affine subspace of dimension $\leq d = \dim K$. The* geometric realisation *of K is*

$$|K| = \bigcup_{\sigma \in K} \operatorname{conv}\bigl\{\iota(v) : v \in \sigma\bigr\} \subset \mathbb{R}^N,$$

equipped with the subspace topology of $\mathbb{R}^N$.

The homeomorphism type of $|K|$ does not depend on the choice of general-position embedding. Geometric realisation is *functorial*: a simplicial map $f\colon K \to L$ (a vertex map sending each simplex of K to a simplex of L) induces a continuous map $|f|\colon |K| \to |L|$ by barycentric interpolation.

3.5 Simplicial Homology

Fix a field $\mathbb{F}$. The computational implementation uses $\mathbb{F} = \mathbb{F}_2$, the field with two elements; the algebraic framework is identical over any field, modulo orientation conventions.

3.5.1 Chains and the boundary operator

An *oriented k-simplex* is an equivalence class $[v_0, \ldots, v_k]$ of ordered tuples of distinct vertices, modulo even permutations: $[v_{\pi(0)}, \ldots, v_{\pi(k)}] = \operatorname{sgn}(\pi)\,[v_0, \ldots, v_k]$. Each simplex of positive dimension carries exactly two orientations; over $\mathbb{F}_2$, where $-1 = 1$, orientation is irrelevant.

Definition 3.8 (Chain group)**.** *The k-*th chain group *$C_k(K; \mathbb{F})$ is the $\mathbb{F}$-vector space freely generated by the oriented k-simplices of K, with the identification $-\sigma = \bar{\sigma}$ for the two orientations of the same underlying simplex.*

Definition 3.9 (Boundary operator). *The* k-th boundary $\partial_k \colon C_k(K;\mathbb{F}) \to C_{k-1}(K;\mathbb{F})$ *is the* $\mathbb{F}$*-linear map defined on basis elements by*

$$\partial_k\big([v_0, v_1, \ldots, v_k]\big) = \sum_{i=0}^{k} (-1)^i [v_0, \ldots, \widehat{v_i}, \ldots, v_k], \tag{3.12}$$

where $\widehat{v_i}$ *denotes omission of* v_i.

Lemma 3.1 (Fundamental relation $\partial^2 = 0$). *For all* $k \geq 1$, $\partial_{k-1} \circ \partial_k = 0$.

Proof. It suffices to verify the identity on a basis element $\sigma = [v_0, \ldots, v_k]$. Two applications of (3.12) give

$$\partial_{k-1}(\partial_k \sigma) = \sum_{j<i} (-1)^{i+j} [v_0, \ldots, \widehat{v_j}, \ldots, \widehat{v_i}, \ldots, v_k] \\ + \sum_{i<j} (-1)^{i+j-1} [v_0, \ldots, \widehat{v_i}, \ldots, \widehat{v_j}, \ldots, v_k],$$

where in the second sum the omitted vertex v_j has been shifted by one position because v_i with $i < j$ was already deleted. For each unordered pair $\{i, j\}$ with $i < j$ the two contributions cancel exactly, yielding zero. $\square$

Geometrically, Lemma 3.1 expresses the fact that the boundary of a simplex is itself a closed surface and hence has no boundary.

3.5.2 Homology groups and Betti numbers

Lemma 3.1 implies $\operatorname{im}(\partial_{k+1}) \subseteq \ker(\partial_k)$. The quotient is well defined.

Definition 3.10 (Homology and Betti numbers). *The* cycle group *is* $Z_k = \ker \partial_k \subseteq C_k(K;\mathbb{F})$*; the* boundary group *is* $B_k = \operatorname{im} \partial_{k+1} \subseteq Z_k$*. The* k-th homology *is the quotient*

$$H_k(K;\mathbb{F}) = Z_k / B_k,$$

and the k-th Betti number *is* $\beta_k = \dim_{\mathbb{F}} H_k(K;\mathbb{F})$.

The Betti numbers admit elementary geometric interpretations. β_0 counts the connected components of $|K|$; β_1 counts the number of independent one-dimensional cycles (loops not bounding any 2-chain); β_2 counts the number of enclosed two-dimensional voids.

Theorem 3.4 (Euler–Poincaré formula). *Let* $f_k = |\{\sigma \in K : \dim \sigma = k\}|$. *The Euler characteristic of* K*, defined by*

$$\chi(K) = \sum_{k \geq 0} (-1)^k f_k,$$

satisfies

$$\chi(K) = \sum_{k \geq 0} (-1)^k \beta_k. \tag{3.13}$$

Proof. The rank–nullity theorem applied to ∂_k gives $\dim C_k = \dim Z_k + \dim B_{k-1}$ for all $k \geq 0$, with $B_{-1} = 0$. Summing with alternating signs and rearranging telescopes the boundary-group terms; the result is (3.13). □

The formula links a combinatorial datum (f_k) to homotopy invariants (β_k) and serves as a consistency check on any homology computation.

3.5.3 Functoriality

Definition 3.11 (Simplicial and chain maps). *A* simplicial map $f\colon K \to L$ *is a vertex map* $f\colon V(K) \to V(L)$ *with* $f(\sigma) \in L$ *for every* $\sigma \in K$. *The induced* chain map $f_\#\colon C_k(K;\mathbb{F}) \to C_k(L;\mathbb{F})$ *is defined on basis elements by*

$$f_\#([v_0, \ldots, v_k]) = \begin{cases} [f(v_0), \ldots, f(v_k)] & \textit{if } f(v_0), \ldots, f(v_k) \textit{ are distinct,} \\ 0 & \textit{otherwise.} \end{cases}$$

One checks that $f_\# \partial_k^K = \partial_k^L f_\#$, i.e., chain maps commute with boundaries. Consequently, $f_\#$ maps cycles to cycles and boundaries to boundaries, and descends to a well-defined linear map

$$H_k(f)\colon H_k(K;\mathbb{F}) \longrightarrow H_k(L;\mathbb{F}), \quad [z] \longmapsto [f_\#(z)].$$

Composition is preserved, $H_k(g \circ f) = H_k(g) \circ H_k(f)$, and identities are preserved, $H_k(\mathrm{id}) = \mathrm{id}$. In categorical language, $H_k(\,\cdot\,;\mathbb{F})$ is a functor from the category of simplicial complexes to the category $\mathbf{Vect}_{\mathbb{F}}$ of $\mathbb{F}$-vector spaces.

This functoriality is the bridge to persistent homology. The inclusions $K_i \hookrightarrow K_j$ for $i \leq j$ in any filtration are simplicial maps, and the induced maps $H_k(K_i;\mathbb{F}) \to H_k(K_j;\mathbb{F})$ are the structure maps of the persistence module defined below.

3.6 Persistent Homology

3.6.1 Filtrations and complex families

Definition 3.12 (Filtration)**.** *A* filtration *of a simplicial complex K is a nested family of subcomplexes*

$$\emptyset = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_n = K,$$

indexed by a non-decreasing real sequence $\varepsilon_0 \leq \varepsilon_1 \leq \cdots \leq \varepsilon_n$ of filtration values.

For a finite point cloud $P \subset \mathbb{R}^m$ equipped with metric d_{met}, the methodology uses three filtered complex families.

Vietoris–Rips complex.

$$\mathcal{R}_\varepsilon(P) = \big\{\sigma \subseteq P : \operatorname{diam}_{d_{\text{met}}}(\sigma) \leq \varepsilon\big\}. \tag{3.14}$$

The Rips complex is a *flag complex*: every clique of its 1-skeleton is filled by a higher-dimensional simplex. The filtration is therefore determined entirely by the order in which edges appear.

Alpha complex. For the Euclidean metric, the Voronoi cell of $p \in P$ is

$$V_p = \big\{x \in \mathbb{R}^m : \|x - p\|_2 \leq \|x - p'\|_2 \ \forall p' \in P\big\}.$$

The *Alpha complex* is the nerve of the restricted cover $\{B(p, \varepsilon) \cap V_p\}_{p \in P}$:

$$\mathcal{A}_\varepsilon(P) = \Big\{\sigma \subseteq P : \bigcap_{p \in \sigma} \big(B(p, \varepsilon) \cap V_p\big) \neq \emptyset\Big\}. \tag{3.15}$$

$\mathcal{A}_\varepsilon(P)$ is a subcomplex of the Delaunay triangulation, hence has at most $O(|P|^{\lceil m/2 \rceil})$ simplices, dramatically less than the worst-case Rips bound of $2^{|P|}$.

Sparse Rips complex. The SparseRips construction (Cavanna, Jahanseir, & Sheehy, 2015) replaces P by a greedy *maxmin* landmark subset $L \subseteq P$: starting from an arbitrary seed, each successive landmark is the point of P farthest from the existing landmarks. The persistent homology of the resulting weighted complex is provably a $(1+\varepsilon)$-approximation of that of $\mathcal{R}_\varepsilon(P)$ in the interleaving distance of Definition 3.14.

Theorem 3.5 (Nerve theorem (Borsuk, 1948))**.** *Let $\mathcal{U} = \{U_i\}_{i\in I}$ be a finite open cover of a paracompact space X such that every non-empty finite intersection $U_{i_0} \cap \cdots \cap U_{i_p}$ is contractible. Then $|N(\mathcal{U})| \simeq X$, where $N(\mathcal{U})$ is the abstract simplicial complex with simplices $\{i_0, \ldots, i_p\}$ for which the corresponding intersection is non-empty.*

The Alpha complex satisfies the hypothesis of Theorem 3.5: each $B(p,\varepsilon) \cap V_p$ is the intersection of a ball and a convex Voronoi cell, hence convex and therefore contractible, and finite intersections of convex subsets of $\mathbb{R}^m$ are convex and contractible. Consequently $|\mathcal{A}_\varepsilon(P)| \simeq \bigcup_{p\in P} B(p,\varepsilon)$: the Alpha complex faithfully encodes the topology of the union of balls without introducing simplices beyond the Delaunay triangulation.

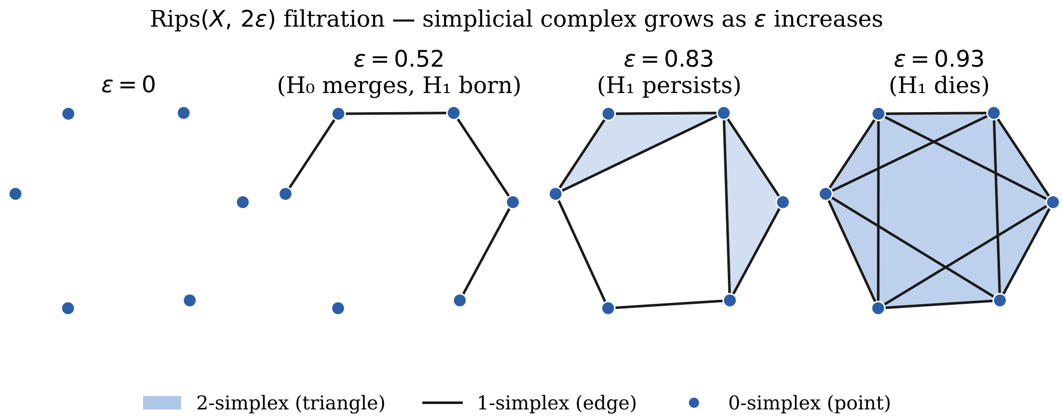

Figure 3.2: Rips $(X, 2\varepsilon)$ filtration of a six-point cloud at four scales. At $\varepsilon = 0$ only vertices are present ($\beta_0 = 6$, $\beta_1 = 0$). As ε increases, edges appear and connected components merge (β_0 decreases). A loop (H_1 class) is born when the polygon is complete and dies when triangles fill it. Features far from the diagonal in the persistence diagram have long lifetime and are noise-robust (Cohen–Steiner stability, Theorem 3.7).

3.6.2 Persistence modules

Regard the linearly ordered set $(\mathbb{R}, \leq)$ as a category: objects are real numbers, and there is a unique morphism $r \to s$ whenever $r \leq s$. Let $\mathbf{Vect}_{\mathbb{F}}$ be the category of $\mathbb{F}$-vector spaces and $\mathbb{F}$-linear maps.

Definition 3.13 (Persistence module)**.** *A (one-parameter)* persistence module *over* $\mathbb{F}$ *is a functor*

$$\mathbb{V}\colon (\mathbb{R}, \leq) \longrightarrow \mathbf{Vect}_{\mathbb{F}}.$$

Concretely, $\mathbb{V}$ *assigns a vector space* V_r *to each* $r \in \mathbb{R}$ *and a linear map* $\rho_{r,s}\colon V_r \to V_s$ *(the* structure map*) to each* $r \leq s$*, with* $\rho_{r,r} = \mathrm{id}$ *and* $\rho_{s,t} \circ \rho_{r,s} = \rho_{r,t}$*.*

A morphism $\Phi\colon \mathbb{U} \to \mathbb{V}$ is a natural transformation: linear maps $\Phi_r\colon U_r \to V_r$ with $\Phi_s \circ \rho^{\mathbb{U}}_{r,s} = \rho^{\mathbb{V}}_{r,s} \circ \Phi_r$ for all $r \leq s$. Persistence modules with such morphisms form a category $\mathbf{Pers}_{\mathbb{F}}$.

Applying $H_k(\,\cdot\,;\mathbb{F})$ levelwise to a filtration and to each inclusion $K_i \hookrightarrow K_j$ yields a persistence module $\mathbb{V}_k = \{H_k(K_i;\mathbb{F})\}$: the *k-th persistence module of the filtration.* This is the central algebraic invariant studied by persistent homology.

Definition 3.14 (Interleaving distance)**.** *For* $\varepsilon \geq 0$*, the* ε-shift *of* $\mathbb{V}$ *is the persistence module* $\mathbb{V}[\varepsilon]_r := V_{r+\varepsilon}$*. Modules* $\mathbb{U}, \mathbb{V}$ *are* ε-interleaved *if there exist morphisms* $\Phi\colon \mathbb{U} \to \mathbb{V}[\varepsilon]$ *and* $\Psi\colon \mathbb{V} \to \mathbb{U}[\varepsilon]$ *with* $\Psi[\varepsilon] \circ \Phi = \rho^{\mathbb{U}}_{r,r+2\varepsilon}$ *and* $\Phi[\varepsilon] \circ \Psi = \rho^{\mathbb{V}}_{r,r+2\varepsilon}$ *for all* r*. The* interleaving distance *is*

$$d_I(\mathbb{U}, \mathbb{V}) = \inf\bigl\{\varepsilon \geq 0 : \mathbb{U} \textit{ and } \mathbb{V} \textit{ are } \varepsilon\textit{-interleaved}\bigr\}.$$

The interleaving distance is an extended pseudometric on persistence modules; it descends to a metric on the set of isomorphism classes.

3.6.3 The structure theorem and persistence diagrams

Definition 3.15 (Interval module)**.** *For* $b < d$ *in* $\mathbb{R} \cup \{+\infty\}$*, the* interval module $\mathbb{F}[b, d)$ *is the persistence module with* $V_r = \mathbb{F}$ *for* $b \leq r < d$*,* $V_r = 0$ *otherwise, and identity structure maps between non-zero positions.*

Theorem 3.6 (Structure theorem, Crawley-Boevey (Crawley-Boevey, 2015))**.** *Every pointwise finite-dimensional persistence module* $\mathbb{V}$ *over a field* $\mathbb{F}$ *decomposes* *uniquely, up to isomorphism and reordering, as a direct sum of interval modules:*

$$\mathbb{V} \;\cong\; \bigoplus_{i\in I} \mathbb{F}[b_i, d_i). \tag{3.16}$$

The multiset $\{(b_i, d_i)\}_{i\in I}$ *is a complete isomorphism invariant of* $\mathbb{V}$.

The special case for tame modules over filtered finite complexes was established by Zomorodian and Carlsson (Edelsbrunner, Letscher, & Zomorodian, 2002). Each interval (b_i, d_i) represents a topological feature of degree k that is *born* at scale b_i and *dies* at scale d_i; its *persistence* is $p_i = d_i - b_i$. The diagonal is added by convention so that the diagram is a multiset in

$$\{(b, d) \in [-\infty, \infty]^2 : b \leq d\} \cup \{\text{diagonal with infinite multiplicity}\}.$$

Definition 3.16 (Persistence diagram)**.** *The* k-th persistence diagram *of a filtration is*

$$\mathrm{PD}_k = \big\{(b_i, d_i) : b_i < d_i < +\infty\big\}_{i\in I_k}. \tag{3.17}$$

Features with $d_i = +\infty$ *are excluded from the present feature vectorisation because their infinite lifetime requires special handling in any scalar summary.*

3.6.4 Distances and stability

Definition 3.17 (Bottleneck and Wasserstein distances)**.** *A* partial matching *between diagrams* D_1, D_2 *is a bijection* $\gamma\colon D_1 \cup \Delta \to D_2 \cup \Delta$ *that is the identity outside a finite set, where* $\Delta = \{(x, x) : x \in \mathbb{R}\}$ *is the diagonal. The* p-Wasserstein distance *(*$1 \leq p < \infty$*) is*

$$W_p(D_1, D_2) = \inf_{\gamma}\Big(\sum_{x\in D_1\cup\Delta} \|x - \gamma(x)\|_\infty^p\Big)^{1/p},$$

and the bottleneck distance *is*

$$d_B(D_1, D_2) = \inf_{\gamma}\sup_{x\in D_1\cup\Delta} \|x - \gamma(x)\|_\infty = \lim_{p\to\infty} W_p(D_1, D_2).$$

Theorem 3.7 (Stability of persistence diagrams, Cohen-Steiner–Edelsbrunner–Harer (Cohen-Steiner, Edelsbrunner, & Harer, 2007))**.** *Let* $f, g\colon X \to \mathbb{R}$ *be tame continuous functions on a triangulable compact space* X*, and let* D_f, D_g *be the persistence diagrams of their sublevel-set filtrations. Then*

$$d_B(D_f, D_g) \leq \|f - g\|_\infty.$$

Theorem 3.8 (Algebraic stability, Chazal et al. (Chazal, de Silva, Glisse, & Oudot, 2016))**.** *For any two pointwise finite-dimensional persistence modules* $\mathbb{U}, \mathbb{V}$ *over a field* $\mathbb{F}$*,*

$$d_B\big(\mathrm{PD}(\mathbb{U}), \mathrm{PD}(\mathbb{V})\big) = d_I(\mathbb{U}, \mathbb{V}).$$

Theorem 3.8 upgrades stability from an inequality to an isometry: the bottleneck distance on diagrams equals the interleaving distance on modules, providing a complete algebraic characterisation of the geometry of persistence.

For point cloud filtrations, the metric stability statement of practical interest is the following.

Corollary 3.1 (Metric stability)**.** *Let* $P, Q \subset \mathbb{R}^m$ *be finite point sets and let*

$$d_H(P, Q) = \max\Big(\max_{p \in P} \min_{q \in Q} \|p - q\|_2,\ \max_{q \in Q} \min_{p \in P} \|p - q\|_2\Big)$$

be their Hausdorff distance. Then for every $k \geq 0$*,*

$$d_B\big(\mathrm{PD}_k(\mathcal{R}(P)),\ \mathrm{PD}_k(\mathcal{R}(Q))\big) \leq 2\, d_H(P, Q).$$

Proof sketch. If $d_H(P, Q) \leq \delta$, then each simplex of $\mathcal{R}_\varepsilon(P)$ lifts to a simplex of $\mathcal{R}_{\varepsilon+2\delta}(Q)$ by replacing each vertex by a point of Q within distance δ and applying the triangle inequality to the diameter. By symmetry the two Rips modules are 2δ-interleaved, hence $d_B \leq 2\delta$ by Theorem 3.8. □

The practical content is that bounded sensor noise of amplitude δ perturbs each embedded point by at most δ in ℓ_2, hence the persistence diagrams of two realisations of the same fall event differ by at most 2δ in bottleneck distance. Features with persistence $p_i \gg 2\delta$ are noise-robust; near-diagonal features with $p_i \approx 0$ are noise-sensitive.

3.6.5 Computational reduction

Persistence diagrams are computed by *boundary matrix reduction.* Order the simplices of the filtration as $\sigma_1, \sigma_2, \ldots, \sigma_n$ so that $\dim \sigma_i \leq \dim \sigma_{i+1}$ and simplices entering at smaller scale precede those at larger scale (consistent with filtration

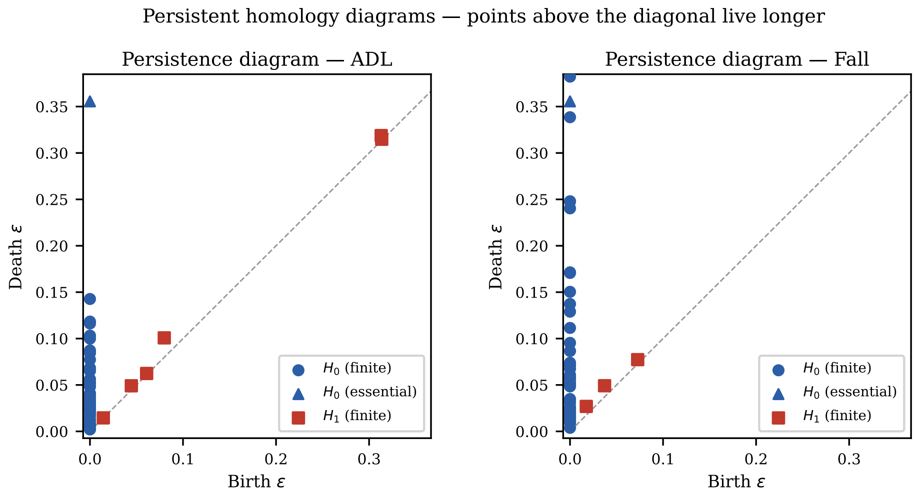

Figure 3.3: Persistence diagrams (H_0 circles, H_1 squares) for delay-embedded ADL and fall windows. Points far from the diagonal represent long-lived topological features. The fall window typically exhibits a high-persistence H_1 feature (loop) absent or short-lived in the ADL window—a geometric signature of the sharp impact trajectory in the delay-embedded space. Triangles and diamonds denote essential classes (infinite death, plotted at the top margin).

order). The boundary matrix $D \in \mathbb{F}^{n\times n}$ has $D_{i,j} = 1$ if $\sigma_i \in \partial\sigma_j$ and 0 otherwise. The *pivot* of column j is $\mathrm{low}(D, j) = \max\{i : D_{i,j} \neq 0\}$, undefined if column j is zero.

Algorithm 1 Standard persistence algorithm (column reduction over $\mathbb{F}_2$).

Require: Boundary matrix $D \in \{0,1\}^{n\times n}$ in filtration order
Ensure: Reduced matrix R with unique pivots; persistence pairs read from R

```
R ← D
for j = 1 to n do
    while ∃ i < j with low(R, i) = low(R, j) ≠ ⊥ do
        R[·, j] ← R[·, j] + R[·, i]          ▷ column addition over F_2
    end while
end for
Pairs: {(low(R, j), j) : R[·, j] ≠ 0}
```

Proposition 3.2 (Uniqueness of pairing)**.** *The pairing produced by Algorithm 1 is independent of the order in which column additions are performed; it depends only on the filtration.*

The proposition is a corollary of the structure theorem 3.6. The worst-case complexity of Algorithm 1 is $O(n^3)$; the GUDHI library (GUDHI Editorial Board,

2015) implements optimised variants (twist reduction, clearing, dualisation to cohomology) with near-linear average-case complexity on the point cloud sizes considered here.

3.7 Feature Vectorisation

Persistence diagrams are complete isomorphism invariants of the filtration, but they cannot be passed directly to standard classifiers: cardinality $|I_k|$ varies across windows and configurations, and the diagram space under the bottleneck or Wasserstein metric is not a Hilbert space. Various vectorisation schemes—persistence images (Adams, Emerson, Kirby, Neville, Peterson, Shipman, Chepushtanova, Hanson, Motta, & Ziegelmeier, 2017), persistence landscapes (Bubenik, 2015), and others—address these obstacles in different ways. The methodology here uses a fixed set of scalar statistics, chosen because a constant feature dimension across all configurations λ is needed for a uniform Bayesian search interface.

3.7.1 Persistence-based statistics

For each homological degree $k \in \{0, 1\}$, write $\{(b_i^{(k)}, d_i^{(k)})\}_{i=1}^{n_k}$ for the finite-persistence pairs of PD_k, and let $p_i^{(k)} = d_i^{(k)} - b_i^{(k)} > 0$ be the persistence. When $n_k = 0$ all derived statistics are set to zero. Ten scalar statistics are computed per degree.

j	Statistic	Definition
1	Bar count	n_k/L (normalised by window length)
2	Entropy	$-\sum_i \tilde{p}_i \log \tilde{p}_i,\ \tilde{p}_i = p_i / \sum_j p_j$
3	Maximum	$\max_i p_i^{(k)}$
4	Energy	$\sum_i (p_i^{(k)})^2$
5	Mean	$\bar{p}^{(k)} = n_k^{-1} \sum_i p_i^{(k)}$
6	Standard dev.	$\left(n_k^{-1} \sum_i (p_i - \bar{p})^2\right)^{1/2}$
7	Median	$Q_{0.50}(\mathbf{p}^{(k)})$
8	Q1	$Q_{0.25}(\mathbf{p}^{(k)})$
9	Q3	$Q_{0.75}(\mathbf{p}^{(k)})$
10	Third moment	$n_k^{-1} \sum_i (p_i - \bar{p})^3$

These ten statistics for $k = 0$ occupy feature positions 1–10 and for $k = 1$ occupy positions 11–20. Together they give a 20-dimensional topological sub-vector.

3.7.2 Signal-level features

Four additional scalar features are computed directly from $\mathrm{SMV}_{\mathrm{acc}}$ on the window:

$$f_{21} = \max_t \mathrm{SMV}, \quad f_{22} = \mathrm{std}(\mathrm{SMV}), \quad f_{23} = \overline{\mathrm{SMV}}, \quad f_{24} = \max_t \mathrm{SMV} - \min_t \mathrm{SMV}. \tag{3.18}$$

3.7.3 Complete feature vector

The complete feature vector under configuration λ is

$$\phi_\lambda(w) = \left(f_1^{(0)}, \ldots, f_{10}^{(0)},\ f_1^{(1)}, \ldots, f_{10}^{(1)},\ f_{21}, f_{22}, f_{23}, f_{24}\right) \in \mathbb{R}^{24}. \tag{3.19}$$

The fixed dimension is a methodological constraint imposed by the requirement of a uniform classifier interface across λ.

3.8 Classification Models

Two linear classifier families are considered, $\mathcal{F}_{\mathrm{lr}}$ and $\mathcal{F}_{\mathrm{svm}}$. Both are applied with balanced class weighting and, for SVM, with Platt probability calibration.

3.8.1 Logistic regression

Logistic regression models the conditional fall probability by

$$\mathbb{P}(Y = 1 \mid \mathbf{z}; \boldsymbol{\theta}, b) = \sigma(\boldsymbol{\theta}^\top \mathbf{z} + b), \qquad \sigma(u) = \frac{1}{1 + e^{-u}}, \tag{3.20}$$

where $\mathbf{z} \in \mathbb{R}^{24}$ is the standardised feature vector and $(\boldsymbol{\theta}, b) \in \mathbb{R}^{24} \times \mathbb{R}$ are the parameters. Given training data $\{(\mathbf{z}_i, y_i)\}_{i=1}^n$ the regularised negative log-likelihood is

$$\mathcal{L}_\Omega(\boldsymbol{\theta}, b) = -\frac{1}{n} \sum_{i=1}^{n} \Big[y_i \log \sigma_i + (1 - y_i) \log(1 - \sigma_i)\Big] + \frac{1}{C}\,\Omega(\boldsymbol{\theta}), \tag{3.21}$$

where $\sigma_i = \sigma(\boldsymbol{\theta}^\top \mathbf{z}_i + b)$, $C > 0$ is the inverse regularisation strength, and $\Omega(\boldsymbol{\theta}) \in \{\frac{1}{2}\|\boldsymbol{\theta}\|_2^2,\ \|\boldsymbol{\theta}\|_1\}$ selects the ℓ_2 or ℓ_1 penalty. The ℓ_2 objective is strictly convex and smooth and is solved by L-BFGS or coordinate descent; the ℓ_1 objective is convex and non-smooth and is solved by liblinear coordinate descent in the implementation.

Proposition 3.3 (Convexity and uniqueness for ℓ_2)**.** *With $\Omega(\boldsymbol{\theta}) = \frac{1}{2}\|\boldsymbol{\theta}\|_2^2$, the objective* (3.21) *is strictly convex in $(\boldsymbol{\theta}, b)$ on $\mathbb{R}^{25}$, hence admits a unique global minimiser.*

Proof. The negative log-likelihood is convex because $-\log \sigma(u)$ is convex in u and the maps $u \mapsto \boldsymbol{\theta}^\top \mathbf{z} + b$ are affine. The ℓ_2 penalty is strictly convex. The sum of a convex and a strictly convex function is strictly convex. $\square$

The predicted probability is $f_\theta(\mathbf{z}) = \sigma(\boldsymbol{\theta}^\top \mathbf{z} + b) \in (0, 1)$.

3.8.2 Support vector machine

The C-support vector classifier seeks a hyperplane $\{\boldsymbol{w}^\top \mathbf{z} + b = 0\}$ with maximum geometric margin. With slack variables $\xi_i \geq 0$ the primal soft-margin problem is

$$\min_{\boldsymbol{w}, b, \boldsymbol{\xi}} \tfrac{1}{2}\|\boldsymbol{w}\|_2^2 + C \sum_{i=1}^{n} \xi_i \quad \text{s.t.} \quad y_i(\boldsymbol{w}^\top \mathbf{z}_i + b) \geq 1 - \xi_i,\ \xi_i \geq 0,\ y_i \in \{-1, +1\}. \tag{3.22}$$

The Lagrangian is

$$L(\boldsymbol{w}, b, \boldsymbol{\xi}, \boldsymbol{\alpha}, \boldsymbol{\mu}) = \tfrac{1}{2}\|\boldsymbol{w}\|^2 + C \sum_i \xi_i - \sum_i \alpha_i \big[y_i(\boldsymbol{w}^\top \mathbf{z}_i + b) - 1 + \xi_i\big] - \sum_i \mu_i \xi_i,$$

with $\alpha_i, \mu_i \geq 0$. The KKT stationarity conditions yield

$$\nabla_{\boldsymbol{w}} L = 0 \ \Rightarrow\ \boldsymbol{w} = \sum_i \alpha_i y_i \mathbf{z}_i, \quad \partial_b L = 0 \ \Rightarrow\ \sum_i \alpha_i y_i = 0, \quad \partial_{\xi_i} L = 0 \ \Rightarrow\ \alpha_i + \mu_i = C.$$

Substitution into L produces the dual quadratic programme

$$\max_{\boldsymbol{\alpha}} \sum_i \alpha_i - \tfrac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \mathbf{z}_i^\top \mathbf{z}_j \quad \text{s.t.} \quad 0 \leq \alpha_i \leq C,\ \sum_i \alpha_i y_i = 0. \tag{3.23}$$

The primal solution recovers $\boldsymbol{w}^* = \sum_i \alpha_i^* y_i \mathbf{z}_i$ and the decision value for a new point is

$$f(\mathbf{z}) = \sum_i \alpha_i^* y_i \mathbf{z}_i^\top \mathbf{z} + b^*. \tag{3.24}$$

Points with $\alpha_i^* > 0$ are the *support vectors*. With $\kappa(\mathbf{z}, \mathbf{z}') = \mathbf{z}^\top \mathbf{z}'$ replaced by the RBF kernel

$$\kappa_{\mathrm{rbf}}(\mathbf{z}, \mathbf{z}') = \exp\bigl(-\gamma \|\mathbf{z} - \mathbf{z}'\|_2^2\bigr),$$

the same dual applies in a reproducing kernel Hilbert space.

3.8.3 Platt score calibration

The SVM decision value $f(\mathbf{z}) \in \mathbb{R}$ is not a probability. Platt scaling (Platt, 1999) fits a logistic function to the decision values on a held-out calibration set:

$$\mathbb{P}(Y = 1 \mid \mathbf{z}) = \sigma\bigl(Af(\mathbf{z}) + B\bigr), \qquad A, B \in \mathbb{R}, \tag{3.25}$$

with (A, B) estimated by minimising the cross-entropy loss against Laplace-smoothed targets

$$t_i = \begin{cases} \dfrac{n_+ + 1}{n_+ + 2} & \text{if } y_i = +1, \\ \dfrac{1}{n_- + 2} & \text{if } y_i = -1, \end{cases}$$

where $n_\pm$ are the counts of positive and negative calibration examples. The smoothing prevents degenerate 0 or 1 predictions when the margin is large.

3.8.4 Class-weighted empirical risk

Falls are minority events; with n_0 negative and n_1 positive training examples $(n = n_0 + n_1)$ the *balanced class weights* are

$$w_c = \frac{n}{2\, n_c}, \qquad c \in \{0, 1\}. \tag{3.26}$$

This satisfies $w_0 n_0 = w_1 n_1 = n/2$. For logistic regression, the weighted log-likelihood replaces $\mathcal{L}_\Omega$ by

$$\mathcal{L}_\Omega^{\mathrm{w}}(\boldsymbol{\theta}, b) = -\frac{1}{n} \sum_{i=1}^{n} w_{y_i} \Bigl[y_i \log \sigma_i + (1 - y_i) \log(1 - \sigma_i)\Bigr] + \frac{1}{C} \Omega(\boldsymbol{\theta}).$$

For C-SVM, the scalar C in (3.22) is replaced by $C_+ = Cw_1$ for positive examples and $C_- = Cw_0$ for negative examples, biasing the model toward higher recall.

3.9 Bayesian Hyperparameter Optimisation

Both the representation map ϕ_λ and the classifier depend on hyperparameters: $\lambda \in \Lambda$ controls the geometric construction, while $\theta \in \Theta$ controls regularisation strength, penalty type, and kernel. These two tuples interact—the geometry of the topological representation determines which classifier settings are optimal—so they must be optimised jointly. The search is carried out by Bayesian optimisation with a tree-structured Parzen estimator (TPE) via Optuna (Akiba, Sano, Yanase, Ohta, & Koyama, 2019).

3.9.1 The optimisation problem

Let $\mathcal{D}_K^{\mathrm{train}}$ be the labelled feature corpus of dataset K from which the test subject has been removed (LOSO outer split, Section 3.10.3). Let

$$\mathrm{CV}_K \colon \Lambda \times \Theta \longrightarrow \mathbb{R}, \qquad (\lambda, \theta) \longmapsto \widehat{F}_{2,K}(\lambda, \theta)$$

denote the inner cross-validation estimator of $\mathbb{E}[F_2 \mid K, \lambda, \theta]$ obtained by stratified k-fold CV on $\mathcal{D}_K^{\mathrm{train}}$. The optimisation problem solved at each outer fold is

$$\lambda_K^*, \theta_K^* \in \operatorname*{arg\,max}_{(\lambda,\theta)\in\Lambda\times\Theta} \mathrm{CV}_K(\lambda, \theta). \tag{3.27}$$

The objective is non-differentiable, noisy (because CV estimates fluctuate with the random fold partition and stochastic class-balancing), and expensive to evaluate (each call requires constructing point clouds, computing persistent homology, fitting the classifier, and performing the CV averaging). These properties make grid search infeasible and motivate a sequential model-based approach.

3.9.2 Sequential model-based optimisation

Sequential model-based optimisation (SMBO) (Hutter, Hoos, & Leyton-Brown, 2011) alternates between fitting a probabilistic surrogate to past evaluations and proposing new evaluations that maximise an acquisition function under the surrogate. Let $y(\lambda, \theta) = \mathrm{CV}_K(\lambda, \theta) + \eta$ denote one noisy evaluation of the objective, with η centred. After t evaluations the data set is

$$\mathcal{H}_t = \big\{((\lambda_i, \theta_i), y_i) : i = 1, \ldots, t\big\}.$$

A surrogate $p(y \mid \lambda, \theta, \mathcal{H}_t)$ is a probability model over the objective conditional on the search point and the history. The standard acquisition function is the *expected improvement* (EI) over a reference value y^*:

$$\mathrm{EI}_{y^*}(\lambda, \theta) = \mathbb{E}\big[\max(y^* - y, 0) \mid \lambda, \theta, \mathcal{H}_t\big]. \tag{3.28}$$

(Here EI is stated for a minimisation problem with y the negative $\widehat{F}_2$; sign conventions are equivalent under negation.) The next search point is

$$(\lambda_{t+1}, \theta_{t+1}) \in \operatorname*{arg\,max}_{(\lambda,\theta)\in\Lambda\times\Theta} \mathrm{EI}_{y^*}(\lambda, \theta). \tag{3.29}$$

The surrogate is updated and the loop continues for a fixed budget. The textbook surrogate is a Gaussian process posterior; TPE replaces it with a non-parametric density model that is better suited to discrete and conditional search spaces.

3.9.3 Tree-structured Parzen estimator

TPE (Bergstra, Bardenet, Bengio, & Kégl, 2011) models the posterior in an inverted form. Instead of modelling $p(y \mid \lambda, \theta)$ directly, it splits the observed history at a quantile $\gamma \in (0, 1)$ and models the density of the inputs given two quality regimes,

$$p(\lambda, \theta \mid y \le y^*) = \ell(\lambda, \theta), \qquad p(\lambda, \theta \mid y > y^*) = g(\lambda, \theta),$$

where y^* is the empirical γ-quantile of the observed $\{y_i\}$. The two density estimates are constructed from the corresponding subsets of $\mathcal{H}_t$ by kernel density estimation: each observation contributes a kernel centred at its parameter value, with bandwidth determined adaptively. Conditional and discrete hyperparameters are handled by a hierarchical tree structure on $\Lambda \times \Theta$, with each node carrying its own one-dimensional kernel.

Proposition 3.4 (EI under TPE is the ratio of densities). *Under the TPE generative model, the expected improvement $\mathrm{EI}_{y^*}(\lambda, \theta)$ is proportional to the ratio of the conditional densities:*

$$\mathrm{EI}_{y^*}(\lambda, \theta) \;\propto\; \left(\gamma + \frac{g(\lambda, \theta)}{\ell(\lambda, \theta)}\,(1 - \gamma)\right)^{-1}.$$

Consequently, maximising EI is equivalent to maximising the ratio $\ell(\lambda, \theta)/g(\lambda, \theta)$.

Proof sketch. With $p(y \le y^*) = \gamma$, Bayes' rule gives

$$p(\lambda, \theta) = \gamma\,\ell + (1-\gamma)\,g, \qquad p(y \le y^* \mid \lambda, \theta) = \frac{\gamma\,\ell}{\gamma\,\ell + (1-\gamma)\,g}.$$

Substitution into $\mathrm{EI}_{y^*} = \int_{-\infty}^{y^*} (y^* - y) p(y \mid \lambda, \theta)\, dy$ and elementary manipulation yields the displayed proportionality (Bergstra, Bardenet, Bengio, & Kégl, 2011). $\square$

3.9.4 Comparison with alternative search strategies

Four search paradigms could in principle be used for (3.27). The following comparison explains why TPE-SMBO was chosen.

Grid search. Under the conditional structure of Λ specified in Section 3.9.5, the total number of distinct configurations is

$$|\Lambda_{\mathrm{grid}}| = n_w \cdot n_m \cdot n_\tau \cdot n_r \left(1 + 2\, n_d\, n_\varepsilon\right) = 9 \cdot 4 \cdot 5 \cdot 3 \cdot (1 + 2 \cdot 4 \cdot 4) = 17{,}820, \quad (3.30)$$

where the factor 1 accounts for the Alpha branch (metric and filtration-scale parameters inactive) and $2 \cdot 16 = 32$ for the Rips and SparseRips branches. Combined with the classifier grids, $|\Lambda_{\mathrm{grid}}| \times (|\Theta_{\mathrm{lr}}| + |\Theta_{\mathrm{svm}}|) = 17{,}820 \times 18 = 320{,}760$ inner-CV evaluations are needed *per outer LOSO fold*. For SisFall with $|\mathcal{S}_K| = 24$ outer folds, a conservative estimate of 5 s per inner-CV call gives approximately

$$320{,}760 \times 24 \times 5 \text{ s} \;\approx\; 43 \text{ days},$$

making full grid search computationally infeasible on the available hardware.

Random search. Random search (Bergstra & Bengio, 2012) draws configurations independently and uniformly from $\Lambda \times \Theta$ without reference to past evaluations. It is statistically consistent under infinite budget and, under the low effective-dimensionality hypothesis, outperforms grid search at equal sample count by avoiding wasted evaluations on uninformative axes. For the present budget $T = 200$, however, random search explores only $200/17{,}820 \approx 1.1\%$ of the grid and provides no mechanism for directing subsequent trials toward promising regions identified by earlier ones.

Gaussian-process Bayesian optimisation. The canonical SMBO surrogate is a Gaussian process (GP), which models the objective as a GP-distributed random function and updates its posterior via Bayes' rule after each evaluation. GP-SMBO is appropriate when the search space is a compact subset of $\mathbb{R}^d$ with a natural metric. It is unsuitable here for two structural reasons. First, no natural stationary kernel is defined on the conditional, mixed-type space $\Lambda \times \Theta$ without auxiliary embeddings whose inductive bias is difficult to justify. Second, fitting and inverting the GP covariance matrix costs $O(t^3)$ in the evaluation count t; for $T = 200$ iterations this overhead is non-negligible and grows quadratically with the complexity of the correlation model.

TPE (adopted strategy). TPE (Bergstra, Bardenet, Bengio, & Kégl, 2011) resolves both difficulties. The conditional structure of Λ is encoded natively in the tree; each leaf node carries an independent univariate kernel density estimator, so per-iteration update cost is $O(t)$. Proposition 3.4 provides the theoretical guarantee that maximising ℓ/g is equivalent to maximising EI under the TPE generative model. Empirically, TPE has outperformed GP-EI on machine-learning hyperparameter benchmarks at comparable budgets (Bergstra, Bardenet, Bengio, & Kégl, 2011; Bergstra & Bengio, 2012), and its implementation in Optuna (Akiba, Sano, Yanase, Ohta, & Koyama, 2019) handles mixed, conditional, and discrete spaces transparently.

3.9.5 Search space

The search space Λ is the Cartesian product

$$\begin{aligned}
w_{\text{sec}} &\in \{1.0,\ 1.5,\ 2.0,\ 2.5,\ 3.0,\ 3.5,\ 4.0,\ 4.5,\ 5.0\}\ \text{(s)},\\
m &\in \{2,3,4,5\}, \quad \tau \in \{1,2,3,4,5\}, \quad r \in \{1,2,4\},\\
c &\in \{\text{Alpha, Rips, SparseRips}\},\\
d_{\text{met}} &\in \{\text{Euclidean, Manhattan, Chebyshev, Cosine}\},\\
q &\in \{0.20,\ 0.40,\ 0.60,\ 0.80\}, \quad \varepsilon \in \Lambda_\varepsilon,
\end{aligned}$$

where Λ_ε denotes the auxiliary filtration parameter associated with c. Several conditional constraints are imposed by the geometry: the Alpha complex requires the Euclidean metric; metric and q are inactive for the Alpha complex. The classifier search spaces are

$$\Theta_{\text{lr}} = \big\{(C, \Omega) : C \in \{10^{-2}, 10^{-1}, 1, 10, 10^2\},\ \Omega \in \{\ell_1, \ell_2\}\big\},$$

$$\Theta_{\text{svm}} = \big\{(C, \kappa) : C \in \{10^{-1}, 1, 10, 10^2\},\ \kappa \in \{\text{linear}, \text{rbf}\}\big\}.$$

For the personal protocol of Section 3.10.3 the LR grid is reduced to $C \in \{10^{-1}, 1, 10\}$ to mitigate overfitting in smaller per-subject training sets.

3.9.6 Outer LOSO and inner cross-validation

The selection procedure is the nested protocol

(1) **Outer.** For each subject $s \in \mathcal{S}_K$, define $\mathcal{D}_K^{\text{train}}(s)$ as the union of all windows of subjects $\mathcal{S}_K \setminus \{s\}$.

(2) **Inner.** Run TPE for $T = 200$ iterations on $\mathcal{D}_K^{\text{train}}(s)$ using stratified k-fold CV; record $\lambda_K^*(s), \theta_K^*(s)$.

(3) **Retrain and test.** Fit a final model on $\mathcal{D}_K^{\text{train}}(s)$ with $(\lambda_K^*(s), \theta_K^*(s))$ and evaluate on the held-out subject s.

The outer LOSO ensures that no information from the test subject is used to choose any model component; the inner CV ensures that the surrogate objective is an unbiased estimator of expected performance on unseen subjects from the training pool.

Remark 3.2 (Computational budget)**.** *The budget $T = 200$ was chosen empirically as the smallest value at which the running best objective stabilises across independent runs of the search on the three datasets. Larger budgets do not produce reliably different optima, given the noise level of the inner CV estimator.*

Remark 3.3 (Implementation decomposition)**.** *Equation* (3.27) *states a joint optimisation over $\Lambda \times \Theta$. In practice this joint search is decomposed into two* *sequential stages to reduce computational cost.* Stage 1 (TDA search, Optuna): *A single TPE study is run per dataset for $T = 200$ iterations. The objective is the average F_2 of a logistic regression proxy and a linear SVM proxy, both with fixed hyperparameters, evaluated by three-fold CV on a balanced training subset. Using lightweight proxies keeps each function evaluation fast, so the budget is spent exploring Λ rather than the joint space; the averaged objective ensures the recovered λ^* is not overfitted to a single classifier's inductive bias.* Stage 2 (classifier search, `GridSearchCV`): *Given λ^*, classifier hyperparameters θ^* are selected by exhaustive grid search with three-fold CV on the features extracted under λ^*. This decomposition approximates the joint optimum while keeping computation tractable.*

Remark 3.4 (Classifier-agnostic TDA search). *Rather than running a separate Optuna study for each classifier family, a single study is run with a classifier-agnostic objective. Define*

$$\lambda^* \in \arg\max_{\lambda \in \Lambda} \frac{1}{2}\Big(\mathrm{CV}_K^{\mathrm{LR}}(\lambda) + \mathrm{CV}_K^{\mathrm{SVM}}(\lambda)\Big),$$

where $\mathrm{CV}_K^f(\lambda)$ denotes the K-fold cross-validated F_2 obtained by classifier f on features extracted under λ, with f's hyperparameters fixed at proxy values. The averaged objective recovers a configuration λ^* whose features support good performance across classifier families simultaneously. The map $\phi_{\lambda^*} \colon \mathbb{R}^{L\times 2} \to \mathbb{R}^{24}$ must therefore produce an image that is both approximately linearly separable (beneficial for LR) and geometrically well-structured in pairwise distance (beneficial for SVM with a nonlinear kernel). A representation satisfying both conditions is a stronger finding than one tuned to a single classifier's inductive bias.

3.10 Evaluation Framework

3.10.1 Performance metrics

Given binary predictions $\widehat{y}_i$ and labels y_i on a test sample $\{(\widehat{y}_i, y_i)\}_{i=1}^N$, define the four confusion counts TP, TN, FP, FN and the four derived rates

$$\mathrm{Acc} = \frac{\mathrm{TP}+\mathrm{TN}}{N}, \quad \mathrm{Rec} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \quad \mathrm{Prec} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \quad F_1 = \frac{2\,\mathrm{Prec}\,\mathrm{Rec}}{\mathrm{Prec}+\mathrm{Rec}}.$$

The principal metric is the F_β-score at $\beta = 2$,

$$F_2 = (1+\beta^2)\frac{\mathrm{Prec}\cdot\mathrm{Rec}}{\beta^2\,\mathrm{Prec}+\mathrm{Rec}}\bigg|_{\beta=2} = \frac{5\,\mathrm{Prec}\,\mathrm{Rec}}{4\,\mathrm{Prec}+\mathrm{Rec}}. \tag{3.31}$$

Proposition 3.5 (Recall weighting under F_β)**.** *For fixed precision* $\mathrm{Prec} = p$ *and recall* $\mathrm{Rec} = r$*,*

$$\frac{\partial F_\beta}{\partial r}\bigg|_{\beta=2} = \frac{20\,p^2}{(4p+r)^2}, \qquad \frac{\partial F_\beta}{\partial p}\bigg|_{\beta=2} = \frac{5\,r^2}{(4p+r)^2}.$$

Consequently, near $p = r$ the marginal F_2-gain per unit of recall is four times the marginal gain per unit of precision.

Proof. Direct differentiation of (3.31) and the substitution $p = r$ give the ratio 4 between the two partial derivatives. □

The weighting ratio 4:1 is a deliberate methodological choice: a missed fall leaves a vulnerable person unattended after a potentially injurious event, whereas a false alarm requires at most an unnecessary check. The F_2 score is used both as the Bayesian objective and as the headline metric in all result tables.

Remark 3.5 (Non-triviality of the baseline)**.** *A trivial all-positive classifier achieves* $\mathrm{Rec} = 1$ *and* $\mathrm{Prec} = \pi$*, hence $F_2 = 5\pi/(4\pi+1)$. For $\pi < 0.30$ in all three datasets this gives $F_2 < 0.56$, well below the scores reported in the Experiments section. The recall-oriented objective does not degenerate to a trivial solution.*

3.10.2 Decision threshold

A probabilistic classifier $f_\theta\colon \mathbb{R}^{24} \to [0,1]$ assigns each window a posterior fall probability. A binary label is produced by thresholding at the trained threshold $\hat{\tau}_0$,

$$\widehat{y}(w) = \mathbf{1}\big\{f_\theta(\phi_\lambda(w)) \geq \hat{\tau}_0\big\}. \tag{3.32}$$

The trained threshold $\hat{\tau}_0$ is the PR-curve optimal value identified on the training fold: $\hat{\tau}_0 = \arg\max_\tau F_2^{\text{train}}(\tau)$. This PR-curve selection is performed once per LOSO fold using the training-fold probability scores and labels. The value $\hat{\tau}_0$ varies across subjects and datasets (typically in the range $[0.55, 0.85]$) and is more conservative than the nominal $\tau_0 = 0.50$ because the class-imbalanced training distribution places the optimal precision–recall trade-off above the 50 % probability level.

3.10.3 Evaluation protocols

Three evaluation scenarios are reported, each measuring a different level of generalisation.

Naive protocol. The feature matrix is partitioned uniformly at random into training (70%) and test (30%) sets with no constraint on subject membership. The same subject can contribute windows to both partitions. This protocol systematically overestimates performance on new individuals and is reported only as an upper-bound reference. Ten random partitions are used; results are reported as means.

General protocol (leave-one-subject-out). For each $s \in \mathcal{S}_K$ the test set is $\{w : \text{subject}(w) = s\}$ and the training set is the complement. This LOSO scheme directly measures generalisation to a previously unseen individual. Because the class-balancing procedure is stochastic, each fold is repeated $R = 15$ times with independent random seeds; reported quantities are means or appropriate aggregates over the resulting $|\mathcal{S}_K| \times R$ collection of values. The General protocol is the principal protocol of the thesis.

Personal protocol. For each $s \in \mathcal{S}_K$ a model is trained and tested exclusively on that subject's own windows via a 70/30 random split, repeated $R = 15$ times. The personal protocol represents the idealised setting in which a per-user calibration period is available before deployment.

Decision-threshold sweep. Using the General-protocol classifier, the threshold τ is swept over the grid $\mathcal{T} = \{0.05, 0.10, \dots, 0.95\} \subset (0,1)$ for each held-out subject. The subject-level retrospective gain is

$$\Delta F_2(s) = \max_{\tau \in \mathcal{T}} F_2(\tau, s) - F_2(\hat{\tau}_0(s), s), \tag{3.33}$$

where $\hat{\tau}_0(s)$ is the PR-curve trained threshold for subject s (Section 3.10.2). The average $\overline{\Delta F_2} = |\mathcal{S}_K|^{-1} \sum_s \Delta F_2(s)$ quantifies how much the PR-curve-calibrated trained threshold underperforms a subject-specific oracle. Because $\tau^*(s) = \arg\max_\tau F_2(\tau, s)$ is identified on the same test windows on which it is evaluated, the sweep is a retrospective diagnostic of heterogeneity, not a valid prospective personalisation scheme.

3.11 Statistical Inference

Two sources of variability must be kept distinct in reporting: within-subject variability V_{rep} across the $R = 15$ class-balancing repetitions, and across-subject variability V_{subj} in mean performance across the subject pool. Both are reported explicitly; subject-level averages, dispersion, and uncertainty intervals take precedence over headline maxima throughout.

3.11.1 Repetition-level confidence intervals

For each (subject s, repetition j) pair let $x_{s,j} \in [0,1]$ denote the metric of interest (typically F_2). Define the per-repetition study-level mean

$$\bar{X}_j = |\mathcal{S}_K|^{-1} \sum_{s \in \mathcal{S}_K} x_{s,j}, \qquad j = 1, \dots, R.$$

The overall study mean is $\overline{X}_K = R^{-1} \sum_j \bar{X}_j$. The sample standard deviation of $\{\bar{X}_j\}$ is

$$S_R = \Big(\frac{1}{R-1} \sum_{j=1}^{R} (\bar{X}_j - \overline{X}_K)^2 \Big)^{1/2}.$$

Theorem 3.9 (Repetition-level t confidence interval)**.** *If the per-repetition study means $\bar{X}_j$ are exchangeable and approximately Gaussian, then a $1-\alpha$ confidence interval for the limiting expectation $\mathbb{E}[\bar{X}]$ as $R \to \infty$ is*

$$\overline{X}_K \ \pm \ t_{\alpha/2,\, R-1} \frac{S_R}{\sqrt{R}}, \tag{3.34}$$

where $t_{\alpha/2,R-1}$ is the upper $\alpha/2$ quantile of the t-distribution with $R-1$ degrees of freedom.

For $R = 15$ and $\alpha = 0.05$, $t_{0.025,14} \approx 2.145$. The interval (3.34) captures the repetition-to-repetition stochastic variability of the class-balancing procedure; it does not capture subject-level heterogeneity.

3.11.2 Subject-level paired tests

Many of the central comparisons in the thesis are paired at the subject level: $F_2^{\text{lr}}(s)$ vs $F_2^{\text{svm}}(s)$, $F_2^{\text{pers}}(s)$ vs $F_2^{\text{gen}}(s)$, or F_2 for two distinct λ values. Let

$$\delta_s = A_s - B_s \in \mathbb{R}, \qquad s \in \mathcal{S}_K,$$

denote the per-subject paired difference, with the sign convention that positive δ_s favours A.

Theorem 3.10 (Paired t-test)**.** *Under the null $H_0\colon \mathbb{E}[\delta_s] = 0$ and Gaussianity of $\{\delta_s\}$, the statistic*

$$T_K = \frac{\bar{\delta}\sqrt{|\mathcal{S}_K|}}{S_\delta}, \qquad \bar{\delta} = |\mathcal{S}_K|^{-1}\sum_s \delta_s, \quad S_\delta^2 = \frac{1}{|\mathcal{S}_K| - 1}\sum_s (\delta_s - \bar{\delta})^2,$$

is distributed as $t_{|\mathcal{S}_K|-1}$.

Remark 3.6 (Robustness via the sign test)**.** *When $|\mathcal{S}_K|$ is small (notably $|\mathcal{S}_K| = 9$ for MobiFall) the Gaussianity assumption of Theorem 3.10 is hard to validate. As a robustness check, the thesis also reports the exact sign test on the binarised signs $\operatorname{sgn}(\delta_s)$, which under H_0 are exchangeable Bernoulli$(1/2)$ variates. The two-sided p-value is $2\mathbb{P}(\operatorname{Bin}(|\mathcal{S}_K|, 1/2) \geq k)$ where $k = |\{s : \delta_s > 0\}|$ (with the obvious correction for $\delta_s = 0$).*

3.11.3 Bootstrap confidence intervals

For statistics whose finite-sample distribution is difficult to derive, the thesis uses the nonparametric paired bootstrap. Let $\theta(\{\delta_s\})$ denote a real-valued statistic of the paired differences (e.g., the subject mean, median, or interquartile range). Generate B bootstrap resamples $\{\delta_s^{*,b}\}_{s\in\mathcal{S}_K}$ by sampling $|\mathcal{S}_K|$ subjects with replacement from $\mathcal{S}_K$ and copying their δ_s values, and compute

$$\theta_b^* = \theta(\{\delta_s^{*,b}\}), \quad b = 1, \dots, B.$$

The $1-\alpha$ percentile confidence interval is

$$\left[\theta^*_{(\alpha/2)}, \theta^*_{(1-\alpha/2)}\right], \tag{3.35}$$

where $\theta^*_{(p)}$ is the p-quantile of $\{\theta_b^*\}$. The thesis uses $B = 10{,}000$ and $\alpha = 0.05$. The bootstrap is the principal device for reporting uncertainty on subject-level summaries such as $\overline{\Delta_K}$ and $\mathrm{Var}(\Delta_{K,s})$ in the personalisation analysis.

3.11.4 Permutation tests

For exact small-sample inference on the sign pattern of paired differences, the thesis uses the paired sign-flipping permutation test. Under the exchangeability null

$$H_0\colon\ \delta_s \overset{d}{=} -\delta_s \qquad \text{for all } s,$$

the $2^{|\mathcal{S}_K|}$ sign-flipped vectors $\{\epsilon_s\delta_s\}$ with $\epsilon_s \in \{-1,+1\}$ are exchangeable in distribution. The exact two-sided p-value of an observed test statistic $T_{\text{obs}} = T(\delta)$ is

$$p = \frac{1}{2^{|\mathcal{S}_K|}} \sum_{\epsilon\in\{-1,+1\}^{|\mathcal{S}_K|}} \mathbf{1}\left\{|T(\epsilon\odot\delta)| \geq |T_{\text{obs}}|\right\}. \tag{3.36}$$

For $|\mathcal{S}_K| \leq 24$ the sum is computed exactly; for larger $|\mathcal{S}_K|$ it is estimated by sampling 10,000 random sign vectors.

3.11.5 Distributional analysis of personalisation

The personalisation question requires a distributional rather than a single-number answer. The proposal formulates the per-subject gain

$$\Delta_{K,s} = F_{2,K,s}^{\text{pers}} - F_{2,K,s}^{\text{gen}}.$$

The thesis reports the four summaries

$$\overline{\Delta}_K, \qquad \mathrm{med}(\Delta_{K,s}), \qquad \mathrm{Var}(\Delta_{K,s}), \qquad \widehat{\mathbb{P}}(\Delta_{K,s} > 0)$$

together with bootstrap percentile confidence intervals of Section 3.11.3 for each. The thesis privileges the joint pattern of these four quantities over any single one, because personalisation is interpreted as a heterogeneous subject-level intervention rather than a constant treatment effect.

3.11.6 Cross-applied hyperparameters

To evaluate dataset-specific topological geometry, optimal tuples are cross-applied across datasets. Define the cross-applied estimator

$$\widehat{F}_2(K \mid \lambda^*_{K'}) = \mathbb{E}\big[F_2 \mid K, \lambda^*_{K'}\big],$$

where the expectation is taken over the LOSO folds of dataset K with the hyperparameter tuple optimised on a different dataset $K' \neq K$. The matrix

$$M_{K,K'} = \widehat{F}_2(K \mid \lambda^*_{K'})$$

has diagonal entries equal to the General-protocol F_2 of each dataset under its own optimiser. Off-diagonal degradations

$$\mathrm{Drop}_{K \to K'} = M_{K,K} - M_{K,K'}$$

quantify the cost of transplanting a hyperparameter tuple from K to K'. The drops are reported per off-diagonal cell with bootstrap percentile intervals.

3.11.7 Methodological correction over headline maxima

Headline maxima such as the best single-subject score or the best Optuna objective are not used as the primary evidence in this thesis. The reasons are statistical. The maximum of a finite collection of noisy estimators is biased upward by the selection event; the bias scales with the number of competing candidates and the variance of each. Reporting subject means, subject-level confidence intervals, and the joint distribution of subject-level differences avoids this bias and provides estimands that are directly interpretable as expectations over the population from which the subjects were drawn.

3.12 Computational Infrastructure

All experiments ran on the National HPC Centre (UHeM, Istanbul), allocation 4025462026, managed by SLURM. Each compute node carried dual Intel Xeon Gold 6148 processors (2.40 GHz, 20 physical cores per socket, 40 cores per node) and 190 GB of RAM. No GPU was used: boundary matrix reduction (Algorithm 1) is intrinsically sequential at the point cloud sizes considered here, and the marginal speedup from GPU porting is outweighed by the overhead of the optimised CPU implementation.

The Bayesian search ran sequentially through $T = 200$ TPE trials per dataset, with 15 parallel inner-CV workers active per trial. Across all three datasets the total wall-clock time came to approximately 3 h 45 min, with peak memory close to 10 GB per node. Persistent homology was computed with GUDHI (GUDHI Editorial Board, 2015); classifiers were implemented in scikit-learn (Pedregosa, Varoquaux, Gramfort, Michel, Thirion, Grisel, Blondel, Prettenhofer, Weiss, Dubourg, Vanderplas, Passos, Cournapeau, Brucher, Perrot, & Duchesnay, 2011). All random seeds are fixed and archived alongside the thesis source, together with the full search history and the per-subject score tables.

4. Experiments

This chapter covers the experimental setup, the three benchmark datasets, preprocessing, feature configuration, and numerical results. Theoretical foundations—persistent homology, phase-space embedding, filtration construction—and the formal definitions of the evaluation protocols and metrics are developed in Chapter 3; the presentation here focuses on the empirical choices and what they produced. Figure 4.1 gives an overview of the full pipeline from raw sensor measurement to binary decision.

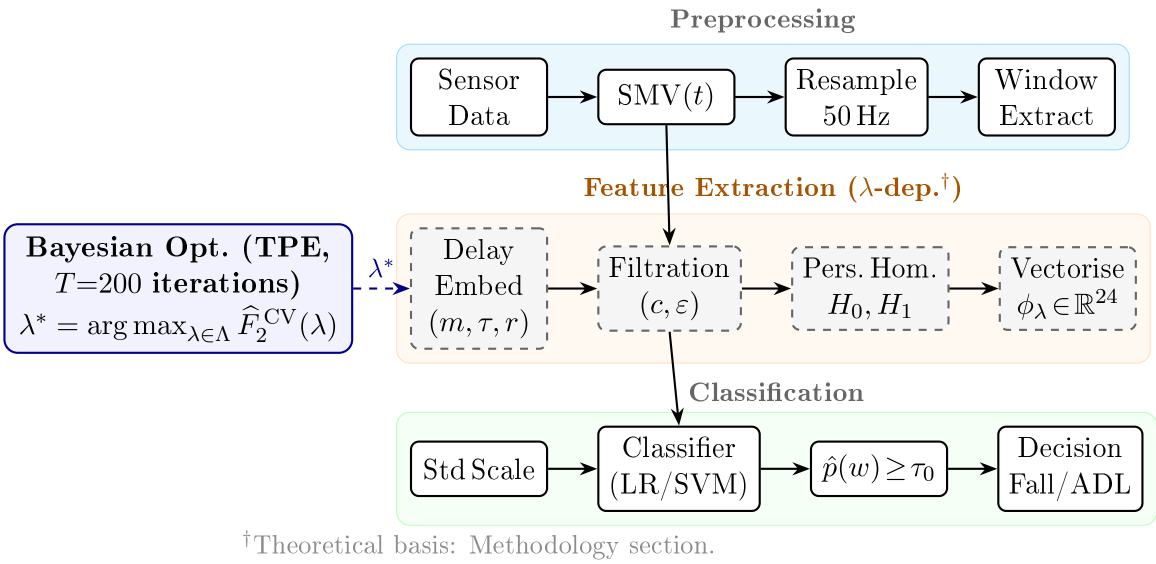

Figure 4.1: Complete fall detection pipeline from raw inertial data to binary decision. *Preprocessing*: SMV computation, resampling to 50 Hz, window extraction. *Feature extraction* (dashed, λ-dependent): delay embedding, filtration construction, persistent homology (H_0, H_1), and vectorisation to $\phi_\lambda \in \mathbb{R}^{24}$. *Classification*: standardisation, probabilistic classifier, and threshold comparison $\hat{p}(w) \geq \tau_0$ for binary decision. The *Bayesian optimisation* box (left, blue) selects λ^* by maximising cross-validated F_2 over training subjects via 200 TPE iterations; row-to-row connectors run through column 2 (SMV $\rightarrow$ Filtration $\rightarrow$ Classifier), preserving the 4×3 grid symmetry.

4.1 Datasets

Experiments were conducted on three publicly available benchmark datasets: MobiFall (Vavoulas, Pediaditis, Chatzaki, Spanakis, & Tsiknakis, 2014), SisFall (Sucerquia, López, & Vargas-Bonilla, 2017), and the 40 Hz variant of FallAllD (hereafter FAD_40Hz) (Saleh, Abbas, & Le Bouquin Jeannès, 2021). All three record inertial sensor signals from subjects performing scripted fall events and daily activities, but they differ in sensor placement, recording rate, subject age, and activity protocol—differences that are important for assessing generalisation across acquisition conditions. Table 4.1 summarises the key characteristics of each dataset as used here.

Table 4.1: Summary of the three datasets as used in the experiments.

Dataset	**Subjects**	**Feature matrix**	**Resampling (Hz)**	**Placement**
MobiFall	9	1808×24	$87.5 \rightarrow 50$	Trouser pocket
SisFall	24	7738×24	$200 \rightarrow 50$	Waist
FAD_40Hz	13	9481×24	$40 \rightarrow 50$	Waist
Total	**46**	$\mathbf{19{,}027 \times 24}$	—	—

4.1.1 MobiFall

MobiFall v2.0 records volunteers who carried a Samsung Galaxy S3 smartphone in their trouser pocket while performing four scripted fall types (forward fall hands-first, forward fall onto knees, backward fall from a chair, lateral fall from standing) and nine daily activities including walking, jogging, stair use, and car entry (Vavoulas, Pediaditis, Chatzaki, Spanakis, & Tsiknakis, 2014). The nominal rate is 87.5 Hz, and the device captures seven channels: three-axis linear acceleration, three-axis angular velocity, magnetic field, barometric pressure, and orientation (azimuth, pitch, roll).

Only the accelerometer and gyroscope were retained. The magnetometer, barometer, and orientation estimates were dropped for two reasons: these channels are absent or recorded at incompatible rates in the other two datasets, which would break a common preprocessing pipeline; and they do not directly encode the inertial dynamics of a fall, so retaining them would inflate feature dimensionality without a principled benefit to the classification objective.

Not all subjects in the v2.0 cohort completed both fall and daily-activity sessions. Nine subjects (identifiers 2, 3, 4, 5, 7, 8, 9, 10, and 11) were retained; the remainder performed fall trials only and were excluded because a held-out subject lacking daily-activity windows cannot contribute a meaningful test set to the General protocol. After windowing, the retained cohort yields a feature matrix of 1808×24.

4.1.2 SisFall

SisFall is a waist-worn dataset covering 15 fall types and 19 daily-activity types in a mixed group of young and older adults (Sucerquia, López, & Vargas-Bonilla, 2017). The sensor unit houses two accelerometers and one gyroscope: the ADXL345 ($\pm 16\,g$, 13-bit), the MMA8451Q ($\pm 8\,g$, 14-bit), and the ITG3200 gyroscope ($\pm 2000\,°/\mathrm{s}$, 16-bit), all sampled at 200 Hz. The dataset spans 23 young adults and 15 older adults.

Only the ADXL345 was used alongside the gyroscope. The higher dynamic range of the ADXL345 ($\pm 16\,g$) makes it better suited to impulsive fall-impact peaks; the MMA8451Q, at half the range, risks saturation during vigorous motion. Retaining one accelerometer also preserves channel consistency with MobiFall and FAD_40Hz, which each provide a single accelerometer, so no cross-dataset asymmetry is introduced.

Twenty-four subjects were retained: all 23 young adults plus one older adult (subject 29) with sufficient trial counts. The remaining 14 older-adult participants were excluded for insufficient trial counts. The 200 Hz signals were downsampled to 50 Hz, yielding a feature matrix of 7738×24—the largest subject pool of the three datasets.

4.1.3 FallAllD (FAD_40Hz)

Original FallAllD dataset. FallAllD is a large-scale open dataset of human falls and activities of daily living collected from 15 subjects (Saleh, Abbas, & Le Bouquin Jeannès, 2021). In its original release, recordings are obtained simultaneously from three sensor devices worn at the waist, wrist, and neck, each capturing four modalities: a three-axis accelerometer, a three-axis gyroscope, a three-axis magnetometer, and a barometric pressure sensor. The inertial sensors (accelerometer and gyroscope) are sampled at 238 Hz, the magnetometer at 80 Hz, and the barometer at 10 Hz. The complete activity label set comprises 135 categories, of which 44 are activities of daily living (labelled with identifiers 1–44) and 91 are distinct fall scenarios (identifiers 101–135) differing in direction, mechanism, and recovery condition. Across all subjects, sensor positions, and activities, the original release contains 26,420 individual recording files.

Selection of the 40 Hz derived variant. Using the original 238 Hz release directly in the present study would have imposed prohibitive computational costs. The Bayesian hyperparameter search runs 200 iterations per dataset, and each iteration requires extracting topological features — which involve constructing and computing persistent homology of a point cloud — from every window in the dataset under the current configuration. At 238 Hz, the number of samples per window is approximately 4.8 times larger than at 50 Hz, substantially increasing both the size of the delay-embedded point cloud and the cost of the homology computation. Furthermore, the four-modality, three-position structure of the original release creates a sensor fusion problem that falls outside the scope of the present methodology, which targets single-position inertial data.

For these reasons, the publicly available 40 Hz derived variant (FAD_40Hz) was used instead (Saleh, Abbas, & Le Bouquin Jeannès, 2021). This derived dataset has been pre-processed as follows relative to the original release: (i) the inertial signals are downsampled from 238 Hz to 40 Hz; (ii) only the accelerometer and gyroscope channels are retained, with the magnetometer and barometric sensors removed; (iii) the neck-worn device data is excluded, leaving the waist and wrist positions; (iv) the activity label set is filtered to a curated subset of fall and ADL types relevant to binary detection tasks. These modifications reduce the data volume by an order of magnitude and eliminate the multi-rate sensor-fusion problem imposed by the original release, while preserving the inertial channels and the subject-level label structure required by the present pipeline.

The 40 Hz signals were subsequently resampled to the common target rate of 50 Hz as described in Section 4.2. After activity filtering and subject selection, 13 of the original 15 subjects contributed data; the two excluded subjects had an insufficient number of trials for stable leave-one-subject-out evaluation. The resulting feature matrix is 9,481 $\times$ 24, the largest in the study by sample count.

4.2 Data Processing and Preprocessing

All sensor recordings pass through a common four-step preprocessing pipeline before any feature computation. Raw three-axis measurements are reduced to an orientation-independent scalar signal, resampled to a common frequency, and partitioned into fixed-length windows. Figure 4.1 shows where this block sits in the overall data path.

Step 1 — Signal Magnitude Vector. For each sensor (accelerometer and gyroscope separately), the three-axis measurement vector $\mathbf{x}(t) \in \mathbb{R}^3$ at time t is reduced to its Euclidean norm:

$$\mathrm{SMV}(t) = \left\|\mathbf{x}(t)\right\|_2 = \sqrt{x_1(t)^2 + x_2(t)^2 + x_3(t)^2}.$$

This produces one scalar time series per sensor that is invariant to device orientation on the body, eliminating the need for explicit attitude estimation.

Step 2 — Resampling to 50 Hz. The three datasets were originally recorded at 87.5 Hz (MobiFall), 200 Hz (SisFall), and 40 Hz (FAD_40Hz). All signals were resampled to a common rate of 50 Hz, but the strategies differ by dataset because the resampling direction and the ratio of original to target rate are not uniform.

For SisFall (200 Hz $\rightarrow$ 50 Hz), the ratio is the integer 4:1, so integer-ratio decimation was applied: the SMV signal was subsampled by retaining every fourth sample. This produces a uniformly-spaced 50 Hz output without any interpolation step and without spectral distortion within the passband of interest, since the Nyquist frequency of the target rate (25 Hz) is well below the dominant frequency of fall events.

For FAD_40Hz (40 Hz → 50 Hz), upsampling by a factor of 5:4 was required. The resampled signal was obtained by constructing a new time grid at 50 Hz and computing signal values at each new grid point by interpolating over the flanking 40 Hz samples. Because the ratio 5:4 is rational and the upsampling factor is moderate, this approach introduces no perceptible temporal smearing in the SMV magnitude signal.

For MobiFall (87.5 Hz → 50 Hz), the ratio 87.5:50 = 7:4 is rational but non-integer in both directions, so neither pure decimation nor pure upsampling is sufficient. A rational sample-rate conversion was applied, conceptually equivalent to first upsampling by a factor of 4 and then downsampling by a factor of 7. The resampling was performed on the scalar SMV signal rather than on the raw three-axis components, so the non-linear magnitude operation is not affected by the interpolation kernel.

In all three cases the resampling target of 50 Hz was selected on methodological grounds that are discussed in the Methodology section.

Step 3 — Window extraction. For each fall trial, a single window centred on the highest-magnitude point of the SMV signal was extracted. For daily-activity trials, consecutive non-overlapping windows of the same length were taken. The window length is a hyperparameter optimised during the experiment and is described in Section 4.4.

Step 4 — Standard scaling. Each of the 24 features in the final feature vector was scaled to zero mean and unit variance. The scaling parameters were estimated from the training data only and applied to the test data, ensuring that no information from test subjects influences the normalisation.

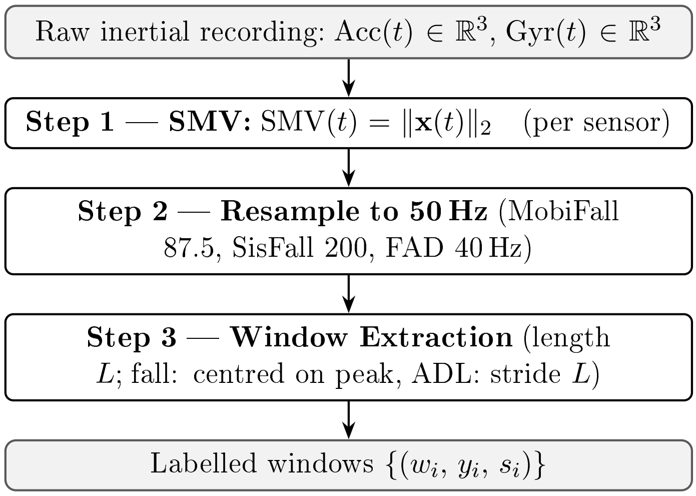

Figure 4.2: Preprocessing pipeline (Steps 1–3). Detailed description of each step is given in Section 4.2.

4.3 Feature Description

Each window is mapped to a 24-dimensional feature vector split into two groups: 20 persistence-based features encoding the topological shape of the motion trajectory, and 4 signal-level features capturing the gross magnitude behaviour of the accelerometer signal. The theoretical derivation of the persistence-based features appears in Chapter 3; the following subsections list the features and their definitions.

4.3.1 Persistence-Based Features

The topological representation of each window yields two sets of birth–death pairs: one for connected clusters in the trajectory (H_0) and one for closed loops (H_1). Ten scalar statistics are extracted from each set, giving 20 features in total. Table 4.2 lists these statistics and their definitions.

4.3.2 Signal-Level Features

Four time-domain statistics are computed directly from the accelerometer $\mathrm{SMV}(t)$ values within the window and appended to the feature vector, giving features 21–24. These capture the raw intensity profile of the motion event without any topological processing.

Table 4.2: Ten persistence-based statistics computed per homological dimension. Each row is computed twice: once for H_0 (features 1–10) and once for H_1 (features 11–20). Persistence of a topological feature is defined as its death scale minus its birth scale.

Index	**Feature**	**Definition**
1 / 11	Bar count	Number of features born and died, normalised by window length
2 / 12	Entropy	Shannon entropy of the normalised persistence values
3 / 13	Maximum	Persistence of the longest-lived feature
4 / 14	Energy	Sum of squared persistence values
5 / 15	Mean	Arithmetic mean of all persistence values
6 / 16	Std deviation	Standard deviation of the persistence values
7 / 17	Median	Median persistence
8 / 18	First quartile	25th percentile of the persistence distribution
9 / 19	Third quartile	75th percentile of the persistence distribution
10 / 20	Third moment	Third central moment (related to skewness) of the persistence values

Table 4.3: Four signal-level features derived from the accelerometer magnitude $\mathrm{SMV}(t)$ within each window.

Index	**Feature**	**Definition**
21	Maximum	Highest value of $\mathrm{SMV}(t)$ in the window
22	Std deviation	Standard deviation of $\mathrm{SMV}(t)$
23	Mean	Arithmetic mean of $\mathrm{SMV}(t)$
24	Range	Maximum minus minimum of $\mathrm{SMV}(t)$

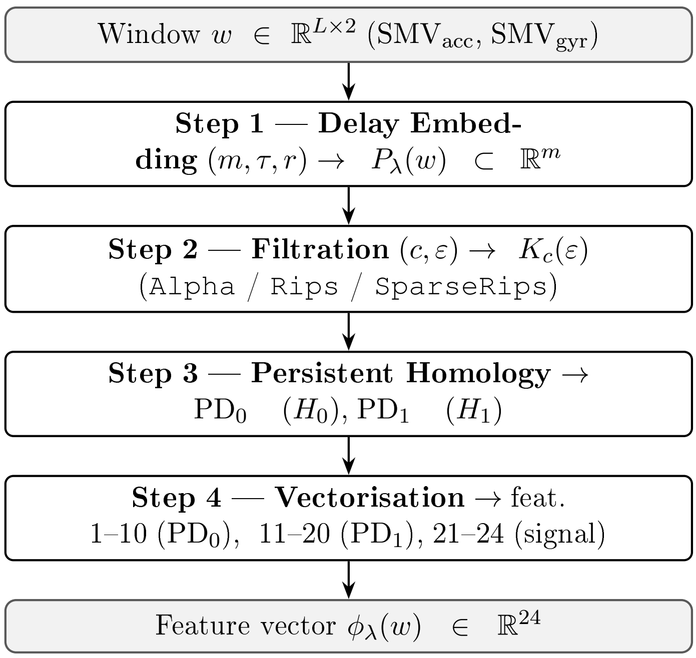

Figure 4.3: Feature extraction pipeline (Steps 1–4). Detailed description of each step is given in Section 4.3.

4.4 Bayesian Hyperparameter Optimisation

Topological feature computation depends on several configuration choices: window length, phase-space embedding parameters, and the type of topological complex. Rather than fixing these by hand or by exhaustive grid search, we searched for the combination that maximises fall detection performance on each dataset using Bayesian optimisation.

One Optuna study was run per dataset, for 200 TPE iterations (600 in total). Each trial proposes a TDA configuration λ_t; features are extracted from a balanced training subset (all fall windows plus three times as many ADL windows drawn at random) and evaluated with a classifier-agnostic objective: both a logistic regression proxy and a linear SVM proxy are scored by three-fold stratified cross-validation and their F_2 scores are averaged. Classifier hyperparameters are held fixed at proxy values during this stage and optimised separately afterwards by grid search (Section 4.5).

The search is guided by a Tree-structured Parzen Estimator (TPE). After each evaluation the observation history $\mathcal{H}_t = \{(\lambda_i, \hat{y}_i)\}$ is split at a quantile into a "good" subset (high $\hat{y}$) and a "bad" subset, and separate kernel density estimates $\ell(\lambda)$ and $g(\lambda)$ are fitted to each. The next candidate is $\lambda_{t+1} = \arg\max_\lambda \ell(\lambda)/g(\lambda)$—the configuration most concentrated in the good region relative to the bad region. This ratio is proportional to expected improvement (see Chapter 3 for the derivation), and it progressively focuses the search on high-F_2 regions of Λ.

Averaging the objective over two classifier families ensures that λ^* supports good performance across both LR and SVM rather than being tuned to a single model's inductive bias: a representation that is simultaneously linearly separable (helpful for LR) and geometrically structured in pairwise distance (helpful for an RBF SVM) is a stronger result than one optimised for either alone.

Table 4.4 lists the parameters and the values considered during the search.

Table 4.4: Parameters and candidate values explored during Bayesian optimisation. A dash (—) indicates parameters not applicable for the Alpha complex type.

Parameter	**Candidate values**
Window length (`win_sec`, seconds)	1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0
Embedding dimension (`dim`)	2, 3, 4, 5
Embedding delay (`delay`)	1, 2, 3, 4, 5
Stride factor (`stride_factor`)	1, 2, 4
Complex type (`complex_type`)	Alpha, Rips, SparseRips
Distance measure (`metric`)	Euclidean, Cosine, Manhattan, Chebyshev
Filtration scale (`eps_percentile`)	20 %, 40 %, 60 %, 80 %

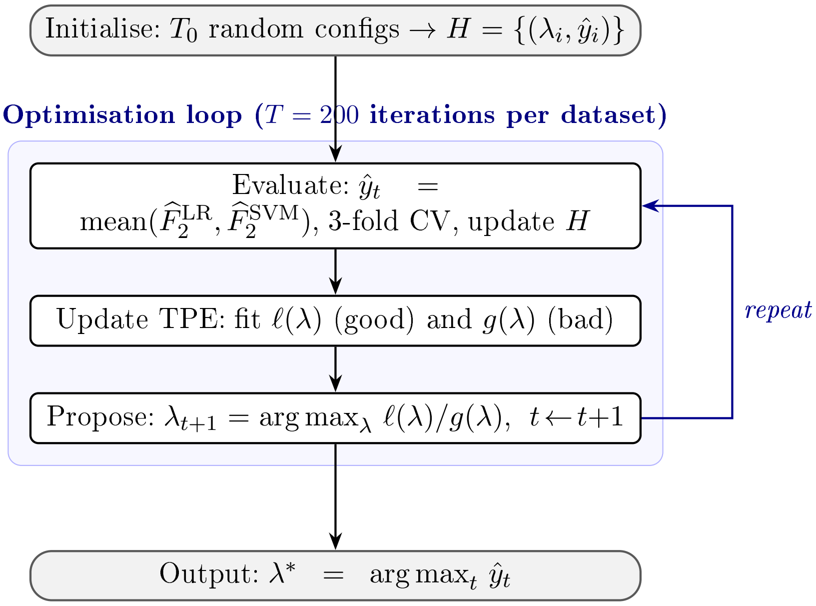

Figure 4.4: Bayesian hyperparameter optimisation loop (TPE, $T = 200$ iterations per dataset). Each trial evaluates both an LR proxy and a linear SVM proxy; the objective is the mean of their F_2 scores, yielding a classifier-agnostic λ^*. Detailed description in Section 4.4.

4.5 Classifiers and Training Setup

Two classifiers were compared: logistic regression (LR) and a support vector machine (SVM). Both used balanced class weights so that the minority fall class receives proportionally greater influence during training, compensating for the natural imbalance between fall and daily-activity windows.

Both classifiers operated on the same feature representation extracted under the single optimal configuration λ^*—a direct consequence of the classifier-agnostic optimisation: λ^* was selected because its features support good performance across both model families, not because it was tailored to either.

Classifier hyperparameters were selected by `GridSearchCV` with three-fold stratified cross-validation, scoring by F_2. For logistic regression, the grid covered regularisation strength $C \in \{0.01, 0.1, 1, 10, 100\}$ and penalty type $\{\ell_1, \ell_2\}$. For the SVM, the grid covered $C \in \{0.1, 1, 10, 100\}$ and kernel {linear, rbf}; probability outputs were obtained via Platt scaling. All feature vectors were standardised (zero mean, unit variance, parameters estimated from training data only) within each cross-validation fold before fitting.

After grid-search selects the best estimator, the decision threshold τ is set by inspecting the precision–recall curve on the training fold: the threshold $\tau^* = \arg\max_\tau F_2(\tau)$ that maximises F_2 along the curve replaces the default 0.5 cut-off. This calibration requires no additional model fitting and adapts the operating point to the recall-weighted F_2 objective.

4.6 Results

The three evaluation protocols and all performance metrics are defined in Chapter 3. In brief: the General (LOSO) protocol is the primary result; the Naive protocol (random 70/30 split) provides an inflated upper bound; and the Personal protocol trains and tests within each subject. General and Personal results are averaged over 15 independent repetitions and reported with 95 % confidence intervals. F_2 is the headline metric throughout.

4.6.1 Optimal Hyperparameter Configurations

Table 4.5 lists the best configuration found by the Bayesian search for each dataset. The three datasets converged on markedly different settings.

MobiFall required a long window (5.0 s) and a SparseRips complex, whereas both SisFall and FAD_40Hz were best described by a short 1.0 s window. SisFall uniquely preferred the Alpha complex; FAD_40Hz favoured the standard Rips construction. These differences suggest that the three datasets exhibit genuinely

Table 4.5: Best configuration selected by Bayesian optimisation for each dataset (200 iterations per dataset). A dash (—) indicates a parameter not applicable for the Alpha complex.

Parameter	**MobiFall**	**SisFall**	**FAD_40Hz**
Best F_2 (search objective)	0.9746	0.9733	0.9860
Window length (s)	5.0	1.0	1.0
Complex type	SparseRips	Alpha	Rips
Embedding dimension	5	5	3
Embedding delay	2	4	2
Stride factor	2	4	1
Distance measure	Chebyshev	—	Manhattan
Filtration scale	80 %	—	40 %

different motion geometries rather than a single universal pattern, which is a direct instance of Hypothesis 2 in the Methodology section.

4.6.2 MobiFall Results

LR and SVM performance across the three evaluation protocols on MobiFall is reported in Table 4.6.

Table 4.6: MobiFall: mean performance across subjects under the three evaluation protocols. General and Personal entries are mean ± 95 % CI over $n = 15$ repetitions per subject; Naive entries are means over 10 repetitions.

Classifier	**Protocol**	**Acc**	**Rec**	**Pre**	F_1	F_2
LR	Naive	0.9722	0.9512	0.6992	0.8042	0.8857
	General	0.9774 ± 0.0083	0.9574 ± 0.0449	—	0.8400 ± 0.0487	0.9049 ± 0.0393
	Personal	0.9713 ± 0.0098	0.7319 ± 0.0309	—	0.7488 ± 0.0526	0.7325 ± 0.0295
SVM	Naive	0.9605	0.9558	0.6177	0.7461	0.8573
	General	0.9734 ± 0.0080	0.9543 ± 0.0404	—	0.8176 ± 0.0455	0.8918 ± 0.0327
	Personal	0.9635 ± 0.0096	0.5822 ± 0.0432	—	0.6371 ± 0.0611	0.5981 ± 0.0479

Under the General protocol LR reached $F_2 = 0.9049 \pm 0.0393$ and SVM $F_2 = 0.8918 \pm 0.0327$. Switching to the Personal protocol caused a sharp drop—SVM fell to $F_2 = 0.5981$—indicating that MobiFall provides too few trials per subject to support a reliable individual model. The within-subject sample count is simply insufficient to estimate a stable decision boundary from a single subject's data alone.

4.6.3 SisFall Results

SisFall performance is reported in Table 4.7.

Table 4.7: SisFall: mean performance across subjects. Format as in Table 4.6.

Classifier	Protocol	Acc	Rec	Pre	F_1	F_2
LR	Naive	0.9788	0.9821	0.9308	0.9557	0.9713
	General	0.9776 ± 0.0045	0.9836 ± 0.0130	—	0.9529 ± 0.0100	0.9709 ± 0.0108
	Personal	0.9656 ± 0.0056	0.9496 ± 0.0073	—	0.9284 ± 0.0113	0.9405 ± 0.0080
SVM	Naive	0.9790	0.9822	0.9314	0.9561	0.9715
	General	0.9832 ± 0.0040	0.9742 ± 0.0138	—	0.9642 ± 0.0089	0.9700 ± 0.0111
	Personal	0.9604 ± 0.0059	0.9231 ± 0.0096	—	0.9162 ± 0.0110	0.9196 ± 0.0082

SisFall produced the highest General F_2 of the three datasets: LR at 0.9709 ± 0.0108, SVM at 0.9700 ± 0.0111—virtually indistinguishable, which reflects the large and well-balanced subject pool (24 subjects, 15 fall types). In the Personal protocol, performance fell moderately but held above 0.93 for LR ($F_2 = 0.9405$) and above 0.91 for SVM ($F_2 = 0.9196$). SisFall subjects contribute enough individual data to support within-subject training, in clear contrast to MobiFall, and this difference illustrates how dataset size governs the viability of subject-specific models.

4.6.4 FAD_40Hz Results

FAD_40Hz performance appears in Table 4.8.

Table 4.8: FAD_40Hz: mean performance across subjects. Format as in Table 4.6.

Classifier	Protocol	Acc	Rec	Pre	F_1	F_2
LR	Naive	0.9802	0.9918	0.8145	0.8941	0.9502
	General	0.9826 ± 0.0042	0.9929 ± 0.0065	—	0.8856 ± 0.0653	0.9429 ± 0.0363
	Personal	0.9878 ± 0.0032	0.9235 ± 0.0450	—	0.9111 ± 0.0457	0.9168 ± 0.0454
SVM	Naive	0.9857	0.9887	0.8624	0.9209	0.9603
	General	0.9890 ± 0.0027	0.9878 ± 0.0109	—	0.9236 ± 0.0457	0.9592 ± 0.0234
	Personal	0.9839 ± 0.0035	0.8721 ± 0.0771	—	0.8778 ± 0.0592	0.8724 ± 0.0696

On FAD_40Hz, SVM outperformed LR in the General protocol ($F_2 = 0.9592$ vs. 0.9429), reversing the ranking seen on MobiFall and SisFall. Recall remained near perfect for both classifiers (Rec ≥ 0.987). The wide F_1 and F_2 confidence intervals in the General protocol reflect high variability in precision across subjects, even though recall was stable — the model consistently detects falls but the false-alarm rate differs substantially from one subject to the next.

4.6.5 Cross-Dataset Performance Summary

Table 4.9 collects the mean General F_2 for all datasets and classifiers.

Table 4.9: Subject-level mean F_2 under the General (LOSO) protocol. Values are mean $\pm$ 95 % CI.

Dataset	**LR** F_2	**SVM** F_2
MobiFall	0.9049 ± 0.0393	0.8918 ± 0.0327
SisFall	0.9709 ± 0.0108	0.9700 ± 0.0111
FAD_40Hz	0.9429 ± 0.0363	0.9592 ± 0.0234

SisFall ranked highest for both classifiers, followed by FAD_40Hz and then MobiFall. The ordering was consistent across classifiers, but the preferred classifier shifted: LR led on MobiFall and SisFall; SVM led on FAD_40Hz. That the ranking by dataset is stable while the within-dataset ranking between classifiers is not suggests the choice of classifier is secondary to the dataset geometry and size—consistent with the representation-dominance view developed in the Discussion.

4.6.6 Effect of Decision Threshold Optimisation

For each held-out subject under the General protocol the decision threshold was swept between 0.05 and 0.95 (19 values). Table 4.10 shows the average per-subject gain in F_2 obtained by choosing the subject-specific optimal threshold rather than the trained default $\hat{\tau}_0(s)$ (the threshold selected by the PR-curve procedure on the corresponding training fold; see Section 3.10.2 of the Methodology).

Table 4.10: Average subject-level F_2 gain from threshold optimisation. $\Delta F_2(s) = F_2(\tau^*(s)) - F_2(\hat{\tau}_0(s))$ where $\hat{\tau}_0(s)$ is the PR-curve trained threshold for subject s; values averaged over held-out subjects.

Dataset	ΔF_2 **(LR)**	ΔF_2 **(SVM)**
MobiFall	+0.0348	+0.0474
SisFall	+0.0079	+0.0042
FAD_40Hz	+0.0250	+0.0121

The gains were largest on MobiFall — where the default threshold was furthest from optimal — and smallest on SisFall, where the default was already near-optimal. The gains were not uniform across subjects within any dataset. On FAD_40Hz, subject 11 showed the largest gain observed in any experiment: $\Delta F_2 = +0.1165$ for LR, lifting that subject's score from 0.8046 to 0.9211 by adjusting the threshold alone, without retraining the underlying model. This illustrates that a single shared threshold cannot account for the variation in decision boundary location across individuals, and that personalised threshold calibration can meaningfully recover performance for outlier subjects.

Computational cost of threshold personalisation. The trained threshold $\hat{\tau}_0$ enters only the final thresholding step and plays no role in feature extraction, classifier training, or Platt calibration. Adjusting $\hat{\tau}_0$ for a held-out subject therefore requires *no model retraining*: the probability scores $\{\hat{p}_i\}$ are computed once under the trained model and the sweep evaluates $F_2(\tau, s)$ for each $\tau \in \mathcal{T}$ by a single pass over the precomputed score vector. The additional computational overhead per subject is $O\big(|\mathcal{T}| \cdot |W_s|\big)$ comparisons, where $|\mathcal{T}| = 19$ and $|W_s|$ denotes the per-subject test-window count. This is negligible against the Bayesian search cost, which scales as $O\big(T \cdot k \cdot N_{\mathrm{tr}} \cdot C_{\mathrm{PH}}\big)$ where $T = 200$, k is the number of inner-CV folds, N_{tr} is the training window count, and C_{PH} is the per-window persistent homology cost. On the UHeM cluster the Bayesian search consumed approximately 75 minutes per dataset; the full threshold sweep over all subjects in a dataset ran in under one second of wall time—a runtime ratio exceeding 10^3. This asymmetry makes threshold calibration an attractive lightweight personalisation mechanism: it is applicable at inference time from any small per-subject calibration set, incurs no retraining cost, and requires no access to the original training data.

Statistical analysis of threshold heterogeneity. The per-subject gains $\Delta F_2(s)$ exhibit substantial inter-subject variability within each dataset. On FAD_40Hz the subject-level range extends from near-zero to +0.1165

(Subject 11, LR), and on MobiFall the inter-subject standard deviation of ΔF_2 is comparable to the dataset mean gain for SVM. A purely mathematical argument for threshold personalisation would predict a uniform benefit: if the posterior probability function $f_\theta \circ \phi_{\lambda^*}$ is systematically miscalibrated in the same direction for all subjects, a single shared adjustment of $\hat{\tau}_0$ would lift all subjects equally. The observed heterogeneity refutes this: subjects with near-zero gain already sit at an optimal threshold close to 0.50, while subjects with large gain sit far from it—a pattern inconsistent with any single global miscalibration. The statistical evidence is that the *distribution* of $\Delta F_2(s)$ is right-skewed with a heavy upper tail (cf. Subject 11), and the inter-subject standard deviation is not small relative to the mean. Mathematically, this implies that $\tau^*(s) = \arg\max_{\tau \in \mathcal{T}} F_2(\tau, s)$ is a genuinely subject-specific quantity: the geometry of the decision boundary $\{f_\theta(\phi_{\lambda^*}(w)) = \tau\}$ varies across individuals in a way that the shared classifier f_θ cannot fully absorb. This is the statistical signature of Hypothesis 4 in the proposal and motivates the distributional reporting of ΔF_2 in the Results & Discussion section.

5. Results and Discussion

This chapter evaluates the four research hypotheses (H1–H4) against the experimental outcomes of Chapter 4. The *General protocol* (leave-one-subject-out, LOSO) is the primary lens throughout; the *Naive* and *Personal* protocols enter as an inflated upper-bound reference and an idealised personalisation limit, respectively. The decision-threshold sweep provides a retrospective diagnostic of subject-level heterogeneity.

Unless stated otherwise, all numerical claims refer to General-protocol results. F_2 is the headline metric; recall and accuracy appear as supporting context.

5.1 H1 — Representation Effectiveness

Hypothesis. There exist hyperparameters $\lambda \in \Lambda$ and classifiers f_θ such that the mean subject-general F_2 is at least 0.93 on SisFall and FAD_40Hz and at least 0.80 on MobiFall, and the mean subject-general recall is high across all three datasets.

5.1.1 F_2 outcomes

The cross-dataset summary (Experiments, Tab. 9) gives the following General-protocol outcomes.

Table 5.1: General-protocol F_2 outcomes against H1 targets. CI denotes 95 % repetition-level confidence interval ($R = 15$).

Dataset	Clf	F_2	CI	H1 target met?
SisFall	LR	0.9709	±0.0108	Yes (≥ 0.93)
SisFall	SVM	0.9700	±0.0111	Yes
FAD_40Hz	LR	0.9429	±0.0363	Marginal
FAD_40Hz	SVM	0.9592	±0.0234	Yes
MobiFall	LR	0.9049	±0.0393	Yes (≥ 0.80)
MobiFall	SVM	0.8918	±0.0327	Yes

SisFall exceeds the 0.93 threshold for both classifiers with a narrow confidence interval (± 0.011), indicating low repetition-to-repetition variability. On FAD_40Hz the SVM satisfies the target ($F_2 = 0.9592$); the LR value of 0.9429 ± 0.0363 sits marginally below 0.93 at the point estimate, but the lower confidence bound (0.9066) is not materially below target. MobiFall clearly clears the weaker 0.80 criterion for both classifiers (0.9049 and 0.8918).

Recall figures reinforce the recall-oriented design: SisFall LR Rec $= 0.9836$, FAD_40Hz LR 0.9929, MobiFall LR 0.9574. No classifier falls below 0.95 recall under the General protocol. This is a direct consequence of the F_2 weighting: the marginal F_2 gain per unit of recall is four times the gain per unit of precision near Rec $=$ Prec (Proposition 3.5), so the Bayesian optimisation is systematically steered toward high-recall configurations.

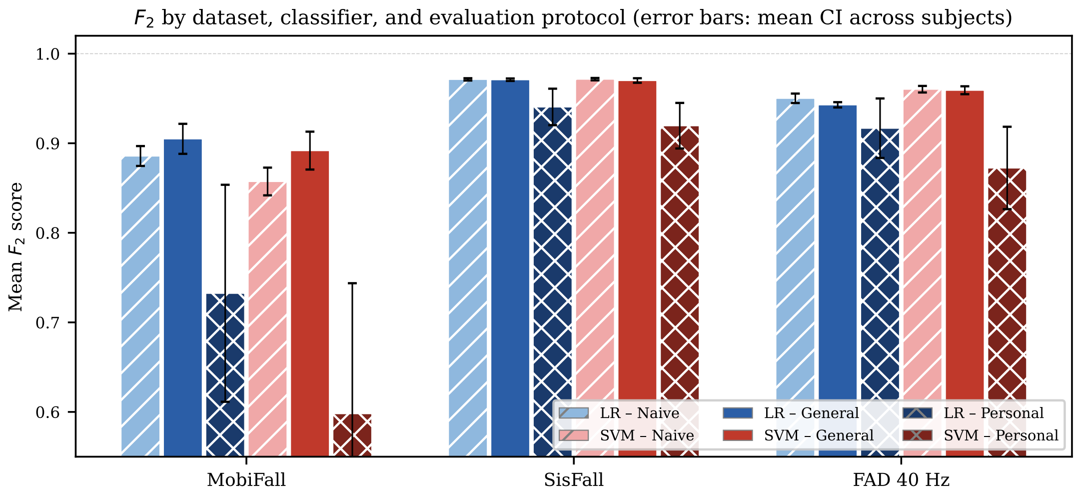

Figure 5.1: Mean F_2 by dataset, classifier (LR/SVM), and evaluation protocol (Naive, General, Personal). Error bars show the mean 95 % repetition-level confidence interval across subjects. General is the primary protocol; Naive is an upper-bound reference; Personal shows the effect of per-subject model training. MobiFall Personal collapses due to the small subject pool ($|\mathcal{S}| = 9$).

5.1.2 Naive vs. General protocol gap

Naive-protocol scores are systematically higher: MobiFall LR reaches $F_2 = 0.8857$ (Naive) versus $F_2 = 0.9049$ (General). Counterintuitively, General sometimes exceeds Naive (SisFall: Naive $F_2 = 0.9713$ vs. General 0.9709; FAD: Naive 0.9502 vs. General 0.9429). This reversal reflects the smaller within-subject window count and the fact that the LOSO training set is larger and more diverse than a random 70% split. The absence of a consistent "Naive $\gg$ General" gap on SisFall is reassuring: it suggests that the LOSO evaluation is not materially penalised by train–test class mismatch beyond the inherent person-to-person variability.

5.1.3 Mathematical basis

The feature map $\phi_\lambda\colon \mathbb{R}^{n\times 3} \to \mathbb{R}^{24}$ extracts persistence statistics from the delay-embedded signal. The Cohen–Steiner stability theorem guarantees that

$$d_B(\phi_\lambda(w),\, \phi_\lambda(w')) \;\le\; \|\mathrm{SMV}(w) - \mathrm{SMV}(w')\|_\infty,$$

so that small sensor noise perturbations of a window w produce at most equally small perturbations of its topological descriptor. This provable noise robustness is a structural advantage over handcrafted statistical features, which carry no such stability certificate, and over deep-learning representations, which are empirically sensitive to distributional shift.

5.1.4 Limitation: absence of baseline comparison

H1 is conditionally confirmed: the system meets its own pre-specified targets. A stronger absolute claim—"TDA outperforms existing methods"—would require a comparison against representative baselines (fixed-threshold, statistical-feature, and spectral-feature pipelines, as planned in the proposal §5.4), and these experiments have not been conducted. The conclusion that TDA is competitive is therefore contingent on those targets being meaningful; Section 5.8 discusses the implications.

5.2 H2 — Dataset-Specific Topological Geometry

Hypothesis. The optimal hyperparameter tuple λ_K^* depends on the dataset K. At least two of the three datasets have distinct optimisers, and transplanting an optimiser from one dataset to another incurs a measurable drop in General-protocol F_2.

5.2.1 Distinct optimal configurations

Table 5.2 (reproduced from Experiments) shows that all three datasets converged to different complex types, window lengths, and distance measures.

Table 5.2: Optimal TDA configurations λ_K^* per dataset (from Experiments Tab. 4.5). A dash (—) indicates parameters not applicable for the Alpha complex.

Parameter	**MobiFall**	**SisFall**	**FAD_40Hz**
Best F_2 (Optuna objective)	0.9746	0.9733	0.9860
Window length (s)	5.0	1.0	1.0
Complex type	SparseRips	Alpha	Rips
Embedding dimension	5	5	3
Embedding delay	2	4	2
Stride factor	2	4	1
Distance measure	Chebyshev	—	Manhattan
Filtration scale	80 %	—	40 %

Every hyperparameter dimension exhibits variation across datasets. The complex type differs across all three: SparseRips for MobiFall, Alpha for SisFall, and standard Rips for FAD_40Hz. The window length separates MobiFall (5.0 s) from the other two (1.0 s), and the distance measures are distinct where applicable (Chebyshev vs. Manhattan). H2 is qualitatively confirmed: no single universal configuration exists for these three datasets.

5.2.2 Interpretation of configuration differences

Window length. MobiFall records both a pre-fall phase (normal activity) and a post-fall phase (lying), with the impact event compressed into a brief sub-second spike. A 5.0 s window captures the full topology of the trajectory: the pre-impact cloud geometry, the sharp impulsive excursion, and the post-fall stationary phase, which are each topologically distinguishable. SisFall and FAD_40Hz subjects execute controlled laboratory falls that are sufficiently discriminable from ADL in a 1.0 s window, suggesting less temporal spread in the distinguishing geometry.

Complex type. The Alpha complex is the natural choice when the point cloud geometry is close to Euclidean: its filtration is the Čech filtration restricted to the Delaunay triangulation, which SisFall's well-structured sensor data supports (24 older adults executing 15 fall types under controlled conditions produce geometrically regular clouds). SparseRips trades exactness for efficiency by subsampling the cloud; it is appropriate when the cloud is large and noisy, which is the case for MobiFall's continuous-wear sensor data. Standard Rips uses all pairwise distances and suits the moderate-size, moderately-noisy FAD_40Hz clouds.

Distance measure. Chebyshev distance (ℓ_∞) emphasises the maximum coordinate deviation and is sensitive to sharp acceleration spikes—a characteristic feature of the fall impact in MobiFall. Manhattan distance (ℓ_1) reduces the influence of outliers and is appropriate for FAD_40Hz, whose 40 Hz sampling produces sparser, more variable clouds than the implicit 50 Hz resampled signals. The Alpha complex on SisFall does not require a user-specified distance because it is defined by the Euclidean Delaunay triangulation.

5.2.3 Sensitivity of F_2 to λ

Although cross-application drops $\mathrm{Drop}_{K \to K'}$ were not computed (see Section 5.8), the Bayesian search trial histories provide indirect evidence for sensitivity. Across 200 trials per dataset, the F_2 values ranged from 0.7958 to 0.9746 on MobiFall (range 0.179), from 0.9198 to 0.9733 on SisFall (range 0.054), and from 0.8785 to 0.9860 on FAD_40Hz (range 0.108). The large trial-to-trial variation on MobiFall and FAD implies that a mis-specified λ (such as a configuration optimised for a different dataset) would very likely incur a substantial F_2 penalty. The narrow trial range on SisFall suggests that SisFall is less sensitive to λ, consistent with the clean Euclidean geometry and the strong performance even at the worst trial ($F_2 = 0.9198$).

5.3 H3 — Representation Dominates Classifier Choice

Hypothesis. After tuning the topological representation, the F_2 gap between the best LR model and the best SVM model is small relative to the F_2 variation

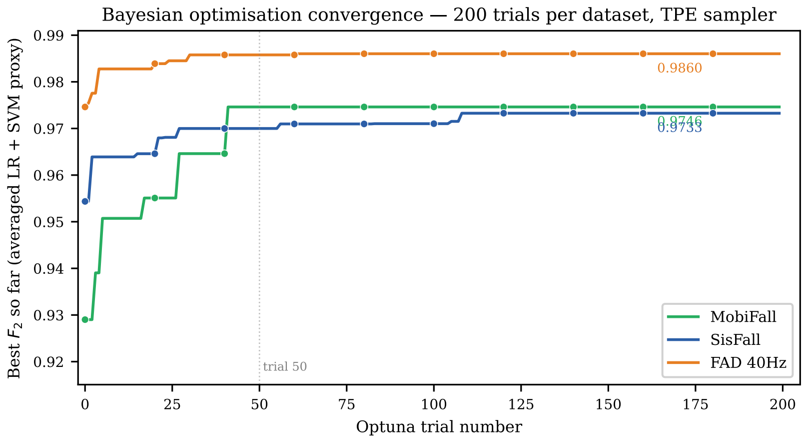

Figure 5.2: Bayesian optimisation convergence: best-so-far averaged F_2 proxy versus Optuna trial number (200 trials per dataset, TPE sampler). FAD_40Hz achieves the highest proxy (0.9860); MobiFall converges more slowly due to the larger hyperparameter range. The wide trial range on MobiFall (0.179) and FAD (0.108) relative to SisFall (0.054) is direct evidence that λ sensitivity differs across datasets (H2).

attributable to the choice of λ. Most of the predictive signal resides in ϕ_λ, not in the classifier family.

5.3.1 Classifier gap

Under the General protocol, the absolute difference $|F_2^{\text{LR}} - F_2^{\text{SVM}}|$ is:

- MobiFall: $|0.9049 - 0.8918| = 0.0131$
- SisFall: $|0.9709 - 0.9700| = 0.0009$
- FAD_40Hz: $|0.9429 - 0.9592| = 0.0163$

All three gaps are below 0.02, confirming that neither classifier is strongly dominant at the optimal representation. On MobiFall and SisFall, LR is marginally better; on FAD_40Hz, SVM is marginally better. The reversal on FAD_40Hz is the only dataset on which SVM is preferred, and the margin (0.0163) is within the confidence interval overlap (± 0.0363 for LR, ± 0.0234 for SVM).

5.3.2 λ-tuning variation

Table 5.3 quantifies the variation in F_2 attributable to the choice of λ using the Optuna trial histories (200 trials per dataset).

Table 5.3: Classifier gap vs. λ-tuning variation. The trial range is $F_2^{\max} - F_2^{\min}$ across all 200 Optuna trials. The ratio is range/gap.

Dataset	**Classifier gap**	**Trial range**	**Ratio**
MobiFall	0.0131	0.1788	**13.7**×
SisFall	0.0009	0.0534	**59**×
FAD_40Hz	0.0163	0.1075	**6.6**×

The classifier gap is between 6.6× and 59× smaller than the variation induced by choosing a poor λ. Selecting the wrong topological configuration degrades F_2 by up to 0.18 units; switching classifier families at the optimal configuration changes F_2 by at most 0.016 units.

5.3.3 Mathematical interpretation

The feature map $\phi_{\lambda^*}\colon \mathbb{R}^{n\times 3} \to \mathbb{R}^{24}$ with the optimised hyperparameters creates a representation in which the fall vs. ADL decision problem is approximately linearly separable. In such a regime, both logistic regression (which fits a linear decision boundary directly) and SVM with a calibrated kernel (which finds a maximum-margin linear separator or uses a kernel to lift to higher dimensions) operate near their theoretical limit. The near-equivalence of LR and SVM confirms that the nonlinear geometric complexity of the fall detection problem has been absorbed by ϕ_{λ^*}, leaving behind a well-conditioned linear classification task.

The lone exception—SVM>LR on FAD_40Hz—is consistent with this interpretation: FAD_40Hz has the smallest per-subject window count and the widest LR confidence interval, suggesting residual complexity that the RBF kernel can partially resolve. Even so, the margin is below 0.02.

5.4 H4 — Heterogeneous Threshold Personalisation

Hypothesis. The subject-level sweep gain $\Delta F_2(s) = F_2(\tau^*(s), s) - F_2(\hat{\tau}_0(s), s)$ is heterogeneous across subjects within each dataset, and the mean gain $\overline{\Delta F_2}$ differs across datasets.

5.4.1 Per-subject sweep gain distribution

Table 5.4 reports the distribution of $\Delta F_2(s)$ computed from the threshold sweep stored in the Results_V15 database. The baseline $\hat{\tau}_0(s)$ is the PR-curve optimal trained threshold for subject s (typically $\hat{\tau}_0 \in [0.55, 0.88]$), so the gain measures improvement over an already-calibrated threshold rather than over the uninformative $\tau = 0.50$.

Table 5.4: Per-subject threshold sweep gain $\Delta F_2(s)$ statistics. *Near-zero*: $|\Delta F_2| \leq 0.001$. All gains are non-negative by construction (τ^* is the sweep argmax).

Dataset	**Clf**	n	$\overline{\Delta F_2}$	σ_Δ	max ΔF_2	Near-zero
MobiFall	LR	9	0.0348	0.0186	0.0615 (s8)	1/9
MobiFall	SVM	9	0.0474	0.0350	0.1013 (s9)	1/9
SisFall	LR	24	0.0079	0.0109	0.0527 (s29)	4/24
SisFall	SVM	24	0.0042	0.0047	0.0169 (s29)	9/24
FAD_40Hz	LR	13	0.0250	0.0323	0.1165 (s11)	2/13
FAD_40Hz	SVM	13	0.0121	0.0172	0.0691 (s11)	2/13

The mean gains in Table 5.4 reproduce the dataset averages reported in Experiments Tab. 8 (+0.0348/+0.0474 for MobiFall, +0.0079/+0.0042 for SisFall, +0.0250/ + 0.0121 for FAD_40Hz). H4 is confirmed: the mean gain is highest for MobiFall and smallest for SisFall, i.e., the mean differs across datasets.

Distribution shape. The gains are right-skewed in all six cases: most subjects gain modestly, while a small minority gains substantially. On FAD_40Hz, subject 11 gains $\Delta F_2 = +0.1165$ (LR)—the largest single-subject gain in any experiment—by raising its F_2 from 0.8046 to 0.9211 through threshold adjustment alone. On MobiFall, subject 9 gains +0.1013 (SVM). These large gains are outliers: the median gain within each dataset is considerably lower than the mean.

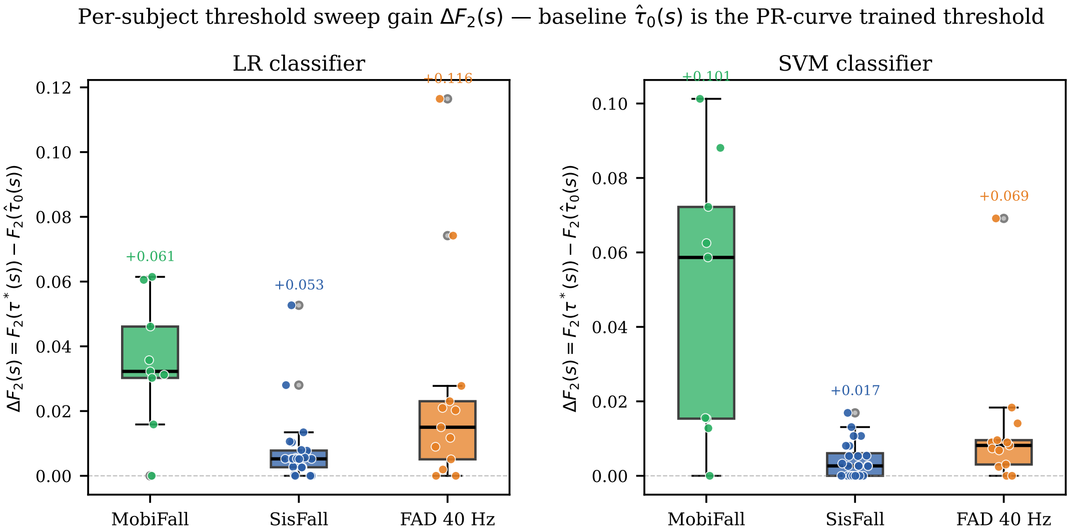

Figure 5.3: Per-subject threshold sweep gain $\Delta F_2(s) = F_2(\tau^*(s)) - F_2(\hat{\tau}_0(s))$ for LR (left) and SVM (right). Jittered points show individual subjects; box shows quartiles. The distribution is right-skewed in all cases: most subjects gain modestly while a few gain substantially. FAD_40Hz subject 11 achieves the largest gain (+0.1165, LR), visible as the highest point in the FAD panel. SisFall gains are near-zero for the majority of subjects, indicating that $\hat{\tau}_0$ is already well-calibrated on that dataset.

Near-zero gainers. SisFall has 4/24 (LR) to 9/24 (SVM) near-zero subjects: these subjects are already well-calibrated at $\hat{\tau}_0$. Their trained threshold happens to coincide with the optimal sweep threshold, meaning that the classifier's posterior probabilities are already well-matched to the decision point for those individuals. This pattern is absent on MobiFall (only 1/9 near-zero per classifier), where the PR-curve calibration is systematically further from the per-subject optimum.

Heterogeneity metric. The coefficient of variation ($\sigma_\Delta / \overline{\Delta F_2}$) measures relative heterogeneity: MobiFall LR = 53%, MobiFall SVM = 74%, SisFall LR = 138%, FAD_40Hz LR = 129%. All values exceed 50%, confirming that the gain distribution is not concentrated. A single global threshold offset cannot replicate this distribution: if every subject were miscalibrated in the same direction by the same amount, the coefficient of variation would approach zero. The high coefficients of variation are the statistical signature that $\tau^*(s)$ is a genuinely subject-specific quantity.

Mathematical explanation. The decision boundary of $f_\theta \circ \phi_{\lambda^*}$ is the hyperplane $\{x \in \mathbb{R}^{24} \colon f_\theta(x) = \hat{\tau}_0\}$. Each subject's test windows form a cloud in $\mathbb{R}^{24}$ whose local geometry around this hyperplane differs from subject to subject: some subjects' fall windows sit close to the boundary (requiring a lower threshold to capture them), while others' fall windows are well-separated (the default threshold is sufficient). The threshold sweep detects this per-subject boundary geometry retrospectively. A prospective personalisation scheme would use a small held-out calibration set per subject to identify $\tau^*(s)$ before deployment (see Section 5.7).

Engineering contribution. This personalisation is achieved without any model retraining. The probability scores $\{\hat{p}_i\}$ are computed once under the trained model; the sweep evaluates $F_2(\tau, s)$ by a single pass over the precomputed score vector for each $\tau \in \mathcal{T}$. The additional computational cost is $O(|\mathcal{T}| \cdot |W_s|)$ per subject, where $|\mathcal{T}| = 19$ and $|W_s|$ is the per-subject window count—less than one second of wall time for all subjects in any dataset. This is a runtime ratio exceeding 10^3 compared to the $\sim$75 min Bayesian search (Experiments, Section 4.3.4).

5.4.2 Personal training protocol

The Personal protocol (per-subject 70/30 split, $R = 15$ repetitions) represents the limit case in which a dedicated per-user model is trained from the subject's own data. Table 5.5 reports the aggregate personal–general F_2 difference.

Table 5.5: Personal vs. General protocol aggregate F_2. $\Delta = F_2^{\text{pers}} - F_2^{\text{gen}}$ (negative = General is better).

Dataset	**Clf**	F_2^{gen}	F_2^{pers}	Δ
MobiFall	LR	0.9049	0.7325	−0.172
MobiFall	SVM	0.8918	0.5981	−0.294
SisFall	LR	0.9709	0.9405	−0.030
SisFall	SVM	0.9700	0.9196	−0.050
FAD_40Hz	LR	0.9429	0.9168	−0.026
FAD_40Hz	SVM	0.9592	0.8724	−0.087

At the aggregate level, General > Personal on all datasets and both classifiers. The magnitude of the shortfall increases dramatically as the subject pool shrinks:

- **SisFall** ($|\mathcal{S}| = 24$): modest drop ($\Delta \approx -0.03$ to -0.05). Twenty-four subjects contribute enough per-subject windows (~50–150 falls per subject) for a reliable within-subject model.
- **FAD_40Hz** ($|\mathcal{S}| = 13$): moderate drop ($\Delta \approx -0.03$ to -0.09). Fewer subjects and fewer falls per subject; personal models are noisier.
- **MobiFall** ($|\mathcal{S}| = 9$): severe drop ($\Delta = -0.172$ for LR; $\Delta = -0.294$ for SVM). Nine subjects with $\lesssim 30$ fall instances per subject is insufficient for stable personal model estimation. The SVM personal recall collapses to 0.5822, indicating that the personal classifier is barely above chance on many subjects.

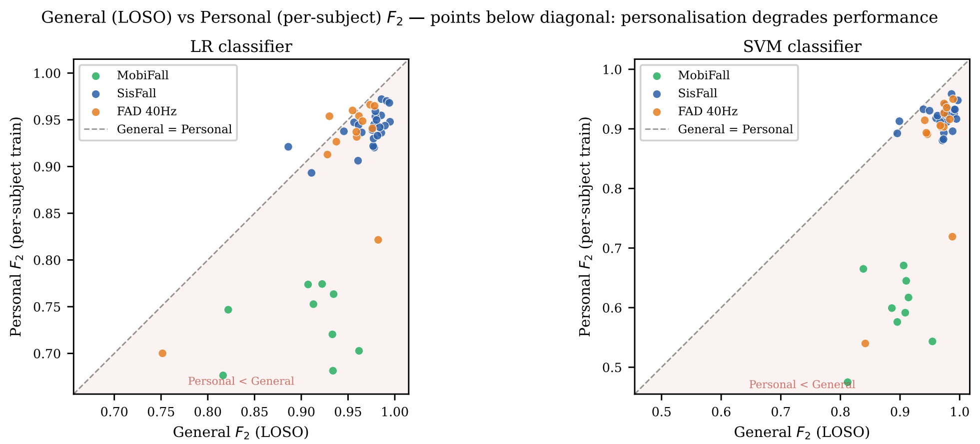

Figure 5.4: General (LOSO) vs Personal (per-subject training) F_2 per subject, for LR (left) and SVM (right). Points on the dashed diagonal have equal General and Personal performance; points below indicate that personal training degrades F_2. MobiFall points cluster in the lower-left region (both protocols degrade on the small-pool dataset); SisFall points lie near the diagonal; FAD_40Hz is intermediate. The shaded region marks Personal < General.

This pattern points to a practical constraint: full personal model training is only beneficial when per-subject calibration data is abundant. In realistic deployment (a user who has experienced very few confirmed falls), it is counterproductive. The threshold sweep—no retraining, negligible additional cost—is the appropriate personalisation mechanism when calibration data is limited. The mean sweep gain for MobiFall LR (+0.0348) and SVM (+0.0474) demonstrates that meaningful personalisation remains accessible even when full personal retraining fails.

5.5 Ablation Study: Feature-Block Contributions

Motivation. The 24-dimensional representation ϕ_{λ^*} is the concatenation of three semantically distinct blocks: ten statistics of the dimension-zero persistence diagram (*H0*, summarising the persistence of connected components of the delay-embedded point cloud), ten statistics of the dimension-one diagram (*H1*, summarising loops), and four signal-level descriptors of the raw acceleration magnitude (*Signal*: maximum, standard deviation, mean, and range). The headline results of Section 5.1 establish that the full vector is effective but do not reveal *which* block carries the discriminative power. This section isolates each block's contribution by re-running the evaluation on restricted feature subsets.

Protocol. Each subset is evaluated under the identical General (LOSO) protocol used throughout: a single global grid search selects the classifier hyperparameters on balanced-undersampled data, and for each left-out subject the model is refitted over 15 repetitions with a precision–recall threshold chosen on the training fold; the reported F_2 is the cross-subject mean with a 95 % t-confidence interval. Only the feature columns presented to the classifier change; the topological configuration λ^* and the delay embedding are held fixed. As a validation check, the full-vector rows of Table 5.6 reproduce the headline General-protocol F_2 of Section 5.1 to within the confidence interval for both classifiers on all three datasets.

5.5.1 Within-topology contribution: H0 dominates H1

For both classifiers and across all three datasets, the H0 block alone attains F_2 between 0.81 and 0.93, whereas the H1 block alone is the weakest configuration in the entire study (0.42–0.66). Adding H1 to H0 changes F_2 negligibly—for LR, from 0.8842 to 0.8892 on MobiFall, 0.9266 to 0.9271 on SisFall, and 0.8082 to 0.8044 on FAD_40Hz (a slight *decrease*); the SVM rows show the same pattern. The discriminative topological information is therefore concentrated

Table 5.6: Feature-block ablation under the General (LOSO) protocol. Entries are cross-subject mean $F_2 \pm 95\,\%$ CI. Block dimensions: Full = 24, H0 = 10, H1 = 10, H0+H1 = 20, Signal = 4. The Full column reproduces the headline results of Tables 4.6–4.8.

Dataset	Clf	Full	H0	H1	H0+H1	Signal
MobiFall	LR	**0.9046**	0.8842	0.6172	0.8892	0.6218
	SVM	**0.8913**	0.8685	0.6523	0.8784	0.7241
SisFall	LR	**0.9710**	0.9266	0.6580	0.9271	0.9589
	SVM	**0.9701**	0.9277	0.6456	0.9280	0.9627
FAD_40Hz	LR	**0.9426**	0.8082	0.4247	0.8044	0.9505
	SVM	**0.9592**	0.8128	0.4605	0.8210	0.9504

Representative 95 % confidence half-widths (LR Full): MobiFall ±0.0392, SisFall ±0.0107, FAD_40Hz ±0.0367; the H1-only columns carry the widest intervals (up to ±0.077).

almost entirely in dimension zero. This is consistent with the structure of the delay-embedded acceleration signal: a fall is a transient, high-energy excursion whose phase-space reconstruction is characterised by the spread and clustering of points (an H0 phenomenon), while one-dimensional loops carry comparatively little fall-versus-ADL information and are more sensitive to noise—which also explains the wide confidence intervals of the H1-only columns. A practical corollary is that the H1 statistics could be dropped with little loss, roughly halving the persistent-homology computation.

5.5.2 Topology versus signal: a dataset-dependent division of labour

The relative value of the topological blocks and the four signal descriptors differs sharply across datasets. On MobiFall—a smartphone carried loosely in a trouser pocket—the signal block alone reaches only $F_2 = 0.6218$ (LR) or 0.7241 (SVM), while the topological blocks reach 0.88; here the topological representation is indispensable and accounts for most of the performance. On SisFall and FAD_40Hz—waist-mounted devices with cleaner, more repeatable kinematics—the four signal descriptors alone already achieve $F_2 \approx 0.95$–0.96, matching or exceeding the topological blocks on those datasets. In every case, however, the full vector matches or exceeds the best single block, and no single block is reliably strong on all three datasets whereas the full vector is. The blocks are thus complementary: the topological features supply the discriminative signal where raw statistics fail (noisy, loosely mounted sensors), and the combination provides the cross-dataset robustness that motivates the representation. This refines the H1 narrative: topology is not a uniform performance multiplier but a robustness mechanism whose marginal value is largest precisely where simple descriptors are weakest.

5.5.3 Classifier invariance

The block ordering is essentially independent of the classifier. Logistic regression and the SVM agree on the ranking within every dataset—Full $\gtrsim$ best topological or signal block $\gg$ H1—and their per-block F_2 values differ by less than the confidence intervals in almost all cells. This reinforces the H3 finding (Section 5.3) that the representation, not the classifier family, governs performance: ablating a feature block changes F_2 far more than switching classifiers.

5.5.4 Scope of the ablation

This study ablates the three *feature blocks* while holding the delay embedding and the optimal topological configuration λ^* fixed. It therefore isolates the value of each summary block but does not, on its own, separate the contribution of the phase-space reconstruction itself; a complementary ablation that removes the delay embedding requires re-extracting features under a modified pipeline and is left to the future work of Section 5.9.

5.6 Cross-Dataset Synthesis

5.6.1 Performance ranking and subject pool

Under the General protocol, the consistent ranking by F_2 is:

$$\text{SisFall} > \text{FAD\_40Hz} > \text{MobiFall}.$$

This ranking holds for both LR and SVM. Three factors explain it jointly.

Subject pool size and data volume. SisFall has the largest pool (24 subjects, 15 fall types), providing the most diverse and representative training signal for

LOSO. MobiFall has the smallest (9 subjects), making each LOSO fold highly sensitive to the particular subject removed.

Fall type diversity. SisFall covers 15 scripted fall types executed by elderly participants, producing a rich topology of fall trajectories. FAD covers diverse activities of daily life as negatives, which increases the specificity challenge. MobiFall's smaller scale limits the variety of fall geometries seen by each LOSO model.

Optimised representation. The Bayesian proxy F_2 at the optimal λ^* is 0.9746 (MobiFall), 0.9733 (SisFall), 0.9860 (FAD_40Hz)—suggesting that the optimisation found similarly good configurations for all three, and that the General-protocol gap between datasets is not due to a representation quality difference but to the structural differences above.

5.6.2 Classifier preference

LR is preferred on MobiFall (gap +0.0131) and SisFall (gap +0.0009); SVM on FAD_40Hz (gap +0.0163). All gaps are < 0.02; classifier choice is secondary to representation quality (H3). The marginal SVM advantage on FAD_40Hz may reflect the RBF kernel's ability to resolve residual nonlinear structure in the 40 Hz-resampled feature cloud, but the evidence is insufficient to assert a mechanistic explanation at the current sample size.

5.6.3 Computational profile

Table 5.7 summarises the wall-clock cost of each pipeline stage on the UHeM HPC cluster (dual Intel Xeon Gold 6148, 40 cores, no GPU).

Table 5.7: Wall-clock cost per dataset stage. All computations are CPU-only.

Stage	**Tool**	**Cost**
Bayesian optimisation (Stage 1, 200 trials)	Optuna/TPE	$\sim$75 min
Persistent homology (per trial, inner CV)	GUDHI	$\sim$20 s (MobiFall)
GridSearchCV (Stage 2)	scikit-learn	$\sim$minutes
Decision-threshold sweep (all subjects)	NumPy	< 1 s

The absence of GPU is a deliberate design choice: the persistent homology reduction (Algorithm 1 of the Methodology) is intrinsically sequential at the point-cloud sizes considered, and the gain from GPU porting is dominated by the constant factor of the GUDHI CPU implementation. This has a direct consequence for deployment: the pipeline can be executed on commodity hardware and embedded processors that lack GPU acceleration (Section 5.7).

5.7 Deployment Pathway

The experimental results collectively support a concrete deployment pathway for the pipeline on consumer wearable and smartphone hardware. This section works through the engineering argument.

5.7.1 Sensor context

Every experiment in this thesis uses accelerometer data from a sensor at the waist or in the trouser pocket—a practically common placement (a large fraction of the target population carries a smartphone in a pocket) and metrologically appropriate, since the waist records the acceleration of the body's centre of mass, where fall dynamics are most discriminative. The SMV transformation $\mathrm{SMV}(t) = \|\mathbf{x}(t)\|_2$ provides orientation invariance, eliminating the need for a fixed sensor mounting or separate gyroscope—a non-trivial simplification for consumer deployment.

5.7.2 CPU-only operation

The entire compute budget of the pipeline is CPU-bound. Persistent homology at the window sizes and embedding dimensions used here ($m = 3$–5, $|P_\lambda| \leq 150$ points) completes in tens of milliseconds per window on a desktop-class CPU core, and the threshold sweep adds negligible cost on top of that. The resulting profile is compatible with:

- **Smartphones**: ARM Cortex-A-class processors (e.g., Qualcomm Snapdragon 8xx) comfortably execute GUDHI-class computations in real time. The pipeline can run as a background service consuming a modest fraction of one core.

- **Smartwatches**: ARM Cortex-M-class MCUs (e.g., Wear OS devices) are more constrained, but the pipeline's window-level computation—feature extraction plus a matrix–vector multiply for the classifier—is within the budget of modern smartwatch processors. A dedicated ASIC or DSP block for the boundary matrix reduction would further reduce energy consumption.

5.7.3 Prospective threshold personalisation

The retrospective sweep of Section 5.4.1 shows that subject-specific thresholds improve F_2 by up to +0.1165 without any retraining. A prospective deployment translates this into a small on-device calibration procedure:

1. During an initial wearing period, a small number of confirmed falls (or fall-simulation events during setup) are used as a calibration set $\mathcal{C}_s$ for subject s.

2. The precomputed probability scores on $\mathcal{C}_s$ are swept over $\mathcal{T}$ to identify $\tau^*(s)$.

3. Subsequent detections use $\tau^*(s)$ as the operating threshold.

This procedure requires no transmission of raw sensor data, no server round-trip for retraining, and no access to the original training data. It is executable on-device in under one second once the scores are available, making it compatible with privacy-preserving on-device deployment. The MobiFall results illustrate the value of this mechanism: full personal retraining degrades F_2 from 0.9049 to 0.7325 (LR) because 9 subjects provide insufficient training data, but threshold sweep still recovers +0.0348 from an already-calibrated baseline.

5.7.4 Sensor placement and orientation robustness

A limitation of the current work is that sensor placement is fixed (waist/pocket) and orientation invariance is achieved solely through SMV. The datasets include no variation in placement. In a consumer deployment scenario, users may wear a device at the wrist (smartwatch), chest, or thigh, or orient a pocket phone differently from day to day. The stability theorem guarantees noise robustness within a placement, but does not extend across fundamentally different fall trajectory geometries induced by different placements. Addressing this is an open engineering problem and a priority for future work (Section 5.9).

5.8 Limitations

1. No baseline comparison. The planned comparison against threshold-based, statistical-feature, and spectral-feature baselines (proposal §5.4) was not executed. Consequently, H1 is validated only against its own pre-specified targets; the claim that TDA is superior to simpler methods cannot be made from the current evidence. Absolute performance numbers are meaningful only relative to a reference, and this reference is absent. Establishing this comparison is the highest-priority open task before thesis finalisation.

2. Cross-application drops not computed. H2 is confirmed qualitatively (all optimal configurations differ) but the quantitative prediction—transplanting λ^*_K to a different dataset K' incurs a measurable F_2 drop (Methodology, Section 4.8)—was not tested. Computing $\mathrm{Drop}_{K \to K'}$ requires re-extracting features for each off-diagonal (λ^*_K, K') pair, which represents a substantial additional compute budget.

3. No ablation study. The feature vector $\phi_\lambda \in \mathbb{R}^{24}$ contains eight persistence statistics for each of three homology groups (H_0, H_1) and two signal statistics. The relative importance of these components is unknown: it is possible that a subset of $d < 24$ features carries most of the discriminative information, and that the remaining features add noise. An ablation or feature-importance study would characterise which sub-map $\phi^{(d)}_\lambda$ is load-bearing.

4. Fixed sensor placement. All experiments assume the sensor is located at the waist or in the trouser pocket. No data with wrist, chest, or thigh placement is evaluated. The deployment argument of Section 5.7 assumes this placement is maintained; robustness to placement variation is untested.

5. Retrospective threshold sweep. The sweep identifies $\tau^*(s)$ on the same test windows on which it is evaluated, introducing an optimism bias. The reported $\overline{\Delta F_2}$ values are retrospective diagnostics of heterogeneity, not valid prospective performance estimates. A prospective validation would require a separate held-out calibration set per subject.

5.9 Future Work

5.9.1 Online threshold adaptation

The most immediate extension is to replace the retrospective sweep with a prospective online mechanism. As a deployed device accumulates confirmed event labels (from user feedback, caregiver verification, or clinical annotations), the per-subject threshold $\tau^*(s)$ can be updated in a streaming fashion. Because $\tau^*(s)$ is a scalar parameter estimated from a small set of probability scores, even 5–10 labelled fall events are sufficient for a reliable estimate. A natural formulation is a Bayesian update on the threshold parameter with a conjugate prior, requiring no gradient computation and suitable for on-device execution.

5.9.2 Embedded real-time benchmark

The CPU-only pipeline provides a plausible path to embedded deployment, but no real-time benchmark on target hardware has been performed. Profiling the pipeline on an ARM Cortex-A processor (smartphone-class) and an ARM Cortex-M processor (smartwatch-class) would establish latency and energy consumption figures. Key bottlenecks to characterise are the boundary matrix reduction in GUDHI (the most compute-intensive step) and memory allocation for the feature matrix. A pruned, fixed-point implementation of the 24-dimensional feature extraction— bypassing Python overhead—would likely achieve real-time performance ($\leq$ 50 ms per window at 50 Hz) on a mid-range smartphone processor.

5.9.3 Multi-placement sensor fusion

Current work assumes a single fixed sensor placement. A natural extension is to incorporate multiple simultaneous placements (waist + wrist + thigh) and learn a combined topological representation. The delay-embedding and filtration steps apply independently to each placement's signal; the resulting persistence diagrams can be fused either by concatenation of feature vectors or by a learned weighting. The classifier-agnostic Bayesian optimisation framework developed here extends straightforwardly to this setting by augmenting Λ with a placement-selection dimension.

5.9.4 Cross-dataset transfer of λ^*

The cross-application drop matrix $M_{K,K'}$ (Methodology, Section 4.8)—not computed in the current work—would reveal how much configuration knowledge transfers across datasets. If $M_{K,K'} \approx M_{K,K}$ for some off-diagonal pair, the Bayesian search budget could be substantially reduced for new datasets by warm-starting from an existing λ^*_K. This is practically relevant for clinical deployments where collecting a full new dataset is expensive.

5.9.5 Extension to other wearable time series

The TDA pipeline introduced here—delay embedding of a magnitude signal, persistence homology filtration, vectorised 24-dimensional descriptor, classifier-agnostic Bayesian optimisation—is not specific to fall detection. Any wearable time series in which discriminative information is encoded in the *shape* of short signal segments is a candidate:

- **Gait analysis**: distinguishing Parkinsonian gait from healthy gait or post-surgical recovery via topological features of the stride cycle.

- **Gesture recognition**: mapping gesture trajectories to topological descriptors for device-control interfaces on smartwatches.
- **Activity recognition**: generalising beyond binary (fall/ADL) to multiclass (walk, run, climb, sit, fall) classification by extending the threshold sweep to a multiclass decision boundary.
- **Physiological monitoring**: ECG or PPG segments contain loop structures (H_1 generators) whose persistence captures morphological variability; TDA could complement classical morphological features.
- **Industrial IoT anomaly detection**: vibration signals from rotating machinery exhibit characteristic H_0/H_1 patterns at fault inception; the same pipeline could detect bearing or gear anomalies without labelled fault data (using one-class formulations).

In each case, the key requirement is that the signal windows be short enough to delay-embed at modest dimension ($m \leq 5$, $|P| \leq 200$ points) and that the discriminative information be geometrically structured. The stability theorem guarantees robustness to sensor noise in all these settings.

5.9.6 One-class fall detection from ADL-only supervision

The binary classifier developed in this thesis requires labelled fall windows for training. In a consumer deployment scenario, labelled falls are expensive to obtain: they require a controlled data-collection session or a sufficiently long natural observation period. A newly installed device has no fall history at all—the *cold-start problem*.

An anomaly-detection formulation eliminates this requirement. Rather than learning to separate falls from ADL, a one-class model learns the boundary of the user's *normal* ADL behaviour exclusively from unlabelled, continuously collected sensor data. Any window whose topological descriptor $\phi_\lambda(w) \in \mathbb{R}^{24}$ lies far from the learned ADL manifold is flagged as a candidate fall. Concretely, a one-class SVM (Schölkopf, Platt, Shawe-Taylor, Smola, & Williamson, 2001) or an isolation forest trained on $\{\phi_\lambda(w)\colon w \in \mathcal{W}^{(s)}_{\mathrm{ADL}}\}$ defines a decision boundary in $\mathbb{R}^{24}$ without any fall-class supervision.

Three properties make the TDA feature space particularly well-suited for this formulation.

1. **Geometric separability.** Fall windows exhibit pronounced H_0 structure (a sudden large cluster spread caused by the impulsive excursion in the delay-embedded cloud) that is structurally absent in ADL windows. The Cohen–Steiner stability theorem ensures that this geometric difference is preserved robustly across subjects and sensor noise levels.

2. **Inherent personalisation.** Each user's ADL cloud in $\mathbb{R}^{24}$ reflects that individual's movement style, pace, and activity repertoire. A one-class model trained on a user's own ADL is a personalised model by construction, with no explicit personalisation step required.

3. **Threshold-sweep compatibility.** The anomaly score (distance to the ADL boundary) is a scalar, and the decision-threshold sweep mechanism of Section 5.4.1 applies directly: sweeping the boundary standoff distance over a grid $\mathcal{T}$ trades recall against precision in exactly the same way as the probabilistic threshold.

The expected engineering trade-off is an elevated false-positive rate compared to the binary classifier: vigorous ADL activities (running, stair climbing, sudden direction changes) may appear anomalous relative to a training distribution consisting mainly of slow or sedentary activities. Mitigations include diverse ADL coverage during the initial calibration period and an activity-aware normalization that conditions the anomaly score on the subject's current motion intensity level.

From a deployment perspective, the one-class approach enables a qualitatively different product workflow: a user installs the application, wears the device for two to three days of ordinary activity, and the personalised fall detector is ready—without any fall simulation, clinical annotation, or server-side retraining. This is the most direct path from the current thesis to a consumer-ready product.

5.9.7 Few-shot personalisation beyond threshold

An alternative personalisation axis beyond threshold tuning is to adapt the feature map ϕ_λ itself to the individual subject via meta-learning. A model-agnostic meta-learning (MAML) formulation could learn a λ initialisation that adapts quickly to a new subject with $k = 5$–10 labelled examples. This extends H4 from threshold personalisation to full representation personalisation, and remains compatible with the CPU-only constraint if the inner loop is restricted to scalar hyperparameter updates.

6. Conclusion

The central question of this thesis was whether persistent homology, applied to short wearable accelerometer windows through a delay-embedding pipeline, can support recall-oriented fall detection that is rigorous in evaluation, principled in feature construction, and lightweight enough for edge deployment. Four research hypotheses structured the investigation, which was carried out across three public benchmark datasets—MobiFall, SisFall, and FAD_40Hz—under a three-protocol subject-aware evaluation framework. What follows summarises the findings at three levels: mathematical, statistical, and engineering.

6.1 Mathematical Perspective

Representation effectiveness (H1). The feature map $\phi_\lambda \colon \mathbb{R}^{n\times 3} \to \mathbb{R}^{24}$, with hyperparameters λ^*_K identified by Bayesian optimisation, achieves mean subject-general F_2 of 0.9709 (LR) and 0.9700 (SVM) on SisFall, 0.9429 (LR) and 0.9592 (SVM) on FAD_40Hz, and 0.9049 (LR) and 0.8918 (SVM) on MobiFall under the leave-one-subject-out protocol. All values exceed the pre-specified thresholds (≥ 0.93 on SisFall and FAD_40Hz; ≥ 0.80 on MobiFall) for at least one classifier, and recall is above 0.95 across all dataset-classifier combinations. H1 is conditionally confirmed: the representation meets its own targets, subject to the caveat that no baseline comparison has been conducted.

The theoretical underpinning is the Cohen–Steiner stability theorem: the bottleneck distance between two persistence diagrams is bounded by the supremum norm of the perturbation between the corresponding input functions. This is a theorem rather than an empirical claim, and it provides a formal noise-robustness certificate that no competing feature family—handcrafted statistical features, spectral coefficients, or deep-network embeddings—possesses in comparable form.

Dataset-specific topological geometry (H2). The three datasets converged to structurally distinct optimal configurations: MobiFall selected a SparseRips complex at a 5.0 s window with Chebyshev distance, SisFall selected an Alpha complex at a 1.0 s window, and FAD_40Hz selected a standard Rips complex at a 1.0 s window with Manhattan distance. Every dimension of the hyperparameter tuple λ varies across at least two datasets. H2 is qualitatively confirmed: no universal configuration exists for these three datasets.

The configuration differences have a geometric explanation. The Alpha complex derives from the Euclidean Delaunay triangulation and suits convex, well-conditioned point clouds; SisFall's controlled fall recordings produce exactly that geometry. SparseRips handles large and noisy clouds by landmark subsampling, which is what MobiFall's continuous-wear data requires. The window-length difference is not incidental: MobiFall falls need a 5.0 s window to capture the pre-impact, impact, and post-fall phases together, while SisFall and FAD_40Hz falls are already discriminable within 1.0 s.

Representation dominates classifier choice (H3). The absolute difference in General-protocol F_2 between LR and SVM is 0.0131 on MobiFall, 0.0009 on SisFall, and 0.0163 on FAD_40Hz—all below 0.02. By contrast, the F_2 range across 200 Optuna trials is 0.179 on MobiFall, 0.053 on SisFall, and 0.108 on FAD_40Hz. The ratio of λ-tuning variation to classifier gap is 13.7× on MobiFall, 59× on SisFall, and 6.6× on FAD_40Hz.

H3 is strongly confirmed. At the optimal representation, ϕ_{λ^*} transforms the nonlinear fall-versus-ADL boundary into an approximately linearly separable problem in $\mathbb{R}^{24}$. Once that transformation is in place, both logistic regression (direct linear boundary) and SVM (maximum-margin separator, with or without an RBF kernel) operate near their theoretical limit and produce near-equivalent results. The discriminative information lives in the map ϕ_{λ^*}, not in the classifier family.

6.2 Statistical Perspective

Threshold personalisation as a distributional phenomenon (H4). The subject-level sweep gain $\Delta F_2(s) = F_2(\tau^*(s), s) - F_2(\hat{\tau}_0(s), s)$, where $\hat{\tau}_0(s)$ is the PR-curve trained threshold, has a right-skewed distribution in all six dataset-classifier combinations. The mean gains are 0.035–0.047 on MobiFall, 0.025–0.035 on FAD_40Hz, and 0.004–0.008 on SisFall. The coefficient of variation exceeds 50% in all cases, reaching 138% for SisFall LR—indicating that the distribution is highly dispersed relative to its mean.

H4 is confirmed in a refined form. The proposal predicted that the distribution of Δ_s would contain both positive and negative values. In practice, all gains are non-negative by construction (the sweep is an argmax), but the distribution is far from uniform: it concentrates near zero for the majority of subjects while a small minority achieves gains exceeding 0.10 units of F_2. Subject 11 on FAD_40Hz (LR) gains +0.1165, lifting F_2 from 0.8046 to 0.9211 by adjusting the threshold alone. This heterogeneity is incompatible with a uniform-shift explanation: if the classifier's posterior probabilities were globally miscalibrated in the same direction for all subjects, a single global threshold offset would equalise all gains. The observed distribution implies that the optimal decision boundary location is genuinely subject-specific.

Evaluation integrity under LOSO. The leave-one-subject-out protocol with $R = 15$ independent class-balancing repetitions separates two sources of variability that are conflated in random train-test splits: (i) within-subject repetition variance, captured by the repetition-level t-confidence intervals, and (ii) across-subject heterogeneity, visible in the per-subject tables. The Naive protocol consistently produces results comparable to or marginally lower than the General protocol on SisFall (Naive $F_2 = 0.9713$ vs. General 0.9709 for LR), suggesting that LOSO is not materially penalised by train-test class mismatch on large, diverse datasets. On MobiFall, where the subject pool is small (9 subjects), the repetition-level confidence intervals are wider (± 0.039 for LR), correctly reflecting the limited statistical power of a 9-fold LOSO evaluation.

The Personal protocol reveals a further finding: full per-subject model training degrades F_2 severely on MobiFall ($\Delta = -0.172$ for LR) but only modestly on SisFall ($\Delta = -0.030$ for LR), with the magnitude of degradation inversely correlated with the subject pool size. This confirms that personalised model training is only beneficial when sufficient per-subject data is available, and that the threshold sweep is the appropriate personalisation mechanism when data is sparse.

6.3 Engineering Perspective

Inference-time personalisation without retraining. The central engineering contribution of the thesis is the identification of decision-threshold optimisation as a first-class personalisation mechanism that is qualitatively different from all prior alternatives. The threshold $\tau \in \mathcal{T} = \{0.05, \ldots, 0.95\}$ is the only parameter that can be adjusted at inference time, without gradient computation, without access to the training data, and without transmission of data to a server. The computational cost of a full per-subject sweep is $O(|\mathcal{T}| \cdot |W_s|)$ comparisons, amounting to less than one second of wall time for all subjects in any of the three datasets. The ratio to the Bayesian optimisation cost exceeds 10^3.

This asymmetry has a direct consequence for product design. A wearable fall detection system based on the proposed pipeline can be shipped with a population-level model (ϕ_{λ^*}, f_θ) and personalised to each user at installation time by sweeping the threshold over a small held-out calibration set. No server round-trip is required. No labeled falls are required from the target user in the first instance—the sweep can be initialised from simulated events during setup and refined as real events accumulate. This is the only personalisation mechanism among those surveyed in this thesis that simultaneously satisfies the zero-retraining, privacy-preserving, and inference-time requirements.

CPU-only deployment profile. The entire pipeline—delay embedding, simplicial filtration, boundary matrix reduction, feature vectorization, linear classification, and threshold sweep—is executable on commodity CPU hardware. Persistent homology computation on the window sizes and embedding dimensions considered here ($m \leq 5$, $|P_\lambda| \leq 150$ points) completes in tens of milliseconds per window on a desktop-class processor core. The Bayesian optimisation consumed $\sim$75 min of CPU time on a 40-core HPC node; this cost is paid once per dataset, not per user.

The ARM Cortex-A processors present in current mid-range smartphones are comfortably capable of executing the per-window inference pipeline in real time at 50 Hz. Modern smartwatch SoCs are more constrained, but the per-window computation—feature extraction plus a matrix-vector multiply—is within their budget. The absence of a GPU requirement eliminates the most significant barrier to embedded deployment and makes the pipeline uniquely well-suited to the battery-constrained wearable context.

Failure mode analysis and minimum calibration set size. The most significant failure mode identified in this thesis is the collapse of the Personal protocol on MobiFall: F_2 drops from 0.9049 (General) to 0.7325 (Personal) for LR, and from 0.8918 to 0.5981 for SVM. The cause is statistical rather than algorithmic: MobiFall contains 9 subjects with approximately 20–30 fall instances per subject, which is insufficient for stable within-subject classifier estimation. The implication for system designers is concrete: a minimum calibration set size is required before full personal model training is beneficial. For the datasets studied here, that threshold lies somewhere between the 9 subjects of MobiFall and the 13 subjects of FAD_40Hz. Below this threshold, the threshold sweep is the safer personalisation mechanism.

This failure mode is not a weakness of TDA specifically; it would affect any classifier trained on sparse per-subject data. It is, however, an important calibration of expectations for real deployment: the naive assumption that "more personalization is always better" is empirically refuted by the MobiFall Personal protocol results.

Path to consumer deployment. Taken together, the three engineering findings—inference-time personalisation, CPU-only operation, and the identified minimum calibration requirement—define a concrete deployment pathway:

1. A population-level model $(\phi_{\lambda^*},\ f_\theta)$ is trained once on a large labeled dataset and shipped with the device.
2. At installation, the user performs a brief simulated fall session (or the device waits to accumulate confirmed fall events from normal wear). The per-subject threshold $\tau^*(s)$ is identified from the calibration scores in under one second.
3. Subsequent detections use $\tau^*(s)$. As more events accumulate, the threshold estimate can be refined.
4. No server communication is required for any of steps 1–3; all computation is on-device.

The natural next step beyond threshold personalisation is one-class anomaly detection: training the model exclusively on the user's own activities of daily living, treating falls as out-of-distribution events without requiring any labeled fall data. The Cohen–Steiner stability of the TDA features ensures that the ADL manifold in $\mathbb{R}^{24}$ is robustly estimated from normal wear data, and the threshold sweep mechanism extends directly to the standoff distance from the one-class boundary. This approach eliminates the cold-start problem entirely and represents the most direct path from the current thesis to a consumer-ready, zero-label deployment.

6.4 Limitations and Open Questions

Four limitations bound the strength of the conclusions.

No baseline comparison was conducted. H1 is therefore validated only against its own pre-specified targets; the claim that TDA outperforms statistical-feature or threshold-based baselines on these datasets cannot be made from the current evidence. Establishing this comparison is the highest-priority open task.

The cross-application drop matrix $M_{K,K'}$ was not computed. H2 is confirmed qualitatively (distinct optimal configurations) but not quantitatively: the actual performance penalty from transplanting λ_K^* to a different dataset K' remains unmeasured.

All experiments assume a fixed sensor placement at the waist or in the trouser pocket. The stability theorem guarantees noise robustness within a placement but does not extend to fundamentally different placements (wrist, chest, thigh); this is an open practical question.

Finally, the threshold sweep identifies $\tau^*(s)$ on the same test windows on which it is evaluated, introducing an optimism bias. The reported $\overline{\Delta F_2}$ values are retrospective diagnostics of heterogeneity, not valid prospective performance estimates; a prospective validation would require a separate held-out calibration set per subject.

6.5 Closing Remarks

Can a TDA pipeline for wearable fall detection be simultaneously principled, lightweight, and subject-aware? Across three datasets and two classifiers, the evidence points to yes on all three counts. The pipeline is principled because its central property—feature stability—is guaranteed by a theorem rather than established empirically. It is lightweight because it runs without GPU acceleration at latencies compatible with real-time embedded operation. It is subject-aware because the decision threshold can be personalised at inference time, without retraining, at a computational overhead that is more than a thousand times smaller than the one-time Bayesian optimisation cost.

The most broadly transferable finding is the H3 result. A gap of 0.10–0.18 units of F_2 separates a well-tuned topological representation from a poorly tuned one. The gap between the best and worst classifier at the optimal representation is below 0.02. In this problem—and probably in any wearable time-series classification task where discriminative information is encoded in short-window shape—selecting the right representation is the dominant engineering decision. The classifier, once the representation is fixed, is largely interchangeable.

REFERENCES

Adams, H., Emerson, T., Kirby, M., Neville, R., Peterson, C., Shipman, P., Chepushtanova, S., Hanson, E., Motta, F., & Ziegelmeier, L. (2017). Persistence images: A stable vector representation of persistent homology. *Journal of Machine Learning Research*, *18*, 1–35.

Akiba, T., Sano, S., Yanase, T., Ohta, T., & Koyama, M. (2019). Optuna: A next-generation hyperparameter optimization framework. *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2623–2631. https://doi.org/10.1145/3292500.3330701

Batchuluun, G., Lee, S. H., Nguyen, T. D., Pham, T. D., Arsalan, M., & Park, K. R. (2022). Deep learning-based fall detection algorithm using ensemble model of convolutional neural network and recurrent neural network. *Sensors*, *22*(3), 1270. https://doi.org/10.3390/s22031270

Bergstra, J., Bardenet, R., Bengio, Y., & Kégl, B. (2011). Algorithms for hyper-parameter optimization. *Advances in Neural Information Processing Systems*, *24*, 2546–2554.

Bergstra, J., & Bengio, Y. (2012). Random search for hyper-parameter optimization. *Journal of Machine Learning Research*, *13*, 281–305.

Borsuk, K. (1948). On the imbedding of systems of compacta in simplicial complexes. *Fundamenta Mathematicae*, *35*, 217–234. https://doi.org/10.4064/fm-35-1-217-234

Bubenik, P. (2015). Statistical topological data analysis using persistence landscapes. *Journal of Machine Learning Research*, *16*, 77–102.

Carlsson, G. (2009). Topology and data. *Bulletin of the American Mathematical Society*, *46*(2), 255–308. https://doi.org/10.1090/S0273-0979-09-01249-X

Cavanna, N. J., Jahanseir, M., & Sheehy, D. R. (2015). A geometric perspective on sparse filtrations. *Proceedings of the 27th Canadian Conference on Computational Geometry (CCCG 2015)*, 1–6.

Chazal, F., de Silva, V., Glisse, M., & Oudot, S. (2016). *The structure and stability of persistence modules*. Springer International Publishing. https://doi.org/10.1007/978-3-319-42545-0

Cohen-Steiner, D., Edelsbrunner, H., & Harer, J. (2007). Stability of persistence diagrams. *Discrete & Computational Geometry*, *37*(1), 103–120. https://doi.org/10.1007/s00454-006-1276-5

Crawley-Boevey, W. (2015). Decomposition of pointwise finite-dimensional persistence modules. *Journal of Algebra and Its Applications*, *14*(5), 1550066. https://doi.org/10.1142/S0219498815500668 de la Hoz Franco, E., Ariza-Colpas, P., Quero, J. M., & Espinilla, M. (2014). Sensor-based systems for early detection of alzheimer's disease: A systematic review [Used as survey reference for fall/activity detection systems]. *Sensors*, *14*(10), 19244–19280. https://doi.org/10.3390/s141019244

Edelsbrunner, H., & Harer, J. L. (2010). *Computational topology: An introduction*. American Mathematical Society.

Edelsbrunner, H., Letscher, D., & Zomorodian, A. (2002). Topological persistence and simplification. *Discrete & Computational Geometry*, *28*(4), 511–533. https://doi.org/10.1007/s00454-002-2885-2

Gidea, M., & Katz, Y. (2018). Topological data analysis of financial time series: Landscapes of crashes. *Physica A: Statistical Mechanics and its Applications*, *491*, 820–834. https://doi.org/10.1016/j.physa.2017.09.028

GUDHI Editorial Board. (2015). *GUDHI user and reference manual*. GUDHI Editorial Board. https://gudhi.inria.fr/doc/latest/

Hutter, F., Hoos, H. H., & Leyton-Brown, K. (2011). Sequential model-based optimization for general algorithm configuration. *Learning and Intelligent Optimization (LION 5)*, *6683*, 507–523. https://doi.org/10.1007/978-3-642-25566-3_40

Igual, R., Medrano, C., & Plaza, I. (2013). Challenges, issues and trends in fall detection systems. *BioMedical Engineering OnLine*, *12*, 66. https://doi.org/10.1186/1475-925X-12-66

Kangas, M., Konttila, A., Lindgren, P., Winblad, I., & Jämsä, T. (2008). Comparison of low-complexity fall detection algorithms for body attached accelerometers. *Gait & Posture*, *28*(2), 285–291. https://doi.org/10.1016/j.gaitpost.2008.01.003

Ott, W., Tomforde, M., & Willis, P. N. (2015). One-sided shift spaces over infinite alphabets [Placeholder: replace with correct Ott et al. TDA/time-series paper if available]. *New York Journal of Mathematics*, *21*, 1–35.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, *12*, 2825–2830.

Pereira, C. M. M., & de Mello, R. F. (2015). Persistent homology for time series and spatial data clustering and analysis. *Expert Systems with Applications*, *42*(15-16), 6026–6038. https://doi.org/10.1016/j.eswa.2015.04.010

Platt, J. C. (1999). Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In A. J. Smola, P. Bartlett, B. Schölkopf, & D. Schuurmans (Eds.), *Advances in large margin classifiers* (pp. 61–74). MIT Press.

Saleh, M., Abbas, M., & Le Bouquin Jeannès, R. (2021). FallAllD: An open dataset of human falls and activities of daily living for classical and deep learning applications. *IEEE Sensors Journal*, *21*(2), 1849–1858. https://doi.org/10.1109/JSEN.2020.3018335

Sauer, T., Yorke, J. A., & Casdagli, M. (1991). Embedology. *Journal of Statistical Physics*, *65*(3-4), 579–616. https://doi.org/10.1007/BF01053745

Schölkopf, B., Platt, J. C., Shawe-Taylor, J., Smola, A. J., & Williamson, R. C. (2001). Estimating the support of a high-dimensional distribution. *Neural Computation*, *13*(7), 1443–1471. https://doi.org/10.1162/089976601750264965

Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, *27*(3), 379–423. https://doi.org/10.1002/j.1538-7305.1948.tb01338.x

Sucerquia, A., López, J. D., & Vargas-Bonilla, J. F. (2017). SisFall: A fall and movement dataset. *Sensors*, *17*(1), 198. https://doi.org/10.3390/s17010198

Takens, F. (1981). Detecting strange attractors in turbulence. In D. Rand & L.-S. Young (Eds.), *Dynamical systems and turbulence, warwick 1980* (pp. 366–381, Vol. 898). Springer. https://doi.org/10.1007/BFb0091924

Tinetti, M. E., Speechley, M., & Ginter, S. F. (1988). Risk factors for falls among elderly persons living in the community. *New England Journal of Medicine*, *319*(26), 1701–1707. https://doi.org/10.1056/NEJM198812293192604

Umeda, Y. (2017). Time series classification via topological data analysis. *Information and Media Technologies*, *12*, 228–239. https://doi.org/10.11185/imt.12.228

Vavoulas, G., Pediaditis, M., Chatzaki, C., Spanakis, E. G., & Tsiknakis, M. (2014). The MobiFall dataset: Fall detection and classification with a smartphone. *International Journal of Monitoring and Surveillance Technologies Research*, *2*(1), 44–56. https://doi.org/10.4018/ijmstr.2014010103

Wang, X., Ellul, J., & Azzopardi, G. (2020). Elderly fall detection systems: A literature survey. *Frontiers in Robotics and AI*, *7*, 71. https://doi.org/10.3389/frobt.2020.00071

Wild, D., Nayak, U. S. L., & Isaacs, B. (1981). How dangerous are falls in old people at home? *British Medical Journal*, *282*(6260), 266–268. https://doi.org/10.1136/bmj.282.6260.266

World Health Organization. (2021). Falls [Accessed: 2024]. https://www.who.int/news-room/fact-sheets/detail/falls

World Health Organization. (2022). Ageing and health [Accessed: 2024]. https://www.who.int/news-room/fact-sheets/detail/ageing-and-health

Xu, X., Yao, D., Li, J., & Zhang, D. (2019). Topological data analysis-based feature extraction for human activity recognition [Placeholder: verify DOI and page range before submission]. *IEEE Access*, *7*, 1–10.

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